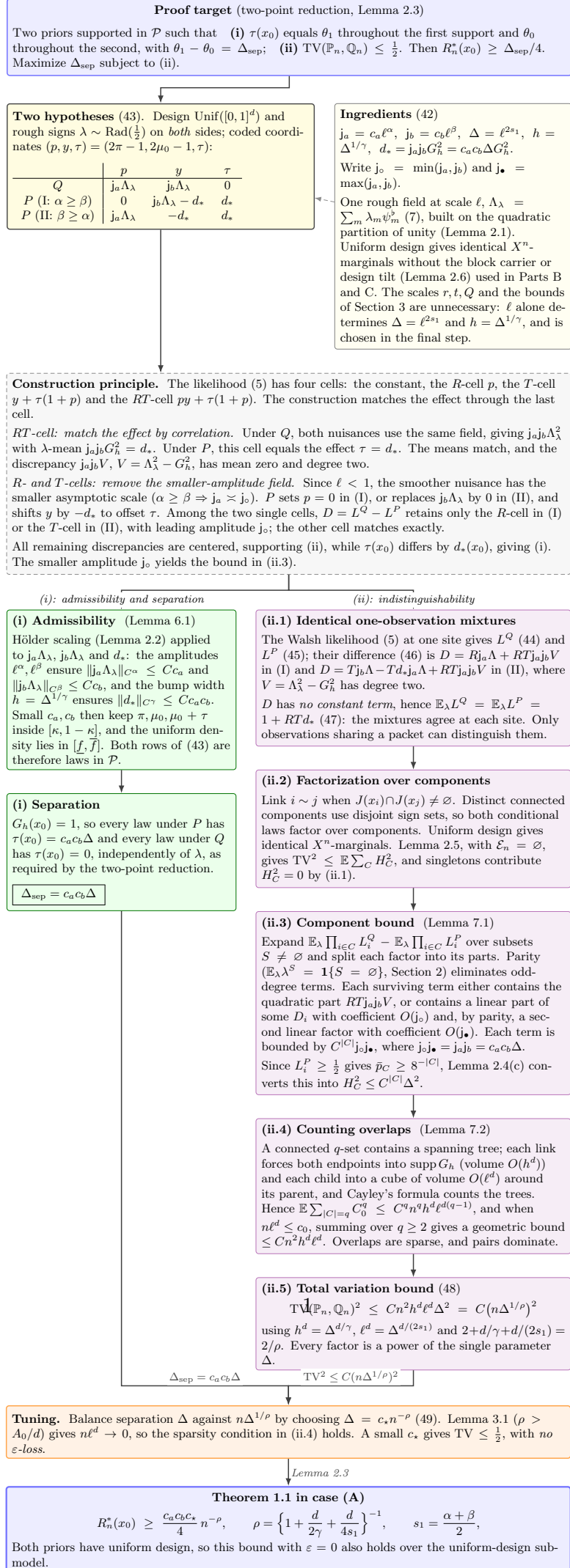


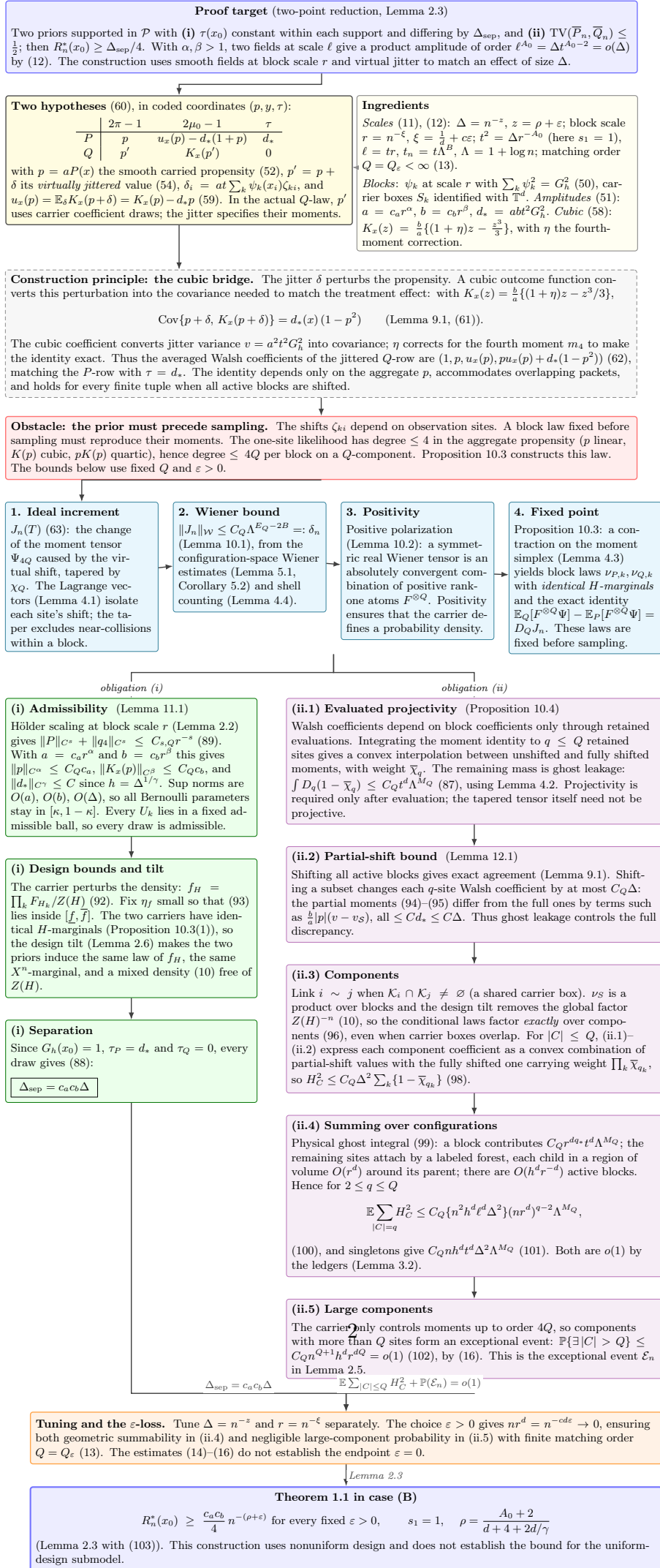
Case (A): both nuisances rough ($\alpha, \beta \leq 1$)

Part A of *Minimax lower bound for pointwise CATE estimation with binary outcomes and rough design*. All references follow the paper.



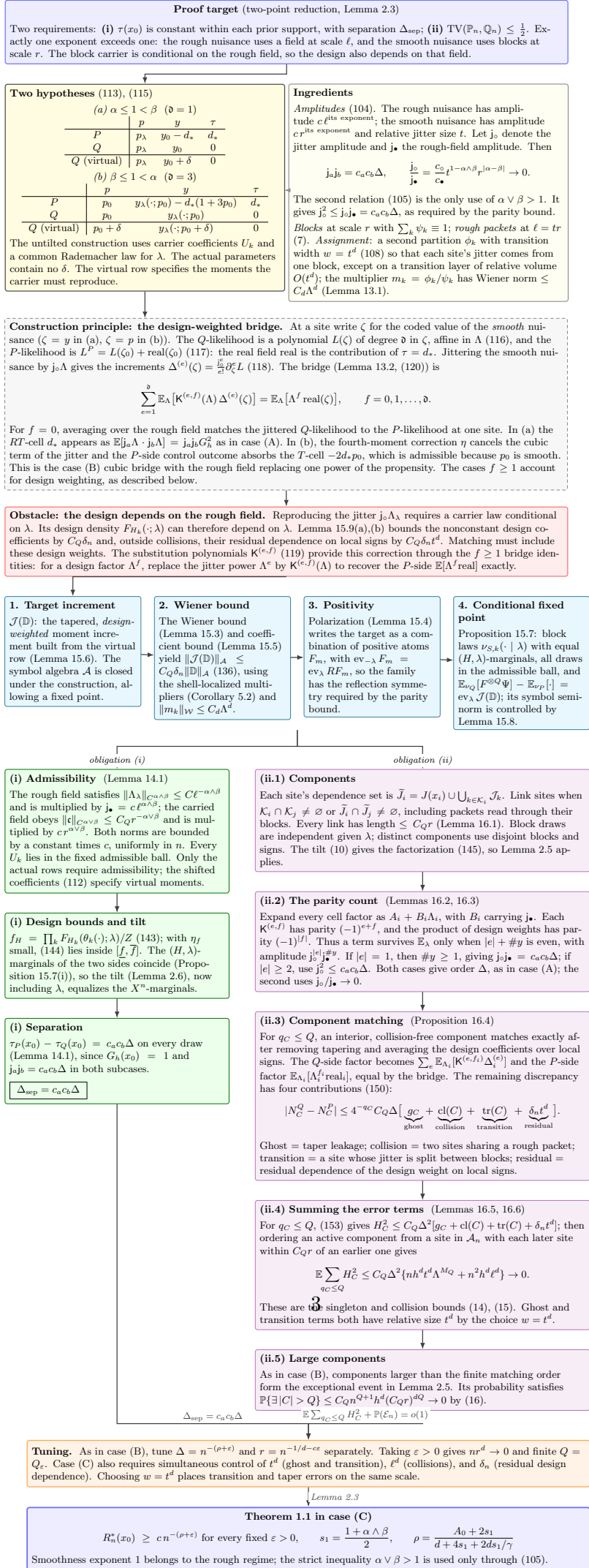
Case (B): both nuisances smooth ($\alpha, \beta > 1$)

Part B. Virtual jitter replaces rough-field averaging; a carrier fixed before sampling realizes the required moments.



Case (C): mixed smoothness

Part C. A carrier conditional on the rough field reproduces design-weighted moments; four error terms control the remaining discrepancy.



Minimax lower bound for pointwise CATE estimation with binary outcomes and rough design: the complete low-smoothness theorem

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Abstract

We prove a pointwise minimax lower bound for conditional average treatment-effect estimation in a Bernoulli-outcome model whose covariate density is assumed only to be bounded above and below. With $A_0 = \alpha + \beta$, $s_1 = \{(\alpha \wedge 1) + (\beta \wedge 1)\}/2$, $D_1 = d + 4s_1 + 2ds_1/\gamma$ and $\rho = (A_0 + 2s_1)/D_1$, in the regime $d > A_0(2 + d/\gamma)$ the minimax risk at an interior point is at least $c_\varepsilon n^{-(\rho+\varepsilon)}$ for every $\varepsilon > 0$; when both nuisance exponents are at most one the bound holds with $\varepsilon = 0$, and it holds over the uniform-design submodel. The proof is organised in three cases according to which nuisance functions are rough, with the shared tools collected in Part I.

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Part I

Model, theorem, and common tools

1 Model and main result

We observe independent copies of $Z = (X, A, Y)$, where

$$X \in [0, 1]^d, \quad A \mid X = x \sim \text{Bernoulli}\{\pi(x)\}, \quad Y \mid (X = x, A = a) \sim \text{Bernoulli}\{\mu_0(x) + a\tau(x)\}. \quad (1)$$

Fix $x_0 \in (0, 1)^d$. For $s > 0$, $\mathcal{H}^s(L)$ denotes the usual isotropic Hölder ball of radius L on $[0, 1]^d$, with the convention that its elements are restrictions of functions defined on a fixed neighbourhood of the cube. Fix constants

$$0 < \underline{f} < 1 < \bar{f} < \infty, \quad 0 < \kappa < \frac{1}{4},$$

and radii L_π, L_0, L_τ such that the constants $1/2$ are interior points of $\mathcal{H}^\alpha(L_\pi)$ and $\mathcal{H}^\beta(L_0)$ (e.g. $L_\pi, L_0 > 1/2$). The class $\mathcal{P} = \mathcal{P}(\alpha, \beta, \gamma)$ consists of all laws (1) with design density f and nuisance functions such that

$$\begin{aligned} \underline{f} &\leq f \leq \bar{f}, & \int f &= 1, \\ \pi &\in \mathcal{H}^\alpha(L_\pi), & \mu_0 &\in \mathcal{H}^\beta(L_0), & \tau &\in \mathcal{H}^\gamma(L_\tau), \\ \kappa &\leq \pi \leq 1 - \kappa, & \kappa &\leq \mu_0 \leq 1 - \kappa, & \kappa &\leq \mu_0 + \tau \leq 1 - \kappa. \end{aligned}$$

No smoothness is imposed on the design density: $\mathcal{P}$ is the *rough-design* class, in which the design is constrained only by the two-sided bound; the design density is one of the objects that the lower-bound constructions perturb, and the rate below is specific to this class. The pointwise absolute-error minimax risk is

$$R_n^*(x_0) = \inf_{\hat{\tau}} \sup_{P \in \mathcal{P}} \mathbb{E}_P |\hat{\tau}(x_0) - \tau_P(x_0)|. \quad (2)$$

Write

$$A_0 = \alpha + \beta, \quad s_1 = \frac{(\alpha \wedge 1) + (\beta \wedge 1)}{2}, \quad D_1 = d + 4s_1 + \frac{2ds_1}{\gamma}, \quad \rho = \frac{A_0 + 2s_1}{D_1}. \quad (3)$$

Theorem 1.1 (Rough-design pointwise lower bound). *Let $\alpha, \beta, \gamma > 0$ satisfy*

$$d > A_0 \left(2 + \frac{d}{\gamma} \right).$$

For every fixed $\varepsilon > 0$ there are $c_\varepsilon > 0$ and $n_\varepsilon < \infty$ such that

$$R_n^*(x_0) \geq c_\varepsilon n^{-(\rho+\varepsilon)}, \quad n \geq n_\varepsilon. \quad (4)$$

If $\alpha, \beta \leq 1$ the bound holds with $\varepsilon = 0$, and the two hypotheses realising it have exactly uniform design, so that (4) holds over the uniform-design submodel as well.

The proof splits according to which nuisance functions are rough (Hölder exponent at most one):

$$(A) \quad \alpha, \beta \leq 1; \quad (B) \quad \alpha, \beta > 1; \quad (C) \quad \alpha \leq 1 < \beta \quad \text{or} \quad \beta \leq 1 < \alpha,$$

proved in Parts A, B and C respectively; every (α, β) falls in exactly one case. The parts share the model, the coded Walsh likelihood, the two-point reduction, the component principle, the design tilt, the scales, the interpolation system and taper on the torus, the finite moment simplex, the configuration-space Wiener estimates, and the polarization identity, all collected in Part I. They differ in the construction of the two hypotheses:

- in case (A) both nuisances are carried by one rough Rademacher field at the fine scale ℓ , the design is uniform, and the proof is a direct Walsh-parity computation on packet-overlap components;
- in case (B) both nuisances are smooth, the outcome is a cubic function of the propensity (the cubic bridge), and the design carries a positive moment carrier whose virtual jitter is made of independent signs attached to the observation sites;
- in case (C) one nuisance is rough and the other is carried by the blocks; the carrier must then be built conditionally on the rough field, which makes the design informative about it, and the two sides are matched after design weighting.

The formula $s_1 = \{(\alpha \wedge 1) + (\beta \wedge 1)\}/2$ is the same in all cases: $s_1 = (\alpha + \beta)/2$ in (A), $s_1 = 1$ in (B), and $s_1 = (1 + \alpha \wedge \beta)/2$ in (C). The regime $d > A_0(2 + d/\gamma)$ is the low-smoothness regime of Kennedy, Balakrishnan, Robins and Wasserman [1]; in case (A) it reads $s_1 < (d/4)/(1 + d/(2\gamma))$ and the rate is $n^{-1/(1+d/(2\gamma)+d/(4s_1))}$, their rate. The exponent 1 is treated as rough in case (C); there the strict inequality $\alpha \vee \beta > 1$ is used only through (105).

Remark 1.2 (Scope). *The theorem is stated for an interior target point, absolute-error risk, and Hölder balls containing the fair-sign baseline $1/2$ as an interior point. The strict inequalities $\underline{f} < 1 < \bar{f}$ place a neighbourhood of the uniform density inside the design class; the block carriers of Parts B and C are perturbations of the uniform density at the block scale, and Part A uses the uniform design itself.*

2 Common tools

2.1 Coded signs and the Walsh likelihood

Let $R = 2A - 1$, $T = 2Y - 1$, $p = 2\pi - 1$, $y = 2\mu_0 - 1$. Relative to independent fair signs, the parameter (p, y, τ) has likelihood

$$L_{p,y,\tau} = 1 + Rp + T\{y + \tau(1 + p)\} + RT\{py + \tau(1 + p)\}. \quad (5)$$

Thus equality of the four Walsh coefficients is equality of all four binary cells. For a function g of iid Rademacher signs $\lambda = (\lambda_m)$, the Walsh norm $\|g\|_W$ is the ℓ^1 norm of its Walsh coefficients; it is an algebra norm, $\|g\|_\infty \leq \|g\|_W$, and $|\mathbb{E}_\lambda g| \leq \|g\|_W$. The *degree* of a Walsh monomial $\lambda^S = \prod_{m \in S} \lambda_m$ is $|S|$; since $\lambda_m^2 = 1$, $\lambda^S \lambda^{S'} = \lambda^{S \Delta S'}$, so the degree of a product of monomials is congruent modulo two to the sum of the degrees. A Walsh polynomial is *even* (*odd*) if all its monomials have even (odd) degree, i.e. if it is invariant (changes sign) under $\lambda \mapsto -\lambda$; an odd Walsh polynomial has zero mean, and more generally $\mathbb{E}_\lambda \lambda^S = \mathbf{1}\{S = \emptyset\}$.

2.2 Partition of unity, bump, and the rough field

Lemma 2.1 (Smooth quadratic partition of unity). *There exists $\phi \in C_c^\infty(\mathbb{R}^d)$ such that*

$$\sum_{m \in \mathbb{Z}^d} \phi(u - m)^2 = 1, \quad u \in \mathbb{R}^d. \quad (6)$$

Proof. Choose a nonnegative $\eta \in C_c^\infty(\mathbb{R})$ which is positive on $(-1, 1)$, so that $\sum_{k \in \mathbb{Z}} \eta(u - k)^2 > 0$ for all u ; the denominator of $\phi_1(u) = \eta(u) \{\sum_k \eta(u - k)^2\}^{-1/2}$ is smooth, strictly positive and periodic, so $\phi_1 \in C_c^\infty(\mathbb{R})$ and $\sum_k \phi_1(u - k)^2 = 1$. Put $\phi(u) = \prod_{j=1}^d \phi_1(u_j)$. $\square$

Fix $G \in C_c^\infty(\mathbb{R}^d)$ with $0 \leq G \leq 1$, $G(0) = 1$, and put $G_h(x) = G((x - x_0)/h)$; since x_0 is interior, $\text{supp } G_h \subset [0, 1]^d$ for small h . For a fine scale $\ell \leq h$ the *rough packets* and the *rough field* are

$$\psi_m^b(x) = G_h(x) \phi\left(\frac{x - x_0}{\ell} - m\right), \quad \Lambda_\lambda(x) = \sum_{m \in \mathbb{Z}^d} \lambda_m \psi_m^b(x), \quad \lambda_m \stackrel{\text{iid}}{\sim} \text{Rad}(\tfrac{1}{2}), \quad (7)$$

where only the $O((h/\ell)^d)$ packets that are not identically zero need be retained. Then $\sum_m \psi_m^b(x)^2 = G_h(x)^2$; at most N_0 packets meet a point; $|\Lambda_\lambda| \leq N_0$; and with $J(x) = \{m : \psi_m^b(x) \neq 0\}$, $J(x) \cap J(x') \neq \emptyset$ implies $x, x' \in \text{supp } G_h$ and $\|x - x'\|_\infty \leq R_0 \ell$, for fixed N_0, R_0 . For fixed x , $\Lambda_\lambda(x)$ is a Walsh polynomial of degree one with $\|\Lambda_\lambda(x)\|_W \leq N_0$; if $J(x_1), \dots, J(x_q)$ are pairwise disjoint, $\Lambda_\lambda(x_1), \dots, \Lambda_\lambda(x_q)$ are independent. The moments $M_j(x) := \mathbb{E} \Lambda_\lambda(x)^j$ are

$$M_0 = 1, \quad M_j = 0 \ (j \text{ odd}), \quad M_2 = G_h^2, \quad M_4 = \mu_4 := 3G_h^4 - 2 \sum_m (\psi_m^b)^4 = G_h^4 \{3 - 2q_4\}, \quad (8)$$

with $q_4(x) = \sum_{m \in \mathbb{Z}^d} \phi((x - x_0)/\ell - m)^4 \in (0, 1]$: indeed $\mathbb{E} \Lambda^4 = \sum_m \psi_m^4 + 3 \sum_{m \neq m'} \psi_m^2 \psi_{m'}^2 = 3(\sum_m \psi_m^2)^2 - 2 \sum_m \psi_m^4$, $\sum_m \psi_m^4 \leq (\sum_m \psi_m^2)^2 = 1$, and odd moments vanish by symmetry. The function q_4 is smooth at scale ℓ , ℓ -periodic, and does not depend on λ ; so $\mu_4 \in [G_h^4, 3G_h^4]$. The centred square $V(x) := \Lambda_\lambda(x)^2 - G_h(x)^2 = \sum_{m \neq m'} \lambda_m \lambda_{m'} \psi_m^b(x) \psi_{m'}^b(x)$ is a Walsh polynomial all of whose monomials have degree two, with $\|V(x)\|_W \leq N_0^2$.

Lemma 2.2 (Hölder scaling). *If $g \in C_c^\infty(\mathbb{R}^d)$ and $0 < q \leq 1$, then for every fixed $t > 0$ $\|g((\cdot - x_c)/q)\|_{C^t} \leq C_{g,t} q^{-t}$, and the same bound holds for finite products and for bounded-overlap sums of translates of such functions. The estimate is valid for noninteger t : derivatives up to $\lfloor t \rfloor$ scale in the displayed way, and the remaining Hölder seminorm gains the fractional power of q^{-1} .*

In particular the rough field satisfies $\|\Lambda_\lambda\|_{C^t} \leq C_t \ell^{-t}$ for every $t > 0$ (it is G_h times a bounded-overlap sum of translates of $\phi(\cdot/\ell)$, and $h \geq \ell$).

2.3 Two-point reduction and the component principle

Lemma 2.3 (Two-point reduction). *Let $\mathbb{P}_n, \mathbb{Q}_n$ be the laws of $(X_i, A_i, Y_i)_{i \leq n}$ obtained by drawing a law from a prior supported in $\mathcal{P}$ and then n iid triples from it, and suppose $\tau_P(x_0) = \theta_1$ for every law in the support of the first prior and $\tau_P(x_0) = \theta_0$ for every law in the support of the second, with $\theta_1 - \theta_0 = \Delta_{\text{sep}} > 0$. Then for every estimator*

$$\sup_{P \in \mathcal{P}} \mathbb{E}_P |\hat{\tau}(x_0) - \tau_P(x_0)| \geq \frac{\Delta_{\text{sep}}}{2} (1 - \text{TV}(\mathbb{P}_n, \mathbb{Q}_n)).$$

Proof. The supremum dominates both mixture risks; with $g = |\hat{\tau}(x_0) - \theta_1|$, $g' = |\hat{\tau}(x_0) - \theta_0|$, $g + g' \geq \Delta_{\text{sep}}$, so $\mathbb{E}_{\mathbb{P}_n} g + \mathbb{E}_{\mathbb{Q}_n} g' \geq \int (g + g') d(\mathbb{P}_n \wedge \mathbb{Q}_n) \geq \Delta_{\text{sep}}(1 - \text{TV})$. $\square$

For probability vectors P', Q' on a finite set write $H^2(P', Q') = \sum_w (\sqrt{P'(w)} - \sqrt{Q'(w)})^2 \in [0, 2]$.

Lemma 2.4 (Hellinger facts). (a) $\text{TV}(P', Q') \leq H(P', Q')$.

(b) $H^2(\otimes_C P'_C, \otimes_C Q'_C) \leq \sum_C H^2(P'_C, Q'_C)$.

(c) If $P'(w) \geq c > 0$ for all w , then $H^2(P', Q') \leq c^{-1} \sum_w |P'(w) - Q'(w)|^2$.

Proof. (a) $2 \text{TV} = \sum |\sqrt{P'} - \sqrt{Q'}|(\sqrt{P'} + \sqrt{Q'}) \leq H \{\sum (\sqrt{P'} + \sqrt{Q'})^2\}^{1/2} \leq 2H$. (b) $1 - H^2/2 = \sum \sqrt{P'Q'}$ is multiplicative and $\prod_C (1 - h_C) \geq 1 - \sum_C h_C$ for $h_C \in [0, 1]$. (c) $(\sqrt{P'} - \sqrt{Q'})^2 = (P' - Q')^2 / (\sqrt{P'} + \sqrt{Q'})^2 \leq (P' - Q')^2 / P'$. $\square$

Lemma 2.5 (Component principle). *Let $\mathbb{P}_n, \mathbb{Q}_n$ have the same X^n -marginal, and let $\mathcal{E}_n$ be an event depending on X^n only. Suppose that for every $x^n \notin \mathcal{E}_n$ there is a partition of $\{1, \dots, n\}$ into components C such that the conditional laws of $w^n = (R_i, T_i)_{i \leq n}$ given $X^n = x^n$ factor as $\mathbb{P}_n(w^n | x^n) = \prod_C P_C(w_C)$, $\mathbb{Q}_n(w^n | x^n) = \prod_C Q_C(w_C)$. Then, with $H_C^2 = H^2(P_C, Q_C)$,*

$$\text{TV}(\mathbb{P}_n, \mathbb{Q}_n) \leq \mathbb{P}(\mathcal{E}_n) + \left\{ \mathbb{E} \left[\mathbf{1}_{\mathcal{E}_n^c} \sum_C H_C^2 \right] \right\}^{1/2}.$$

Proof. Since the X^n -marginals coincide, $\text{TV}(\mathbb{P}_n, \mathbb{Q}_n) \leq \mathbb{E} \text{TV}(\mathbb{P}_n(\cdot | X^n), \mathbb{Q}_n(\cdot | X^n))$. Bound the conditional total variation by one on $\mathcal{E}_n$ and, on $\mathcal{E}_n^c$, by $\{\sum_C H_C^2\}^{1/2}$ (Lemma 2.4(a),(b)); then apply Jensen's inequality to the square root. $\square$

Lemma 2.6 (Design tilt). *Let ν_P, ν_Q be laws of a triple (H, λ, U) with the same (H, λ) -marginal, and let $(H, \lambda) \mapsto \tilde{f}_{H,\lambda}$ be a measurable family of nonnegative functions on $[0, 1]^d$ with $Z(H, \lambda) = \int \tilde{f}_{H,\lambda} \in [z_0, z_1]$, $0 < z_0 \leq z_1 < \infty$; put $f_{H,\lambda} = \tilde{f}_{H,\lambda}/Z(H, \lambda)$. Define the tilted priors*

$$d\Pi_{S,n} = \frac{Z(H, \lambda)^n}{\mathcal{Z}_n} d\nu_S, \quad \mathcal{Z}_n = \int Z^n d\nu_P = \int Z^n d\nu_Q, \quad S \in \{P, Q\}, \quad (9)$$

and let $\mathbb{P}_n, \mathbb{Q}_n$ be the mixtures obtained by drawing $(H, \lambda, U) \sim \Pi_{S,n}$, then $X_1, \dots, X_n$ iid with density $f_{H,\lambda}$, then $W_i = (R_i, T_i)$ from site likelihoods $L_i^S(w_i)$ relative to fair signs, depending on (x_i, H, λ, U) . Then the (H, λ) -marginals of $\Pi_{P,n}$ and $\Pi_{Q,n}$ coincide, hence so do the laws of $f_{H,\lambda}$ and the X^n -marginals of $\mathbb{P}_n$ and $\mathbb{Q}_n$; and the joint density of (X^n, W^n) with respect to Lebesgue measure times counting measure is

$$\bar{p}_S(x^n, w^n) = \frac{4^{-n}}{\mathcal{Z}_n} \int \prod_{i=1}^n \tilde{f}_{H,\lambda}(x_i) \prod_{i=1}^n L_i^S(w_i) d\nu_S(H, \lambda, U) : \quad (10)$$

the design normaliser Z^n cancels against the tilt and the untilted prior ν_S reappears.

Proof. Z is a function of (H, λ) alone, so $\mathcal{Z}_n$ is the same on both sides and the density $Z^n/\mathcal{Z}_n$ of $\Pi_{S,n}$ with respect to ν_S tilts only the common (H, λ) -marginal. The joint density is

$$\int \prod_{i=1}^n f_{H,\lambda}(x_i) \prod_{i=1}^n 4^{-1} L_i^S(w_i) d\Pi_{S,n}, \quad \prod_{i=1}^n f_{H,\lambda}(x_i) = Z(H, \lambda)^{-n} \prod_{i=1}^n \tilde{f}_{H,\lambda}(x_i),$$

and substituting (9) gives (10). $\square$

3 Scales

This section fixes the scales of Parts B and C. (Part A uses only Δ , h and ℓ , which it defines directly in (42) with $\varepsilon = 0$; the quantities r , t , Q and the ledgers below are not used there.) Fix $\varepsilon > 0$ small and put

$$z = \rho + \varepsilon, \quad \Delta = n^{-z}, \quad h = \Delta^{1/\gamma}, \quad r = n^{-\xi}, \quad \xi = \frac{1}{d} + c\varepsilon, \quad 0 < c < \frac{D_1}{dA_0}, \quad \Lambda = 1 + \log n. \quad (11)$$

Define the fine ratio t , the rough scale ℓ , and the taper scale by

$$t^{2s_1} = \Delta r^{-A_0}, \quad \ell = tr, \quad t_n = t\Lambda^B, \quad (12)$$

with a fixed exponent B chosen in each Part (Lemma 10.1, Lemma 15.3). Note that $\ell^{2s_1} = \Delta r^{2s_1 - A_0}$, which reduces to $\ell^{2s_1} = \Delta$ when $A_0 = 2s_1$; this is the relation between ℓ and Δ used in Part A. Let

$$Q = Q_\varepsilon = \max\left\{2, 1 + \left\lfloor \frac{[1 - zd/\gamma]_+}{cd\varepsilon} \right\rfloor\right\}, \quad (13)$$

the finite matching order of Parts B and C; it depends on ε but not on n . Constants denoted by C_Q may depend on the fixed quantities $d, \alpha, \beta, \gamma, \varepsilon, c$, the Hölder radii, and the fixed smooth cutoffs and feature system, but never on n or on dyadic shell indices.

Lemma 3.1 (Ordering). *In the regime of Theorem 1.1: $\gamma > A_0$, $\rho > A_0/d$, $d\rho < A_0 + 2s_1$, $\gamma(d + 4s_1) > dA_0$, and $\rho < 1$. Moreover, for all sufficiently small $\varepsilon > 0$: $t \rightarrow 0$, $\ell < r < h$, $nr^d = n^{-cd\varepsilon} \rightarrow 0$, and $t/t_n = \Lambda^{-B} \rightarrow 0$.*

Proof. The regime reads $\gamma d > A_0(2\gamma + d)$, so $d(\gamma - A_0) > 2\gamma A_0 > 0$ and $\gamma > A_0$; also $\gamma(d + 4s_1) > \gamma d > dA_0$. $\rho > A_0/d$ iff $d(A_0 + 2s_1) > A_0 D_1$ iff $2ds_1 > 4s_1 A_0 + 2ds_1 A_0/\gamma$ iff $d > A_0(2 + d/\gamma)$. $d\rho < A_0 + 2s_1$ because $D_1 > d$, and $\rho < 1$ because $A_0 + 2s_1 < d + 4s_1 < D_1$ (the regime gives $d > 2A_0$). From (12),

$$\log_n t = -\frac{z - \xi A_0}{2s_1} = -\frac{(\rho - A_0/d) + \varepsilon(1 - cA_0)}{2s_1} < 0$$

for small ε , so $t \rightarrow 0$ and $\ell < r$. $r < h$ iff $\xi > z/\gamma$, which for small ε is $\gamma > d\rho$, i.e. $\gamma D_1 > d(A_0 + 2s_1)$, i.e. $\gamma(d + 4s_1) > dA_0$. $\square$

Lemma 3.2 (Rate ledgers).

$$nh^d t^d \Delta^2 = n^{-\varepsilon(D_1 - cdA_0)/(2s_1)} \rightarrow 0, \quad (14)$$

$$n^2 h^d \ell^d \Delta^2 = n^{-\varepsilon\{D_1 - cd(A_0 - 2s_1)\}/(2s_1)} \rightarrow 0, \quad (15)$$

$$n^{Q+1} h^d r^{dQ} = n^{1 - zd/\gamma - cd\varepsilon Q} \rightarrow 0. \quad (16)$$

Fixed powers of Λ multiplying the left sides do not change the conclusions.

Proof. By (12), $d \log_n t = -d(z - \xi A_0)/(2s_1)$, so

$$\log_n(nh^d t^d \Delta^2) = 1 - \frac{zd}{\gamma} - \frac{d(z - \xi A_0)}{2s_1} - 2z = 1 + \frac{A_0}{2s_1} + \frac{cd\varepsilon A_0}{2s_1} - \frac{zD_1}{2s_1}.$$

Since $\rho D_1 = 2s_1 + A_0$, substituting $z = \rho + \varepsilon$ gives (14), negative because $c < D_1/(dA_0)$. Multiply by $nr^d = n^{-cd\varepsilon}$ and use $t^d r^d = \ell^d$ for (15). For (16): if $zd/\gamma \geq 1$ the exponent is negative for $Q \geq 2$; otherwise $Q > [1 - zd/\gamma]/(cd\varepsilon)$ by the choice of Q . $\square$

Nothing in this section distinguishes cases (B) and (C): only A_0 , s_1 , γ , d enter.

4 Interpolation, taper, moment simplex, and shells

4.1 An explicit finite interpolation system

Let

$$\iota(u) = (\cos 2\pi u_1, \sin 2\pi u_1, \dots, \cos 2\pi u_d, \sin 2\pi u_d) \in \mathbb{R}^{2d}.$$

Write

$$d_o(u, v) = \|\iota(u) - \iota(v)\| = 2 \left\{ \sum_{a=1}^d \sin^2(\pi(u_a - v_a)) \right\}^{1/2}.$$

Choose $r_0 > 0$ small and a smooth nondecreasing truncation $\kappa_\delta : [0, \infty) \rightarrow [0, 1]$ such that $\kappa_\delta(r) = r$ for $r \leq r_0$, $\kappa_\delta(r)$ is a fixed positive constant for $r \geq 2r_0$, and $\kappa_\delta(r) > 0$ for $r > 0$. Put $\delta(u, v) = \kappa_\delta\{d_o(u, v)\}$. Thus δ depends only on the torus difference, is smooth off the diagonal, is comparable to $d_o \wedge 1$, and is constant away from a fixed diagonal neighborhood. For a tuple $T = (u_1, \dots, u_q)$ of distinct points, define

$$P_i(T) = \prod_{j \neq i} \delta(u_i, u_j), \quad P_{\min}(T) = \min_i P_i(T). \quad (17)$$

Let $v_Q(u)$ contain all coordinate monomials in $\iota(u)$ of total degree at most $Q-1$, including repetitions and the constant. Its dimension depends only on (d, Q) .

Lemma 4.1 (Uniform Lagrange vectors). *For $q \leq Q$ and any distinct tuple $T = (u_1, \dots, u_q)$, there are vectors $c_i(T)$ satisfying*

$$v_Q(u_j)^\top c_i(T) = \mathbf{1}\{i = j\}, \quad \|c_i(T)\| \leq \frac{C_Q}{P_i(T)}. \quad (18)$$

The construction is equivariant under permutations of the tuple.

Proof. For $i \neq j$, put

$$L_{ij}(u; T) = \frac{\langle \iota(u) - \iota(u_j), \iota(u_i) - \iota(u_j) \rangle}{\|\iota(u_i) - \iota(u_j)\|^2}.$$

Then $L_{ij}(u_j; T) = 0$, $L_{ij}(u_i; T) = 1$, and it is a linear combination of the constant and the coordinates of $\iota(u)$. Set

$$L_i(u; T) = \prod_{j \neq i} L_{ij}(u; T).$$

This lies in the span of v_Q and satisfies $L_i(u_j; T) = \mathbf{1}\{i = j\}$. Each factor has coefficient norm at most $C/\delta(u_i, u_j)$; multiplication in the fixed finite-dimensional feature space therefore gives the second assertion. The formula is permutation equivariant. $\square$

Each coordinate of $c_i(T)$, being a coefficient of $\prod_{j \neq i} L_{ij}$, is a finite sum of products containing exactly one of the *elementary multipliers*

$$\frac{\{\iota(u_i) - \iota(u_j)\}_a}{\|\iota(u_i) - \iota(u_j)\|^2}, \quad \frac{\langle \iota(u_j), \iota(u_i) - \iota(u_j) \rangle}{\|\iota(u_i) - \iota(u_j)\|^2} \quad (19)$$

for every edge $\{i, j\}$ incident to i , each of inverse-distance order one.

4.2 Taper and close-tuple volume

With $\Lambda = 1 + \log n$ and the fine ratio t of (12), the taper scale is

$$t_n = t\Lambda^B, \quad (20)$$

where the fixed exponent $B > 0$ is chosen in each Part. For large n , $t_n < 1/4$. Let $\vartheta \in C^\infty([0, \infty); [0, 1])$ be zero on $[0, 1]$ and one on $[2, \infty)$, and define the symmetric taper

$$\chi_Q(T) = \prod_{i=1}^Q \vartheta\left(\frac{P_i(T)}{t_n}\right). \quad (21)$$

The product is extended by zero on the diagonals. Thus $\chi_Q = 0$ if $P_{\min} \leq t_n$, and $\chi_Q = 1$ if every $P_i \geq 2t_n$.

Lemma 4.2 (Close-tuple volume). *For each fixed $2 \leq q \leq Q$ and $0 < \varrho < 1/4$, the normalized dq -dimensional volume of*

$$\mathcal{B}_q(\varrho) = \{T \in (\mathbb{T}^d)^q : P_{\min}(T) \leq C\varrho\}$$

satisfies

$$|\mathcal{B}_q(\varrho)| \leq C_Q \varrho^d \{1 + \log(1/\varrho)\}^{q-2}. \quad (22)$$

Consequently, the set on which $\chi_Q \neq 1$ has volume at most

$$C_Q t^d \Lambda^{M_Q}, \quad M_Q = Bd + Q - 2. \quad (23)$$

Proof. Fix i and u_i . Up to a fixed multiplicative constant, integration in polar coordinates around u_i reduces the relevant part to

$$\int_{[0,1]^{q-1}} \mathbf{1} \left\{ \prod_{j=1}^{q-1} r_j \leq C\varrho \right\} \prod_{j=1}^{q-1} r_j^{d-1} dr_j.$$

With $w_j = r_j^d$, this is a constant times the volume of $\{w \in [0, 1]^{q-1} : \prod_j w_j \leq C^d \varrho^d\}$, which equals

$$O_Q \left(\varrho^d \{1 + \log(1/\varrho)\}^{q-2} \right)$$

by one integration, or induction on q . Sum over i . Substituting $\varrho = 2t_n = 2t\Lambda^B$ and using $\log(1/t) = O(\log n)$ gives (23). $\square$

4.3 Wiener notation and an interior coefficient law

For $m \geq 1$, let $\mathcal{W}_m$ be the real Wiener algebra on $\mathbb{T}^m$:

$$\|g\|_{\mathcal{W}_m} = \sum_{\lambda \in \mathbb{Z}^m} |\widehat{g}(\lambda)|.$$

For finite-dimensional vector-valued functions, sum the coordinate norms. The Wiener algebra is a Banach algebra and

$$\|gh\|_{\mathcal{W}_m} \leq \|g\|_{\mathcal{W}_m} \|h\|_{\mathcal{W}_m}, \quad \|g\|_{\infty} \leq \|g\|_{\mathcal{W}_m}. \quad (24)$$

Let M be the dimension of v_Q , and let $\Psi_{4Q}(U) \in \mathbb{R}^{N_Q}$ contain every nonconstant monomial in $U \in \mathbb{R}^M$ of total degree at most $4Q$.

Lemma 4.3 (Interior finite moment simplex). *There is a finite law Π_0 , supported in an arbitrarily small open ball $\mathbb{B}_0 \subset \mathbb{R}^M$, and a number $\delta_Q > 0$ such that every vector $v \in \mathbb{R}^{N_Q}$ with $\|v\| \leq \delta_Q$ is realized as*

$$\mathbb{E}_{\Pi_v} \Psi_{4Q}(U) = \mathbb{E}_{\Pi_0} \Psi_{4Q}(U) + v \quad (25)$$

by a probability law Π_v supported on the same finite set as Π_0 .

Proof. The functions 1 and the coordinates of Ψ_{4Q} are linearly independent polynomials on every open ball. Hence one can choose $N_Q + 1$ points $u^0, \dots, u^{N_Q} \in \mathbb{B}_0$ for which the vectors $(1, \Psi_{4Q}(u^j))$ are affinely independent; otherwise a nonzero polynomial of degree at most $4Q$ would vanish on an open set. Give all points strictly positive mass and call the resulting law Π_0 . Its moment vector is an interior point of the resulting N_Q -dimensional simplex, so a fixed Euclidean ball around it is contained in that simplex. This gives (25). $\square$

4.4 Dyadic configuration shells

We fix a dyadic partition that is also used in Section 5. Choose an integer m_0 so large that $2^{-m_0+3} < r_0$, every pair with $\delta(u, v) \leq 2^{-m_0+3}$ has a unique difference lift in $(-1/4, 1/4)^d$, and all rescaled fine-shell lifts lie in a fixed compact set contained in $(-2^{m_0-2}, 2^{m_0-2})^d$. Let $\rho \in C_c^\infty((1/4, 4))$ be nonnegative and satisfy

$$\sum_{m \in \mathbb{Z}} \rho(2^m r) = 1, \quad r > 0.$$

For $m \geq m_0$, set $\rho_m(r) = \rho(2^m r)$, and put

$$\rho_c(r) = \sum_{m < m_0} \rho(2^m r).$$

The latter is smooth on $[0, 1]$, vanishes in a neighborhood of zero, and together with the ρ_m 's forms a nonnegative partition of unity on $(0, 1]$. Write

$$\mathbb{M} = \{c\} \cup \{m : m \geq m_0\}, \quad \mu(c) = 0, \quad \mu(m) = m.$$

There are fixed constants $0 < c_\rho < C_\rho < \infty$ such that

$$c_\rho 2^{-\mu(m)} \leq r \leq C_\rho 2^{-\mu(m)} \quad \text{on } \text{supp } \rho_m, \quad m \in \mathbb{M}. \quad (26)$$

For the edge set

$$\mathfrak{E}_Q = \{\{i, j\} : 1 \leq i < j \leq Q\}, \quad E_Q := |\mathfrak{E}_Q| = \binom{Q}{2},$$

and a shell assignment $\mathbf{m} = (m_e)_{e \in \mathfrak{E}_Q} \in \mathbb{M}^{\mathfrak{E}_Q}$, define

$$\rho_{\mathbf{m}}(T) = \prod_{e=\{i,j\} \in \mathfrak{E}_Q} \rho_{m_e} \{\delta(u_i, u_j)\}. \quad (27)$$

Off the diagonals, $\sum_{\mathbf{m}} \rho_{\mathbf{m}} = 1$, with a locally finite sum.

For a shell assignment $\mathbf{m}$ and a vertex i write

$$M_i(\mathbf{m}) = \sum_{e \in \mathfrak{E}_Q : e \ni i} \mu(m_e). \quad (28)$$

Lemma 4.4 (Admissible shells). *If $\chi_Q \rho_{\mathbf{m}} \not\equiv 0$, then $M_i(\mathbf{m}) \leq \log_2(1/t_n) + C_Q$ and $2^{M_i(\mathbf{m})} \leq C_Q t_n^{-1}$ for every $1 \leq i \leq Q$; and the number of shell assignments with $\chi_Q \rho_{\mathbf{m}} \not\equiv 0$ is at most $C_Q \{1 + \log(1/t_n)\}^{E_Q}$.*

Proof. If $\chi_Q \rho_{\mathbf{m}}$ is not identically zero, there is a point of its support at which every $P_i > t_n$. By (26), throughout that shell $P_i(T) \leq C_Q 2^{-M_i(\mathbf{m})}$, whence the first claim. In particular every fine edge index is at most $\log_2(1/t_n) + C_Q$; since there are E_Q edges, the finitely many coarse choices and the bounded overlap of the dyadic partition give the count. $\square$

4.5 The polarization identity

For elements $f_1, \dots, f_Q$ of a commutative algebra with unit, write $f_1 \odot \dots \odot f_Q = (Q!)^{-1} \sum_{\sigma} f_{\sigma(1)} \otimes \dots \otimes f_{\sigma(Q)}$ for the symmetrised tensor product.

Lemma 4.5 (Real polarization). *Let $g_1, \dots, g_j$ ($1 \leq j \leq Q$) be elements of a commutative Banach algebra with unit and put $g_\epsilon = j^{-1} \sum_{i=1}^j \epsilon_i g_i$, $\epsilon \in \{\pm 1\}^j$. Fix distinct nodes $t_0, \dots, t_Q \in [-1, 1]$ and let λ_{jr} solve the Vandermonde system*

$$\sum_{r=0}^Q \lambda_{jr} t_r^k = \frac{\mathbf{1}\{k=j\}}{\binom{Q}{j}}, \quad 0 \leq k \leq Q. \quad (29)$$

Then for every $\theta \neq 0$,

$$g_1 \odot \dots \odot g_j \odot 1^{\odot(Q-j)} = \frac{j^j}{2^j j! \theta^j} \sum_{\epsilon \in \{\pm 1\}^j} \left(\prod_{i=1}^j \epsilon_i \right) \sum_{r=0}^Q \lambda_{jr} (1 + \theta t_r g_\epsilon)^{\otimes Q}. \quad (30)$$

The identity is algebraic: it only uses commutativity, and the tensor powers on the right are products over distinct slots.

Proof. The real polarization identity gives

$$g_1 \odot \cdots \odot g_j = \frac{1}{2^j j!} \sum_{\epsilon \in \{\pm 1\}^j} \left(\prod_i \epsilon_i \right) \left(\sum_i \epsilon_i g_i \right)^{\otimes j} = \frac{j^j}{2^j j!} \sum_{\epsilon} \left(\prod_i \epsilon_i \right) g_{\epsilon}^{\otimes j}, \quad (31)$$

since expanding $(\sum_i \epsilon_i g_i)^{\otimes j}$ and summing against $\prod_i \epsilon_i$ retains exactly the terms in which every g_i occurs once. Expanding $(1 + \theta t_r g)^{\otimes Q} = \sum_{k=0}^Q (\theta t_r)^k \binom{Q}{k} g^{\otimes k} \odot 1^{\odot(Q-k)}$ and using (29),

$$g^{\otimes j} \odot 1^{\odot(Q-j)} = \theta^{-j} \sum_{r=0}^Q \lambda_{jr} (1 + \theta t_r g)^{\otimes Q}. \quad (32)$$

Combine (31) with (32) for $g = g_{\epsilon}$. □

5 Configuration-space Wiener estimates

The purpose of this section is to justify the shellwise estimate used in Lemma 10.1 in the full dQ -variable Wiener algebra. The argument treats all pair differences simultaneously; no independence of the edge variables is assumed.

Lemma 5.1 (Uniform dyadic pullback on a finite configuration graph). *Let $V = \{1, \dots, Q\}$, let E be any fixed set of unordered pairs from V , and fix an integer m_0 . For every $e = \{i, j\} \in E$, choose an integer $m_e \geq m_0$ and put $L_e = 2^{m_e}$. Let $K \Subset (-2^{m_0-2}, 2^{m_0-2})^d$, and let*

$$H_{\mathbf{m}} \in C^\infty((\mathbb{T}^d)^Q \times (\mathbb{R}^d)^E)$$

be periodic in the first dQ variables and supported, in each Euclidean variable z_e , inside K . Suppose that for an integer

$$R > \frac{dQ + d|E|}{2}$$

one has the uniform derivative bound

$$\sup_{\mathbf{m}} \sum_{|\alpha| \leq 2R} \|\partial^\alpha H_{\mathbf{m}}\|_{L^\infty} \leq C_H. \quad (33)$$

Periodize $H_{\mathbf{m}}$ in the variable z_e with period L_e , and define on $(\mathbb{T}^d)^Q$

$$(\mathfrak{P}_{\mathbf{m}} H_{\mathbf{m}})(u_1, \dots, u_Q) = H_{\mathbf{m}}^{\text{per}}(u_1, \dots, u_Q, \{2^{m_e}(u_i - u_j)\}_{e=\{i,j\} \in E}). \quad (34)$$

The expression is independent of the chosen representatives of the torus points. Moreover,

$$\|\mathfrak{P}_{\mathbf{m}} H_{\mathbf{m}}\|_{\mathcal{W}_{dQ}} \leq C_{d,Q,E,R,K} C_H, \quad (35)$$

uniformly over all dyadic indices $\mathbf{m}$. The conclusion remains true for a finite sum of such profiles and for a family depending on auxiliary parameters ranging over a fixed compact set, provided (33) is uniform in those parameters.

Proof. Because $K \Subset (-L_e/2, L_e/2)^d$ for every e , periodization creates no overlap on the support of $H_{\mathbf{m}}$. Write the Fourier series of the periodized profile as

$$H_{\mathbf{m}}^{\text{per}}(u, z) = \sum_{k \in \mathbb{Z}^{dQ}} \sum_{\nu \in (\mathbb{Z}^d)^E} b_{\mathbf{m}}(k, \nu) e^{2\pi i k \cdot u} \prod_{e \in E} e^{2\pi i \nu_e \cdot z_e / L_e}. \quad (36)$$

Put $p = dQ + d|E|$ and $V_{\mathbf{m}} = \prod_{e \in E} L_e^d$. The Fourier coefficient is normalized by $V_{\mathbf{m}}^{-1}$. Apply $(1 - \Delta_{u,z})^R$, integrate by parts, and use the fixed compact support in the z -variables. The derivative bound (33) gives

$$|b_{\mathbf{m}}(k, \nu)| \leq C_{R,K} C_H V_{\mathbf{m}}^{-1} \left\{ 1 + |k|^2 + \sum_{e \in E} \frac{|\nu_e|^2}{L_e^2} \right\}^{-R}. \quad (37)$$

We now sum this estimate without losing a scale factor. For fixed k , the points $(\nu_e/L_e)_{e \in E}$ form an anisotropic rectangular grid in $\mathbb{R}^{d|E|}$ whose cell volume is $V_{\mathbf{m}}^{-1}$. If y lies in the grid cell adjacent to $(\nu_e/L_e)_e$, then, because all $L_e \geq 1$, the two quantities are uniformly comparable:

$$C_E^{-1} (1 + |k|^2 + |y|^2) \leq 1 + |k|^2 + \sum_e |\nu_e/L_e|^2 \leq C_E (1 + |k|^2 + |y|^2).$$

The left inequality is the direction needed below. Comparison with the integral over the disjoint grid cells gives

$$\begin{aligned} V_{\mathbf{m}}^{-1} \sum_{\nu \in (\mathbb{Z}^d)^E} \left\{ 1 + |k|^2 + \sum_e \frac{|\nu_e|^2}{L_e^2} \right\}^{-R} \\ \leq C_E \int_{\mathbb{R}^{d|E|}} (1 + |k|^2 + |y|^2)^{-R} dy. \end{aligned} \quad (38)$$

The integral on the right is bounded by $C(1 + |k|^2)^{-R+d|E|/2}$. Summing over $k \in \mathbb{Z}^{dQ}$ is finite precisely under the stated condition $R > (dQ + d|E|)/2$. Equations (37)–(38) therefore imply

$$\sum_{k, \nu} |b_{\mathbf{m}}(k, \nu)| \leq C_{d,Q,E,R,K} C_H, \quad (39)$$

uniformly in all m_e .

Substituting (34) into one term of (36) produces

$$e^{2\pi i k \cdot u} \prod_{e=\{i,j\} \in E} e^{2\pi i \nu_e \cdot (u_i - u_j)},$$

because $2^{m_e}/L_e = 1$. This is one character of $(\mathbb{T}^d)^Q$. Edges sharing a vertex merely add their integer frequencies in that vertex coordinate, and different dyadic scales have already disappeared from the character after the matched period choice $L_e = 2^{m_e}$. Every pulled-back Fourier term thus has Wiener norm one. Distinct (k, ν) may produce the same character, but the triangle inequality and (39) give (35).

Finally, changing a representative u_i by an integer changes the argument $2^{m_e}(u_i - u_j)$ by an integer multiple of the period $L_e = 2^{m_e}$. Thus (34) is well defined. Finite sums and compact auxiliary parameter families obey the same estimates. $\square$

Corollary 5.2 (Shell-localized Lagrange and product-taper multipliers). *Use the dyadic partition from (27). Fix a shell assignment $\mathbf{m} \in \mathbb{M}^{\mathbb{Q}}$, and for each edge $e = \{i, j\}$ let a_e be a nonnegative integer bounded by a fixed constant A . Let $M_{\mathbf{m}}(T)$ be any one of the finitely many scalar products obtained from*

1. the shell cutoffs $\rho_{m_e}\{\delta(u_i, u_j)\}$;
2. the product taper $\chi_Q(T)$;

3. a_e factors, at edge e , chosen from the elementary Lagrange coefficient multipliers in (19); and
4. a fixed number of functions from a fixed bounded subset of C^∞ on the torus coordinates.

Then

$$\|M_{\mathbf{m}}\|_{\mathcal{W}_{dQ}} \leq C_{d,Q,A} 2^{\sum_{e \in \mathfrak{E}_Q} a_e \mu(m_e)}. \quad (40)$$

The constant is independent of n , t_n , and all shell indices.

Proof. Call an edge fine when $m_e \neq c$. On a coarse edge, $\rho_c(\delta_e)$ is supported a fixed positive distance from the diagonal. Every multiplier in (19) is therefore a fixed smooth function there, with a Wiener norm bounded independently of all scales. We absorb every coarse-edge factor into the periodic u -dependence of a profile.

Consider a fine edge $e = \{i, j\}$, write $m = m_e$, and use a local lift $w = u_i - u_j \in (-1/4, 1/4)^d$. Put $z = 2^m w$. By (26) and the local comparability of δ with the chordal distance, the shell cutoff restricts z to a fixed compact annulus

$$K_0 \Subset \mathbb{R}^d \setminus \{0\},$$

independently of m . The families

$$\begin{aligned} q_m(u_j, z) &= 2^m \delta(u_j + 2^{-m} z, u_j), \\ B_{m,a}(u_j, z) &= 2^{-m} \frac{\{\iota(u_j + 2^{-m} z) - \iota(u_j)\}_a}{\|\iota(u_j + 2^{-m} z) - \iota(u_j)\|^2}, \\ B_{m,0}(u_j, z) &= 2^{-m} \frac{\langle \iota(u_j), \iota(u_j + 2^{-m} z) - \iota(u_j) \rangle}{\|\iota(u_j + 2^{-m} z) - \iota(u_j)\|^2} \end{aligned}$$

are uniformly C^N on $\mathbb{T}^d \times K_0$, for every fixed N . To verify this, multiply numerator and denominator by the appropriate powers of 2^m and use the Taylor formula for the smooth embedding ι . Its differential has full rank, so the rescaled denominator is bounded above and below on K_0 ; differentiating the resulting quotient gives bounds independent of m . The same argument applies to q_m , and $0 < c_Q \leq q_m \leq C_Q$ on the shell. Thus each order-one multiplier is 2^m times a uniformly smooth function of (u_j, z) , while the shell cutoff itself is the uniformly smooth function $\rho\{q_m(u_j, z)\}$. A product containing a_e order-one factors at edge e therefore contributes the explicit scale $2^{a_e m_e}$ and otherwise has a uniformly smooth rescaled profile.

It remains to verify the product taper. For a vertex i , put

$$M_i(\mathbf{m}) = \sum_{e \ni i} \mu(m_e), \quad \lambda_i = t_n^{-1} 2^{-M_i(\mathbf{m})}.$$

On the fixed shell profile,

$$\frac{P_i(T)}{t_n} = \lambda_i \prod_{e \ni i} q_{e,m_e}(u, z_e), \quad (41)$$

where, for a coarse edge, $q_{e,c}$ denotes the corresponding fixed positive smooth distance factor. The product on the right ranges in a fixed interval $[c_Q, C_Q]$. If $\lambda_i C_Q \leq 1$, the factor $\vartheta(P_i/t_n)$ is identically zero on the shell. If $\lambda_i c_Q \geq 2$, it is identically one. In every remaining case, λ_i belongs to the fixed compact interval $[C_Q^{-1}, 2c_Q^{-1}]$. Hence, by the chain rule applied to (41), the family of all nonconstant transition profiles has uniformly bounded derivatives of every fixed order. Taking the product over the finitely many vertices preserves this uniform bound. This is the required control of the product taper; no derivative bound contains a dyadic index or t_n .

To make the pullback exact, choose once and for all a cutoff $\kappa_0 \in C_c^\infty(\mathbb{R}^d)$ that equals one on a neighborhood of K_0 . For every fine edge, replace its rescaled factor by the same factor multiplied by $\kappa_0(z_e)$; this does not alter the multiplier on the support of the shell cutoff. Products of these extended factors, together with the coarse-edge factors and the taper profiles above, define functions $H_{\mathbf{m}}(u, z)$ supported in one fixed compact set in each z_e . Their derivatives through any prescribed finite order are uniformly bounded. After the explicit factor $2^{\sum_e a_e m_e}$ is extracted, one has the exact identity

$$M_{\mathbf{m}}(T) = 2^{\sum_e a_e m_e} (\mathfrak{P}_{\mathbf{m}} H_{\mathbf{m}})(T).$$

The matched period $L_e = 2^{m_e}$ creates no extraneous copies: modulo L_e , the point $2^{m_e}(u_i - u_j)$ lies in the fixed support of κ_0 exactly when the torus difference has the unique fine-shell lift used above.

The profiles now satisfy the hypotheses of [Lemma 5.1](#). Applying that lemma gives (40). Its Fourier proof also explains why shared vertices do not create an additional loss: all pulled-back edge frequencies add to one configuration-space character, and the Wiener triangle inequality is taken only after that pullback. $\square$

Part II

Case A: two rough nuisances

6 Case A: construction

Throughout Part A, $\alpha, \beta \leq 1$, so $s_1 = (\alpha + \beta)/2 = A_0/2$ and $\rho = 1/\{1 + d/(2\gamma) + d/(4s_1)\}$. Both nuisances are carried by the single rough field Λ_λ of (7), at the fine scale ℓ ; there is no block carrier, the design is exactly uniform, and no ε -loss is needed: the scales Δ, h, ℓ are defined directly in (42), and the quantities r, t, Q and the ledgers of Section 3 are not used. The construction is that of the uniform-covariate note of Hou, written in the coded notation of Section 2.

6.1 Scales and amplitudes

Put

$$\begin{aligned} j_a &= c_a \ell^\alpha, & j_b &= c_b \ell^\beta, & \Delta &:= \ell^{2s_1} = \ell^{A_0}, & h &= \Delta^{1/\gamma}, \\ d_*(x) &= j_a j_b G_h(x)^2 = c_a c_b \Delta G_h(x)^2, \end{aligned} \tag{42}$$

with equal small constants $c_a = c_b = c_0 > 0$ fixed in [Lemma 6.1](#); the scale ℓ (equivalently Δ) is tuned in [Section 7.2](#). Since $\gamma > A_0 = 2s_1$ ([Lemma 3.1](#)), $\ell = \Delta^{1/(2s_1)} \leq \Delta^{1/\gamma} = h$ for small Δ . Write $j_\circ = j_a \wedge j_b$, $j_\bullet = j_a \vee j_b$; then $j_\circ j_\bullet = j_a j_b = c_a c_b \Delta$ and $j_a, j_b \leq 1$.

6.2 The two hypotheses

In coded coordinates $(p, y, \tau) = (2\pi - 1, 2\mu_0 - 1, \tau)$, the rough signs λ being drawn from the Rademacher prior on both sides and the design being $\text{Unif}([0, 1]^d)$ on both sides:

	p	y	τ
Q	$j_a \Lambda_\lambda$	$j_b \Lambda_\lambda$	0
$P \text{ (I: } \alpha \geq \beta)$	0	$j_b \Lambda_\lambda - d_*$	d_*
$P \text{ (II: } \beta \geq \alpha)$	$j_a \Lambda_\lambda$	$-d_*$	d_*

(43)

Under Q the two nuisances are proportional to the same rough field, and their product produces the RT -cell $j_a j_b \Lambda_\lambda^2$ with mean $j_a j_b G_h^2 = d_*$; under P this cell is supplied by the treatment effect d_* ,

the rougher of the two nuisances keeps its field, and the smoother one is shifted by $-d_*$ to absorb the T -cell.

Lemma 6.1 (Legality and separation). *For sufficiently small fixed $c_a, c_b > 0$ and all sufficiently small Δ , every law in either row of (43) belongs to $\mathcal{P}$, and $\tau_P(x_0) - \tau_Q(x_0) = c_a c_b \Delta$.*

Proof. By Lemma 2.2, $\|j_a \Lambda_\lambda\|_{C^\alpha} \leq C c_a$ and $\|j_b \Lambda_\lambda\|_{C^\beta} \leq C c_b$, while $|j_a \Lambda_\lambda| \leq N_0 c_a$, $|j_b \Lambda_\lambda| \leq N_0 c_b$. The treatment effect $d_* = c_a c_b \Delta G_h^2$ has $\|d_*\|_{C^\gamma} \leq C c_a c_b \Delta h^{-\gamma} = C c_a c_b$ and $\|d_*\|_{C^\beta} \leq C \Delta h^{-\beta} = C h^{\gamma-\beta} \rightarrow 0$ since $\gamma > A_0 \geq \beta$. Hence $\pi = (1+p)/2 \in \mathcal{H}^\alpha(L_\pi)$, $\mu_0 = (1+y)/2 \in \mathcal{H}^\beta(L_0)$, $\tau \in \mathcal{H}^\gamma(L_\tau)$ after choosing c_a, c_b small, and $\pi, \mu_0, \mu_0 + \tau \in [\kappa, 1 - \kappa]$ because $|p|, |y|, |\tau| \leq N_0(c_a + c_b) + o(1)$. The uniform density lies in $[\underline{f}, \bar{f}]$. Finally $G_h(x_0) = 1$. $\square$

6.3 Site likelihoods

Fix a site x and write $\Lambda = \Lambda_\lambda(x)$, $G = G_h(x)$, $d = d_*(x) = j_a j_b G^2$, $V = \Lambda^2 - G^2$ (the centred square of Section 2, a Walsh polynomial of degree two with $\|V\|_W \leq N_0^2$). By (5),

$$L^Q = 1 + R j_a \Lambda + T j_b \Lambda + R T j_a j_b \Lambda^2 = [1 + R j_a \Lambda + T j_b \Lambda + R T d] + R T j_a j_b V, \quad (44)$$

$$L^P = 1 + T j_b \Lambda + R T d \quad (\text{I}), \quad L^P = 1 + R j_a \Lambda + T d j_a \Lambda + R T d \quad (\text{II}), \quad (45)$$

(for (II): $y + \tau(1+p) = -d + d(1 + j_a \Lambda) = d j_a \Lambda$ and $p y + \tau(1+p) = -d j_a \Lambda + d(1 + j_a \Lambda) = d$). The difference $D := L^Q - L^P$ is

$$D = R j_a \Lambda + R T j_a j_b V \quad (\text{I}), \quad D = T j_b \Lambda - T d j_a \Lambda + R T j_a j_b V \quad (\text{II}). \quad (46)$$

Thus D has no constant term; its *linear part* (monomials of degree one) has Walsh norm $\leq C N_0 j_\circ$ in both subcases (in (I) $j_a = j_\circ$; in (II) $j_b = j_\circ$ and $d j_a \leq j_a j_b \leq j_\circ$); its *quadratic part* $R T j_a j_b V$ has degree two and Walsh norm $\leq N_0^2 j_a j_b$. Every part of L^P is constant, or linear with Walsh norm $\leq C N_0 j_\bullet$ (in (II) also $d j_a \leq j_\bullet$). In particular

$$\mathbb{E}_\lambda L^Q = \mathbb{E}_\lambda L^P = 1 + R T d \quad \text{in both subcases}, \quad (47)$$

because $\mathbb{E} \Lambda = 0$ and $\mathbb{E} V = 0$: the one-observation mixtures coincide.

7 Case A: proof

7.1 Components

Condition on $X^n = x^n$. Declare $i \sim j$ if $J(x_i) \cap J(x_j) \neq \emptyset$ and let $\{C\}$ be the components of the transitive closure; the sign sets $M_C = \bigcup_{i \in C} J(x_i)$ of distinct components are disjoint, so the signs of distinct components are independent, and every link satisfies $x_i, x_j \in \text{supp } G_h$, $\|x_i - x_j\|_\infty \leq R_0 \ell$. The X^n -marginals coincide (iid uniform), and relative to fair coins the conditional laws factor over components:

$$\begin{aligned} \mathbb{P}_n(w^n | x^n) &= \prod_C \bar{p}_C(w_C), & \mathbb{Q}_n(w^n | x^n) &= \prod_C \bar{q}_C(w_C), \\ \bar{p}_C(w_C) &= 4^{-|C|} \mathbb{E}_\lambda \prod_{i \in C} L_i^P, & \bar{q}_C(w_C) &= 4^{-|C|} \mathbb{E}_\lambda \prod_{i \in C} L_i^Q. \end{aligned}$$

For a singleton $C = \{i\}$, (47) gives $\bar{p}_C = \bar{q}_C$ and $H_C^2 = 0$.

Lemma 7.1 (Component bound). *For every component C with $q = |C|$ and every cell w_C , $|\bar{q}_C(w_C) - \bar{p}_C(w_C)| \leq 4^{-q} C^q j_{ab}$, and $H_C^2 \leq C^q (j_{ab})^2 = C^q \Delta^2$.*

Proof. Expand

$$\mathbb{E}_\lambda \prod_{i \in C} L_i^Q - \mathbb{E}_\lambda \prod_{i \in C} L_i^P = \sum_{\emptyset \neq S \subseteq C} \mathbb{E}_\lambda \left[\prod_{i \in S} D_i \prod_{j \in C \setminus S} L_j^P \right],$$

and expand every D_i into its linear or quadratic part and every L_j^P into its constant or linear part: at most C^q terms, each a product of Walsh polynomials whose absolute mean is at most the product of the Walsh norms of its factors. A term with nonzero mean must have even total degree (Section 2), the degree of a product being the sum of the degrees modulo two. If the term contains a quadratic part, its coefficient product is $\leq C^q j_{ab}$ (all other coefficients are $\leq C$). Otherwise all its nonconstant parts are linear; it contains the linear part of some D_i , $i \in S$ (coefficient $\leq C j_o$), and by parity at least one further linear part (coefficient $\leq C j_\bullet$); so the coefficient product is $\leq C^q j_{o\bullet} = C^q j_{ab}$. Summing over the terms gives the first claim. For the second, $L_i^P \geq 1 - |p| - |y + \tau(1 + p)| - |py + \tau(1 + p)| \geq \frac{1}{2}$ for small c_a, c_b (Lemma 6.1), so $\bar{p}_C \geq 8^{-q}$, and Lemma 2.4(c) gives $H_C^2 \leq 8^q \sum_{w_C} (4^{-q} C^q j_{ab})^2 \leq C^q (j_{ab})^2$. $\square$

Lemma 7.2 (Component counting). *For every $C_0 > 0$ there is C such that*

$$\mathbb{E} \sum_{C: |C|=q} C_0^q \leq C^q n^q h^d \ell^{d(q-1)}, \quad q \geq 2.$$

Consequently, if $n\ell^d \leq c_0$ with c_0 sufficiently small, $\mathbb{E} \sum_{C: |C| \geq 2} C_0^{|C|} \leq C n^2 h^d \ell^d$.

Proof. It suffices to count connected labelled q -sets, which may overcount components. Every such set contains a spanning tree; fix a labelled tree and a root. Each edge forces both endpoints into $\text{supp } G_h$ (volume $O(h^d)$), and each child into a cube of volume $O(\ell^d)$ around its parent; since the X_i are iid uniform, the tree is realised with probability $\leq C h^d \ell^{d(q-1)}$. By Cayley's formula there are q^{q-2} labelled trees, and $\binom{n}{q} q^{q-2} \leq \frac{n^q}{q!} q^{q-2} \leq \frac{e^q}{q^2} n^q \leq C^q n^q$ (using $q! \geq (q/e)^q$); this gives the first bound. Summing over $q \geq 2$ gives $C n^2 h^d \ell^d \sum_{q \geq 2} (C n \ell^d)^{q-2}$, a bounded geometric series when $n\ell^d \leq c_0$. $\square$

7.2 Tuning and conclusion

By Lemmas 2.5 (with $\mathcal{E}_n = \emptyset$), 7.1 and 7.2,

$$\text{TV}(\mathbb{P}_n, \mathbb{Q}_n)^2 \leq \mathbb{E} \sum_{C: |C| \geq 2} H_C^2 \leq C n^2 h^d \ell^d \Delta^2 = C n^2 \Delta^{d/\gamma + d/(2s_1) + 2} = C (n \Delta^{1/\rho})^2, \quad (48)$$

provided $n\ell^d \leq c_0$; here $h^d = \Delta^{d/\gamma}$, $\ell^d = \Delta^{d/(2s_1)}$, and $2 + d/\gamma + d/(2s_1) = 2/\rho$ because $1/\rho = D_1/(A_0 + 2s_1) = 1 + d/(2\gamma) + d/(4s_1)$ when $A_0 = 2s_1$. Choose

$$\Delta = c_\star n^{-\rho}, \quad (49)$$

with a small fixed constant $c_\star > 0$. Then $n\ell^d = c_\star^{d/(2s_1)} n^{1-\rho d/(2s_1)} \rightarrow 0$, because $\rho > A_0/d = 2s_1/d$ (Lemma 3.1); so $n\ell^d \leq c_0$ for large n , and (48) gives $\text{TV}(\mathbb{P}_n, \mathbb{Q}_n) \leq C^{1/2} c_\star^{1/\rho} \leq \frac{1}{2}$ for $c_\star$ small. Every law in the support of the P -prior has $\tau(x_0) = d_\star(x_0) = c_a c_b \Delta$ and every law in the support of the Q -prior has $\tau(x_0) = 0$ (Lemma 6.1), so Lemma 2.3 gives

$$R_n^*(x_0) \geq \frac{c_a c_b c_\star}{4} n^{-\rho}$$

for all large n ; smaller n are absorbed into the constant. This proves Theorem 1.1 in case (A), with $\varepsilon = 0$, and the two priors are supported on laws with uniform design. $\square$

Remark 7.3. *Case (A) needs no design carrier: the shared rough field produces the one-site matching (47) exactly, and $n\ell^d \rightarrow 0$ makes the packet-overlap components sparse.*

Part III

Case B: two smooth nuisances

8 Case B: packets and carrier boxes

Throughout Part B, $\alpha, \beta > 1$, so $s_1 = 1$ and the scales of Section 3 read $t^2 = \Delta r^{-A_0}$, $\ell = tr$, $t_n = t\Lambda^B$; the fine ratio t is the relative size of the virtual jitter below. The rough field of Section 2 is not used in Part B: both nuisances are smooth, and the role of the rough field is played by *virtual* Rademacher signs attached to the observation sites.

Let ϕ be the quadratic partition of unity of Lemma 2.1, (6), and let G_h be the bump of Section 2. Define

$$x_k = x_0 + rk, \quad u_k(x) = \frac{x - x_k}{r}, \quad \psi_k(x) = G_h(x)\phi(u_k(x)).$$

Then

$$\sum_k \psi_k(x)^2 = G_h(x)^2. \quad (50)$$

(In Part B the word *packet* refers to these block bumps at scale r , not to the rough packets of Section 2.) Only a fixed number of packets meet a point, and there are $O(h^d r^{-d})$ active packets. Enlarge the support of each active packet to a cube S_k of side Cr , with C fixed. The cubes have bounded overlap and their union has volume $O(h^d)$. Each S_k is identified, by an affine map θ_k , with the normalized torus $\mathbb{T}^d$. The packet support is contained in a fixed compact subcube of that torus.

9 Case B: the exact aggregate cubic bridge

9.1 Fields and virtual jitter

Let

$$a = c_a r^\alpha, \quad b = c_b r^\beta, \quad d_*(x) = a b t^2 G_h(x)^2. \quad (51)$$

For bounded coefficient vectors U_k , define

$$P(x) = \sum_k \psi_k(x) v_Q(\theta_k(x))^\top U_k, \quad p(x) = a P(x). \quad (52)$$

For a separated tuple of sites in a block, use Lemma 4.1 and put

$$\tilde{U}_k = U_k + t \sum_i c_{ki}(T) \zeta_{ki}, \quad \zeta_{ki} \stackrel{\text{ind}}{\sim} \text{Rad}(\tfrac{1}{2}), \quad (53)$$

where $\text{Rad}(1/2)$ denotes the symmetric $\{\pm 1\}$ law. At an observed point,

$$p'_i = p_i + \delta_i, \quad \delta_i = a t \sum_k \psi_k(x_i) \zeta_{ki}. \quad (54)$$

Conditional on the baseline field, the δ_i 's are independent and symmetric. By (50),

$$v(x) := \mathbb{E}\delta(x)^2 = a^2 t^2 G_h(x)^2, \quad (55)$$

$$m_4(x) := \mathbb{E}\delta(x)^4 = a^4 t^4 G_h(x)^4 \{3 - 2q_4(x)\}, \quad (56)$$

$$q_4(x) = \sum_k \phi(u_k(x))^4. \quad (57)$$

9.2 Cubic identity

Define

$$K_x(z) = \frac{b}{a} \left[\{1 + \eta(x)\}z - \frac{z^3}{3} \right], \quad \eta(x) = \frac{a^2 t^2 G_h(x)^2}{3} \{3 - 2q_4(x)\}. \quad (58)$$

This is also the definition when $G_h = 0$. Put

$$u_x(p) = \mathbb{E}_\delta K_x(p + \delta) = K_x(p) - d_*(x)p. \quad (59)$$

The two fuzzy sides use

	$2\pi - 1$	$2\mu_0 - 1$	τ
P	p	$u_x(p) - d_*(1 + p)$	d_*
Q	p'	$K_x(p')$	$0.$

(60)

Lemma 9.1 (Cubic covariance identity). *For every fixed baseline value p ,*

$$\text{Cov}\{p + \delta, K_x(p + \delta)\} = d_*(x)(1 - p^2). \quad (61)$$

Consequently,

$$\mathbb{E}_\delta(1, p', K_x(p'), p' K_x(p')) = (1, p, u_x(p), pu_x(p) + d_*(1 - p^2)). \quad (62)$$

Proof. Write $K_x(z) = c_1 z + c_3 z^3$, where

$$c_3 = -\frac{b}{3a} = -\frac{d_*}{3v}, \quad c_1 = \frac{b}{a}(1 + \eta) = \frac{d_*}{v} + \frac{d_* m_4}{3v^2}$$

when $v > 0$; the identities extend continuously to $v = 0$. Symmetry of δ gives

$$\text{Cov}\{p + \delta, K_x(p + \delta)\} = c_1 v + c_3(3p^2 v + m_4) = d_*(1 - p^2).$$

It also gives

$$\mathbb{E}K_x(p + \delta) = K_x(p) + 3c_3 p v = K_x(p) - d_* p.$$

Substitution proves (62). □

The right side of (62) is exactly the pair of outcome Walsh coefficients generated by the P -row of (60). Conditional independence of the site jitters therefore gives equality of all 4^q joint cells for every finite tuple when every active block is virtually shifted. The identity uses the aggregate p , so overlapping smooth packets cause no additional term.

10 Case B: a positive pre-data moment carrier

10.1 The tapered ideal increment and its Wiener norm

For a Q -tuple T , write

$$C_T(\zeta) = \sum_{i=1}^Q c_i(T) \zeta_i.$$

On the support of χ_Q , [Lemma 4.1](#) gives

$$t \|C_T(\zeta)\| \leq C_Q \frac{t}{P_{\min}(T)} \leq C_Q \Lambda^{-B}.$$

Choose the support ball in [Lemma 4.3](#) strictly inside a fixed legal coefficient ball. For large n , the displayed virtual shifts stay inside that legal ball wherever $\chi_Q \neq 0$. Define the vector-valued symmetric tensor

$$J_n(T) = \chi_Q(T) [\mathbb{E}_{\Pi_0, \zeta} \Psi_{4Q}\{U + tC_T(\zeta)\} - \mathbb{E}_{\Pi_0} \Psi_{4Q}(U)], \quad (63)$$

with the product extended by zero across the diagonals.

Lemma 10.1 (Inverse-square Wiener estimate). *With $E_Q = \binom{Q}{2}$,*

$$\|J_n\|_{\mathcal{W}_{4Q}} \leq C_Q \{1 + \log(1/t_n)\}^{E_Q} \sum_{j=2}^{4Q} \left(\frac{t}{t_n}\right)^j. \quad (64)$$

Consequently,

$$\|J_n\|_{\mathcal{W}_{4Q}} \leq C_Q \Lambda^{E_Q - 2B} =: \delta_n. \quad (65)$$

In particular, any fixed choice

$$B > \frac{E_Q + 2}{2} \quad (66)$$

makes $\delta_n = o(1)$. All constants in this lemma are independent of n and of the shell indices, with the fixed-dependence convention stated after [\(13\)](#).

Proof. Fix one coordinate monomial U^ν of Ψ_{4Q} , with $1 \leq |\nu| \leq 4Q$, and expand $(U + tC_T(\zeta))^\nu - U^\nu$ by the multinomial theorem. After integrating $U \sim \Pi_0$, every coefficient is bounded by a constant depending only on Q and the fixed support of Π_0 . A term containing j shift factors is a finite linear combination of expressions

$$t^j \prod_{r=1}^j c_{i_r, a_r}(T), \quad (67)$$

where $c_{i,a}$ denotes a coordinate of c_i . The expectation over the independent symmetric signs is zero unless every site index occurs an even number of times among $i_1, \dots, i_j$. Thus every surviving term has $j \geq 2$; this is what gives an inverse-square rather than an inverse-first-order gain.

Insert the partition [\(27\)](#) into a surviving term, and let $M_i(\mathbf{m})$ be as in [\(28\)](#). Since every coordinate of $c_i(T)$ is a finite sum of products containing exactly one elementary multiplier [\(19\)](#) for every edge incident to i , in the product in [\(67\)](#) edge e occurs with some inverse order a_e , and

$$\sum_{e \in \mathfrak{E}_Q} a_e \mu(m_e) = \sum_{r=1}^j M_{i_r}(\mathbf{m}). \quad (68)$$

Apply [Corollary 5.2](#) to the product of the shell cutoffs, the product-distance taper, and the elementary multipliers. It gives

$$\|\chi_Q \rho_{\mathbf{m}} \prod_{r=1}^j c_{i_r, a_r}\|_{\mathcal{W}_{dQ}} \leq C_Q 2^{\sum_r M_{i_r}(\mathbf{m})}. \quad (69)$$

There is no additional scale loss from shared vertices, different edge scales, or the product taper; those points are exactly what [Corollary 5.2](#) proves. On a shell with $\chi_Q \rho_{\mathbf{m}} \neq 0$, [Lemma 4.4](#) yields

$$2^{M_i(\mathbf{m})} \leq C_Q t_n^{-1},$$

and hence

$$\|\chi_Q \rho_{\mathbf{m}} t^j \prod_{r=1}^j c_{i_r, a_r}\|_{\mathcal{W}_{dQ}} \leq C_Q \left(\frac{t}{t_n}\right)^j. \quad (70)$$

There are only finitely many coordinates, monomials, and multinomial terms, with their number depending on Q but not on n . Sum (70) over them, over $j = 2, \dots, 4Q$, and over the $\leq C_Q \{1 + \log(1/t_n)\}^{E_Q}$ shell assignments of [Lemma 4.4](#). This proves (64). Finally, $t_n = t\Lambda^B$, so

$$\sum_{j=2}^{4Q} (t/t_n)^j = \sum_{j=2}^{4Q} \Lambda^{-Bj} \leq C_Q \Lambda^{-2B},$$

and $1 + \log(1/t_n) \leq C_\epsilon \Lambda$ because t is a fixed negative power of n . This proves (65). Under (66), $E_Q - 2B < -2$, and therefore $\delta_n = o(1)$. $\square$

10.2 A fixed positive polarization operator

Lemma 10.2 (Positive Wiener polarization). *Fix Q and $0 < \eta_f < 1/2$. There is a countable dictionary $\{F_m : m \geq 0\} \subset \mathcal{W}_d$, fixed independently of the tensor to be decomposed, with*

$$\int_{\mathbb{T}^d} F_m = 1, \quad 1 - \eta_f \leq F_m \leq 1 + \eta_f, \quad \sup_m \|F_m\|_{\mathcal{W}_d} < \infty, \quad (71)$$

and bounded linear maps $A \mapsto a_m(A)$ from the symmetric real Wiener tensors on $(\mathbb{T}^d)^Q$ to any fixed finite-dimensional real vector space V , such that

$$A = \sum_{m \geq 0} a_m(A) F_m^{\otimes Q}, \quad (72)$$

$$\sum_{m \geq 0} \|a_m(A)\| \leq C_{Q, \eta_f} \|A\|_{\mathcal{W}_{dQ}}, \quad (73)$$

$$\sum_{m \geq 0} \|a_m(A) - a_m(A')\| \leq C_{Q, \eta_f} \|A - A'\|_{\mathcal{W}_{dQ}}. \quad (74)$$

The series converges absolutely in Wiener norm.

Proof. Use the real trigonometric basis consisting of 1, sines, and cosines, normalized so that every nonconstant basis element w has mean zero, $\|w\|_\infty \leq 1$, and $\|w\|_{\mathcal{W}_d} = 1$. Passing from the complex Fourier basis to this real basis changes the ℓ^1 coefficient norm by only a fixed factor depending on Q .

It suffices to decompose a symmetrized basis tensor

$$w_1 \odot \dots \odot w_j \odot 1^{\odot(Q-j)}, \quad 0 \leq j \leq Q,$$

where each w_i is nonconstant. For $j \geq 1$, apply [Lemma 4.5](#) in the Wiener algebra $\mathcal{W}_d$ with $g_i = w_i$ and a fixed $0 < \theta \leq \eta_f/2$: every symmetrized basis tensor is a fixed finite linear combination of $F^{\otimes Q}$ with $F = 1 + \theta t_r g_\epsilon$, $g_\epsilon = j^{-1} \sum_i \epsilon_i w_i$. Since $\int g_\epsilon = 0$, $\|g_\epsilon\|_\infty \leq 1$ and $\|g_\epsilon\|_{\mathcal{W}_d} \leq 1$, such F has integral one, lies between $1 - \eta_f$ and $1 + \eta_f$, and has uniformly bounded Wiener norm. The case $j = 0$ uses $F = 1$.

Apply this fixed stencil coefficientwise to the absolutely summable real Fourier expansion of A . Index all resulting atoms by a single countable index m . The total absolute coefficient generated by one basis tensor is bounded by a constant depending only on (Q, η_f) . Summing the Fourier coefficients proves [\(73\)](#). The construction is linear, so applying the same estimate to $A - A'$ proves [\(74\)](#). Absolute Wiener convergence gives [\(72\)](#). $\square$

10.3 The self-consistent positive carrier

Proposition 10.3 (Positive pre-data carrier). *For all sufficiently large n , there are joint block laws $\nu_{P,k}(dH_k, dU_k)$ and $\nu_{Q,k}(dH_k, dU_k)$ with the following properties.*

1. *Their full H_k -marginals are identical. The label H_k selects a factor F_{H_k} satisfying*

$$\int_{\mathbb{T}^d} F_{H_k} = 1, \quad 1 - \eta_f \leq F_{H_k} \leq 1 + \eta_f. \quad (75)$$

All coefficient draws lie in one fixed legal ball.

2. *If*

$$D_Q(T) = \mathbb{E} F_{H_k}(u_1) \cdots F_{H_k}(u_Q), \quad (76)$$

then the same density law that generates D_Q satisfies the exact moment identity

$$\begin{aligned} \mathbb{E}_{\nu_{Q,k}} \left[F_{H_k}^{\otimes Q}(T) \Psi_{4Q}(U_k) \right] - \mathbb{E}_{\nu_{P,k}} \left[F_{H_k}^{\otimes Q}(T) \Psi_{4Q}(U_k) \right] \\ = D_Q(T) J_n(T). \end{aligned} \quad (77)$$

The laws are chosen before observing the data and are not tuple-indexed.

Proof. For a symmetric scalar tensor D , set

$$A(D) = D J_n.$$

Apply [Lemma 10.2](#) coordinatewise with $V = \mathbb{R}^{N_Q}$:

$$A(D) = \sum_m a_m(D) F_m^{\otimes Q}.$$

Let C_{pol} be a constant valid in both [\(73\)](#) and [\(74\)](#). Choose $K_0 > 2/\delta_Q$, where δ_Q is from [Lemma 4.3](#), and define

$$\omega_m(D) = K_0 \|a_m(D)\|, \quad \Phi(D) = \left(1 - \sum_m \omega_m(D) \right) 1^{\otimes Q} + \sum_m \omega_m(D) F_m^{\otimes Q}.$$

Set

$$C_F = \sup_m \|F_m^{\otimes Q} - 1^{\otimes Q}\|_{\mathcal{W}_{dQ}} < \infty, \quad C_0 = K_0 C_F C_{\text{pol}}, \quad C_1 = K_0 C_{\text{pol}}.$$

For any symmetric D, D' , the reverse triangle inequality, (74), and the Wiener algebra inequality give

$$\begin{aligned}\|\Phi(D) - \Phi(D')\|_{\mathcal{W}_{dQ}} &\leq K_0 C_F \sum_m \|a_m(D) - a_m(D')\| \\ &\leq C_0 \|(D - D')J_n\|_{\mathcal{W}_{dQ}} \\ &\leq C_0 \delta_n \|D - D'\|_{\mathcal{W}_{dQ}}.\end{aligned}\tag{78}$$

Define the closed ball in the symmetric Wiener tensors

$$\mathbb{B}_n = \{D = D^{\text{sym}} : \|D - 1^{\otimes Q}\|_{\mathcal{W}_{dQ}} \leq R_n\}, \quad R_n = 2C_0 \delta_n.\tag{79}$$

For $D \in \mathbb{B}_n$,

$$\sum_m \omega_m(D) \leq C_1 \|DJ_n\|_{\mathcal{W}_{dQ}} \leq C_1(1 + R_n)\delta_n\tag{80}$$

$$\|\Phi(D) - 1^{\otimes Q}\|_{\mathcal{W}_{dQ}} \leq C_F \sum_m \omega_m(D) \leq C_0(1 + R_n)\delta_n.\tag{81}$$

Since $\delta_n = o(1)$, choose n so large that

$$R_n \leq 1, \quad C_0 \delta_n \leq \frac{1}{2}, \quad 2C_1 \delta_n < 1.\tag{82}$$

Then (81) is at most $2C_0 \delta_n = R_n$, so $\Phi(\mathbb{B}_n) \subseteq \mathbb{B}_n$. Equation (78) has contraction constant $q_n = C_0 \delta_n \leq 1/2$, and (80) gives $\sum_m \omega_m(D) < 1$ throughout the ball. The weights are nonnegative by definition. The ball $\mathbb{B}_n$ is a closed subset of a Banach space, so Banach's theorem yields a unique fixed point

$$D_Q = \Phi(D_Q) \in \mathbb{B}_n.\tag{83}$$

Use the probabilities $\omega_m(D_Q)$ on labels carrying F_m , and put the remaining probability on a label carrying $F = 1$. This common label law is the H_k -marginal on both sides and, by (83), its Q -fold density moment is exactly D_Q . Under P , conditionally on every label, let $U_k \sim \Pi_0$. If $\omega_m > 0$, define

$$v_m = \frac{a_m(D_Q)}{\omega_m}, \quad \|v_m\| = K_0^{-1} < \delta_Q,$$

and under Q , conditionally on label m , use the law Π_{v_m} from Lemma 4.3; on the remaining label use Π_0 . If $\omega_m = 0$, no conditional law for that null label is needed. All positive-mass conditional laws have the same fixed legal support. Therefore

$$\mathbb{E}_Q[F^{\otimes Q}\Psi] - \mathbb{E}_P[F^{\otimes Q}\Psi] = \sum_m \omega_m F_m^{\otimes Q} v_m = \sum_m a_m(D_Q) F_m^{\otimes Q} = D_Q J_n,$$

which is (77). There is no circularity: the fixed point determines one common label law, and this law both generates D_Q and carries the moment perturbation. $\square$

10.4 Evaluated projectivity and ghost leakage

For $q \leq Q$, let

$$D_q(T_q) = \mathbb{E} F_{H_k}(u_1) \cdots F_{H_k}(u_q).$$

Since every density atom has integral one,

$$D_q(T_q) = \int D_Q(T_q, Z) dZ, \quad (84)$$

where $Z \in (\mathbb{T}^d)^{Q-q}$ and Haar measure is normalized. Define

$$\bar{\chi}_q(T_q) = \frac{\int D_Q(T_q, Z) \chi_Q(T_q, Z) dZ}{D_q(T_q)}. \quad (85)$$

Because every F is between fixed positive constants, D_q is bounded above and below, and $0 \leq \bar{\chi}_q \leq 1$.

A likelihood evaluation polynomial for one block means a polynomial in U_k of degree at most $4Q$ that depends on U_k only through the retained linear evaluations

$$\psi_k(x_i) v_Q(\theta_k(x_i))^\top U_k, \quad i = 1, \dots, q.$$

Such polynomials are exactly the block factors obtained after expanding a component Walsh coefficient.

Proposition 10.4 (Evaluated lower-order projectivity and ghost leakage). *Let $\mathcal{L}_{T_q}$ be any likelihood evaluation functional as above. If $b_{\mathcal{L},q}(T_q)$ denotes the corresponding ideal virtual moment increment at the retained sites, then*

$$\mathcal{L}_{T_q} \{ \mathbb{E}_Q[F^{\otimes q} \Psi] - \mathbb{E}_P[F^{\otimes q} \Psi] \} = D_q(T_q) \bar{\chi}_q(T_q) b_{\mathcal{L},q}(T_q). \quad (86)$$

Moreover,

$$\int D_q(T_q) \{1 - \bar{\chi}_q(T_q)\} dT_q \leq C_Q t^d \Lambda^{M_Q}. \quad (87)$$

Projectivity is used only after evaluation; the tapered coefficient tensor itself need not be projective.

Proof. Integrating (77) in the last $Q - q$ coordinates gives the exact lower-order coefficient moment. For $T = (T_q, Z)$ with $\chi_Q(T) > 0$, the Lagrange identities imply

$$v_Q(u_j)^\top \left\{ U + t \sum_{i=1}^Q c_i(T) \zeta_i \right\} = v_Q(u_j)^\top U + t \zeta_j, \quad 1 \leq j \leq q.$$

Thus, after applying $\mathcal{L}_{T_q}$, the ideal increment in (63) is independent of the ghost tuple Z and is exactly $b_{\mathcal{L},q}(T_q)$. Pull it through the integral and use (85); this proves (86).

For (87), use Fubini and (84):

$$\begin{aligned} \int D_q(1 - \bar{\chi}_q) &= \iint D_Q(T_q, Z) \{1 - \chi_Q(T_q, Z)\} dZ dT_q \\ &\leq C_Q |\{T \in (\mathbb{T}^d)^Q : \chi_Q(T) \neq 1\}|. \end{aligned}$$

The result follows from Lemma 4.2. □

11 Case B: global priors and parameter legality

11.1 The two parameter fields

Take independent copies of the local carrier from Proposition 10.3 over all active blocks. For a coefficient realization, form p by (52), and define the two sides by (60). On the Q -side the actual coefficient draws are those supplied by $\nu_{Q,k}$; the virtual shifts are used only to identify the finite moments of the mixed likelihood.

Lemma 11.1 (Hölder legality and positivity). *For sufficiently small fixed $c_a, c_b > 0$, every parameter draw on both sides belongs to $\mathcal{P}(\alpha, \beta, \gamma)$, apart from the design density, which is handled in the next subsection. Moreover,*

$$\tau_P(x_0) - \tau_Q(x_0) = c_a c_b \Delta. \quad (88)$$

Proof. By the Hölder scaling rule (Lemma 2.2), all packet fields used below satisfy, for each fixed t ,

$$\|P\|_{C^t} + \|P'\|_{C^t} + \|q_4\|_{C^t} \leq C_{t,Q} r^{-t}, \quad \|G_h\|_{C^t} \leq C_t h^{-t} \leq C_t r^{-t}. \quad (89)$$

Since $a = c_a r^\alpha$, (89) gives

$$\|p\|_{C^\alpha} + \|p'\|_{C^\alpha} \leq C_Q c_a.$$

Thus the two propensity fields lie in the prescribed Hölder ball after choosing c_a sufficiently small.

Next write $p' = aP'$. From (58),

$$K_x(p') = b \left[\{1 + \eta(x)\} P' - \frac{a^2 (P')^3}{3} \right]. \quad (90)$$

The fixed-overlap packet calculus and (89) give

$$\|P'\|_{C^\beta} + \|(P')^3\|_{C^\beta} \leq C_Q r^{-\beta}.$$

Moreover,

$$\|\eta\|_\infty \leq C_Q a^2 t^2, \quad \|\eta\|_{C^\beta} \leq C_Q a^2 t^2 r^{-\beta},$$

because $r < h$, q_4 varies at scale r , and G_h varies at scale h . The Hölder product inequality therefore yields

$$\begin{aligned} \|K_x(p')\|_{C^\beta} &\leq C_Q b r^{-\beta} \{1 + a^2 + a^2 t^2\} \\ &\leq C_Q c_b, \end{aligned}$$

since $b = c_b r^\beta$, $a \leq 1$, and $t \rightarrow 0$. The same estimate holds for $K_x(p)$.

The identity $a b t^2 = c_a c_b \Delta$ gives

$$d_*(x) = c_a c_b \Delta G_h(x)^2, \quad \|d_*\|_{C^\gamma} \leq C c_a c_b \Delta h^{-\gamma} = C c_a c_b. \quad (91)$$

Since $\gamma > A_0 > \beta$, also

$$\|d_*\|_{C^\beta} \leq C c_a c_b \Delta h^{-\beta} = C c_a c_b h^{\gamma-\beta}.$$

Although a draw of p is smooth, its C^β norm can grow when $\beta > \alpha$; the small target amplitude compensates for this. Indeed, using the Hölder product inequality and (89),

$$\begin{aligned} \|d_* p\|_{C^\beta} &\leq C_Q \left\{ \Delta a r^{-\beta} + a \Delta h^{-\beta} \right\} \\ &\leq C_Q \left\{ c_a t^2 r^{2\alpha} + a h^{\gamma-\beta} \right\} = o(1). \end{aligned}$$

Here $\Delta a r^{-\beta} = c_a t^2 r^{2\alpha}$, because $t^2 = \Delta r^{-A_0}$ and $A_0 = \alpha + \beta$. Consequently

$$u_x(p) - d_*(1+p) = K_x(p) - d_*(1+2p)$$

has C^β norm at most $C_Q c_b + o(1)$. Choose c_b first, then n large, so that this lies within the remaining radius of the prescribed outcome ball about 1/2. This proves the required smoothness of both baseline outcome regressions, while (91) proves the C^γ bound for the treatment effect.

All coded fields have sup norm tending to zero with c_a, c_b fixed and small: $p = O(a)$, $K_x(p) = O(b)$, and $d_* = O(\Delta)$. First choose c_a, c_b small enough to fit the fixed Hölder radii, and then take n large enough. The treatment propensity and both Bernoulli outcome means are then contained in $[\kappa, 1 - \kappa]$ on every draw. Finally, $G_h(x_0) = 1$, $\tau_P = d_*$, and $\tau_Q = 0$, so

$$\tau_P(x_0) - \tau_Q(x_0) = c_a c_b \Delta.$$

□

11.2 Design density and tilt

Transfer each torus factor to its physical carrier cube and set it equal to one outside that cube. Write this physical factor again as $F_{H_k}(x)$. It satisfies

$$\int_{S_k} \{F_{H_k}(x) - 1\} dx = 0.$$

Define

$$\tilde{f}_H(x) = \prod_{k:x \in S_k} F_{H_k}(x), \quad Z(H) = \int_{[0,1]^d} \tilde{f}_H(x) dx, \quad f_H = \tilde{f}_H / Z(H). \quad (92)$$

Let N_1 be the maximal number of carrier boxes S_k containing a point (a fixed number). Then

$$\frac{(1 - \eta_f)^{N_1}}{(1 + \eta_f)^{N_1}} \leq f_H(x) \leq \frac{(1 + \eta_f)^{N_1}}{(1 - \eta_f)^{N_1}}. \quad (93)$$

Choose $\eta_f > 0$ sufficiently small that these bounds lie inside $[\underline{f}, \bar{f}]$. Let $\nu_S = \bigotimes_k \nu_{S,k}$, $S \in \{P, Q\}$; their H -marginals are identical (Proposition 10.3). Apply Lemma 2.6 (with λ trivial): the tilted priors (9) induce the same law of f_H and the same X^n -marginal, and the mixed joint density is given by (10) with the product law ν_S restored.

12 Case B: fixed-sample information bound

12.1 Partial-jitter stability

For a subset $\mathcal{S}$ of the active blocks at a site, define

$$\delta_{\mathcal{S}} = a t \sum_{k \in \mathcal{S}} \psi_k \zeta_k, \quad v_{\mathcal{S}} = \mathbb{E} \delta_{\mathcal{S}}^2 \leq v, \quad m_{4,\mathcal{S}} = \mathbb{E} \delta_{\mathcal{S}}^4 \leq 3v_{\mathcal{S}}^2 \leq 3v^2.$$

The exact partial moments are

$$\mathbb{E} K_x(p + \delta_{\mathcal{S}}) = K_x(p) - \frac{b}{a} p v_{\mathcal{S}}, \quad (94)$$

$$\mathbb{E} \{(p + \delta_{\mathcal{S}}) K_x(p + \delta_{\mathcal{S}})\} = p K_x(p) + c_1 v_{\mathcal{S}} + c_3 \{6p^2 v_{\mathcal{S}} + m_{4,\mathcal{S}}\}. \quad (95)$$

Lemma 12.1 (Uniform stability under partial block shifts). *Fix $q \leq Q$, a baseline coefficient realization, and any subset of the blocks meeting the q sites. Average only the virtual jitters in that subset. Every joint q -site Walsh coefficient differs from the corresponding P -side coefficient by at most $C_Q \Delta$. If all active blocks are shifted, the difference is zero.*

Proof. The propensity coefficient at a site remains p . Since $v = a^2 t^2 G_h^2$, $d_* = (b/a)v$, and $|p| \leq Ca$, (94) differs from the full-jitter value by at most

$$\frac{b}{a}|p|(v - v_S) \leq Cd_*.$$

Using $c_1 = (b/a)(1 + \eta)$, $c_3 = -b/(3a)$, and $m_{4,S} \leq 3v^2$, the difference between (95) and the full-jitter value is also at most Cd_* . Here the terms are bounded respectively by $(b/a)v = d_*$, $(b/a)p^2v \leq Ca^2d_*$, and $(b/a)v^2 = d_*v$. Since $d_*(x) \leq C\Delta$, each one-site outcome Walsh coefficient differs from its target value by $C\Delta$.

Conditional on the baseline field, site signs are independent. Hence a joint Walsh coefficient is a product of one-site coefficients, and a finite telescoping bound gives $C_Q\Delta$. A product containing only propensity characters is unchanged exactly: each site propensity appears only once, and every nonempty jitter term has an odd site sign and averages to zero. When all active blocks are shifted, Lemma 9.1 gives exact equality. $\square$

12.2 Component factorization and the local discrepancy

Condition on the common design $X^n = x^n$, and write $\mathcal{K}_i = \{k : x_i \in S_k\}$. Relative to counting measure on the binary signs, the tilted mixed joint density has the exact form

$$\bar{p}_S(x^n, w^n) = \frac{4^{-n}}{\mathbb{E}Z(H)^n} \int \prod_k \left\{ \prod_{i:k \in \mathcal{K}_i} F_{H_k}(x_i) \right\} \prod_{i=1}^n L_{S,i}(w_i | x_i, U_{\mathcal{K}_i}) d\nu_S(H, U), \quad (96)$$

where $S \in \{P, Q\}$, and the common X^n density is obtained by removing the likelihood factors. Formula (96) follows directly from (10); in particular, no residual $Z(H)$ -dependence remains.

Form the bipartite incidence graph whose vertices are the observations and the active block labels, with an edge $i \sim k$ when $k \in \mathcal{K}_i$. Equivalently, connect observations i, j when $\mathcal{K}_i \cap \mathcal{K}_j \neq \emptyset$. Distinct connected components have disjoint block-label sets. Since ν_S is a product over blocks, both the integral in (96) and the analogous design integral factor exactly over the connected components. The common global factor $\{\mathbb{E}Z(H)^n\}^{-1}$ cancels when conditioning on $X^n = x^n$. Hence the conditional mixed laws of (R_i, T_i) factor exactly over the component graph, including when physical carrier boxes overlap.

Consider a component of size $q \leq Q$. Expand each of its finitely many Walsh coefficients in block coefficient monomials. At one observation the degree in the aggregate propensity is at most four: p is linear, $K(p)$ is cubic, and $pK(p)$ is quartic. Therefore the degree in every block is at most $4q \leq 4Q$, so Proposition 10.3 applies.

For a block meeting q_k sites in the component, Proposition 10.4 says that its lower-order moment vector is exactly the convex interpolation, with coefficient $\bar{\chi}_{q_k}$, between the baseline moment and the ideal shifted moment. Multiplying over the finitely many blocks shows that each component Walsh coefficient under the Q -mixture is a convex combination of the partial-shift Walsh coefficients from Lemma 12.1. The weight of the fully shifted term is the product of the block $\bar{\chi}_{q_k}$'s. Consequently, for every binary cell w ,

$$|p_{Q,C}(w | x_C) - p_{P,C}(w | x_C)| \leq C_Q\Delta \sum_{k \in \mathcal{K}(C)} \{1 - \bar{\chi}_{q_k}(T_k)\}. \quad (97)$$

All fixed-parameter binary cells are bounded below by a positive constant, so every q -site mixed cell is bounded below by a constant depending only on Q . The Walsh transform is finite-dimensional

and orthogonal. Therefore

$$H_C^2(x_C) \leq C_Q \Delta^2 \sum_{k \in \mathcal{K}(C)} \{1 - \bar{\chi}_{q_k}(T_k)\}. \quad (98)$$

We used $(\sum b_k)^2 \leq C_Q \sum b_k$, valid because the number of blocks in a q -component is bounded and $0 \leq b_k \leq 1$.

12.3 Geometry and low components

The common mixed design is not necessarily iid, but every fixed q -coordinate marginal density is at most $\bar{f}^q$, because conditionally on H the density is bounded by $\bar{f}$. A labelled connected q -tuple has one point in the $O(h^d)$ active region and, after choosing a spanning tree, every other point in an $O(r^d)$ neighborhood of its parent.

We insert the ghost factor explicitly. Fix a labelled connected q -tuple, one of its incident blocks k_* , and let $I_* \subseteq \{1, \dots, q\}$ be the $q_* \geq 1$ sites lying in S_{k_*} . There are $O(h^d r^{-d})$ possible active choices of k_* . In affine coordinates on S_{k_*} , the integral of the corresponding leakage factor satisfies

$$\begin{aligned} \int_{S_{k_*}^{q_*}} \{1 - \bar{\chi}_{q_*}(T_{k_*})\} dx_{I_*} \\ \leq C_Q r^{dq_*} t^d \Lambda^{M_Q}. \end{aligned} \quad (99)$$

Indeed, the affine Jacobian is $C r^{dq_*}$, and (87) applies after using the fixed positive lower bound on D_{q_*} . Because the observation component is connected, the remaining $q - q_*$ sites can be attached to the set I_* by a labelled forest with exactly $q - q_*$ edges. Each attachment forces the child into an $O(r^d)$ neighborhood of its parent. Thus, for fixed k_* , the remaining integration contributes at most $C_Q r^{d(q-q_*)}$. Multiplying by the number $O(h^d r^{-d})$ of choices of k_* gives exactly

$$C_Q h^d r^{d(q-1)} t^d \Lambda^{M_Q}.$$

The number of labelled forests and the number of incident blocks in a q -component are bounded in terms of Q , and the mixed-design q -marginal is bounded by $\bar{f}^q$. Summing (98) therefore gives, for $2 \leq q \leq Q$,

$$\begin{aligned} \mathbb{E} \sum_{C: |C|=q} H_C^2 &\leq C_Q n^q h^d r^{d(q-1)} t^d \Delta^2 \Lambda^{M_Q} \\ &= C_Q n^q h^d r^{d(q-2)} \ell^d \Delta^2 \Lambda^{M_Q} \\ &= C_Q \{n^2 h^d \ell^d \Delta^2\} (nr^d)^{q-2} \Lambda^{M_Q} = o(1). \end{aligned} \quad (100)$$

For singleton components, fix an incident block k_* , sum over its $O(h^d r^{-d})$ possible locations, and use (99) with $q_* = 1$. There is no forest edge, so the same calculation gives

$$\mathbb{E} \sum_{C: |C|=1} H_C^2 \leq C_Q n h^d t^d \Delta^2 \Lambda^{M_Q} = o(1). \quad (101)$$

The final equalities use Lemma 3.2 and $nr^d = o(1)$. Fixed powers of Λ are absorbed by the strictly negative powers of n .

12.4 High components

Every component larger than Q contains a connected labelled subset of size $Q+1$. The spanning-tree volume bound and the mixed-design upper bound give

$$\mathbb{P}\{\text{some component has size } > Q\} \leq C_Q n^{Q+1} h^d r^{dQ} = o(1) \quad (102)$$

by (16). These configurations are handled as the exceptional event in Lemma 2.5; no sum of uncontrolled high-component Hellinger distances is needed.

12.5 Total variation and Le Cam

Let $\bar{P}_n, \bar{Q}_n$ denote the two prior mixtures. Their X^n -marginals coincide, and given the design the conditional laws factor over the component graph; on the event of no high component every component has at most Q sites. Lemma 2.5 with (100), (101), and (102) gives

$$\text{TV}(\bar{P}_n, \bar{Q}_n) \leq \mathbb{P}\{\text{a high component}\} + \left\{ \mathbb{E} \sum_{C: |C| \leq Q} H_C^2 \right\}^{1/2} = o(1).$$

In particular,

$$\text{TV}(\bar{P}_n, \bar{Q}_n) \leq \frac{1}{2} \quad (103)$$

for all sufficiently large n .

Every draw from the P -prior has target value $c_a c_b \Delta$, and every draw from the Q -prior has target value zero (Lemma 11.1). Lemma 2.3 gives

$$\sup_{P \in \mathcal{P}} \mathbb{E}_P |\hat{\tau}(x_0) - \tau_P(x_0)| \geq \frac{c_a c_b \Delta}{2} \{1 - \text{TV}(\bar{P}_n, \bar{Q}_n)\} \geq \frac{c_a c_b}{4} n^{-(\rho+\varepsilon)}.$$

This proves Theorem 1.1 in case (B) for all sufficiently small fixed $\varepsilon > 0$. For any prescribed $\varepsilon > 0$, choose a fixed $0 < \varepsilon_0 \leq \varepsilon$ in that range and use $n^{-(\rho+\varepsilon_0)} \geq n^{-(\rho+\varepsilon)}$, changing the constant if necessary.

Part IV

Case C: mixed smoothness

13 Case C: construction

Throughout Part C exactly one of α, β exceeds one: case (a) $\alpha \leq 1 < \beta$, case (b) $\beta \leq 1 < \alpha$; thus $s_1 = (1 + \alpha \wedge \beta)/2$ and the scales are those of Section 3. The degree

$$\mathfrak{d} = \begin{cases} 1 & \text{in case (a),} \\ 3 & \text{in case (b)} \end{cases}$$

is the degree of the likelihood cells in the carried coefficients and the degree of the carrier algebra of Section 15.

13.1 Amplitudes

In both cases the rough nuisance has amplitude $c\ell^{(\text{its exponent})}$, the smooth nuisance has amplitude $c r^{(\text{its exponent})}$ and receives a virtual jitter of relative size t :

	rough amplitude	smooth amplitude	jitter	
(a) $\alpha \leq 1 < \beta$	$j_a = c_a \ell^\alpha$	$b = c_b r^\beta$	$j_b = bt = c_b tr^\beta$	$j_\circ = j_b, j_\bullet = j_a$
(b) $\beta \leq 1 < \alpha$	$j_b = c_b \ell^\beta$	$a = c_a r^\alpha$	$j_a = at = c_a tr^\alpha$	$j_\circ = j_a, j_\bullet = j_b$

(104)

Here $j_\circ$ (the *shift amplitude*) is the jitter of the smooth nuisance and $j_\bullet$ (the *field amplitude*) the amplitude of the rough one. In both cases

$$j_a j_b = c_a c_b \Delta, \quad d_*(x) := j_a j_b G_h(x)^2 = c_a c_b \Delta G_h(x)^2, \quad \frac{j_\circ}{j_\bullet} = \frac{c_\circ}{c_\bullet} t^{1-(\alpha \wedge \beta)} r^{|\alpha - \beta|} \rightarrow 0, \quad (105)$$

where $G_h(x) = G((x - x_0)/h)$ for a fixed $G \in C_c^\infty$ with $G(0) = 1$, $0 \leq G \leq 1$; the first identity is $\ell^{\alpha \wedge \beta} t r^{\alpha \vee \beta} = t^{1+\alpha \wedge \beta} r^{A_0} = t^{2s_1} r^{A_0} = \Delta$, and the last uses $\alpha \vee \beta > 1 \geq \alpha \wedge \beta$: the factor $r^{|\alpha - \beta|}$ tends to zero because $\alpha \neq \beta$, and $t^{1-\alpha \wedge \beta} \leq 1$. Hence, for large n ,

$$j_\circ \leq j_\bullet, \quad j_\circ^2 \leq j_\circ j_\bullet = c_a c_b \Delta. \quad (106)$$

13.2 Blocks and rough packets

Blocks. Let $\phi_{\text{lin}} \in C_c^\infty(\mathbb{R}^d)$ be a nonnegative partition-of-unity bump: $\phi_{\text{lin}} \geq 0$, $\text{supp } \phi_{\text{lin}} \subseteq [-1, 1]^d$, $\phi_{\text{lin}} > 0$ on $(-1, 1)^d$, and $\sum_{k \in \mathbb{Z}^d} \phi_{\text{lin}}(u - k) \equiv 1$ (take $\phi_{\text{lin}} = \vartheta_0 / \sum_k \vartheta_0(\cdot - k)$ for a nonnegative $\vartheta_0 \in C_c^\infty$ positive on $(-1, 1)^d$ and supported in $[-1, 1]^d$; the denominator is bounded below by ϑ_0 at the nearest lattice point). By compactness there is a constant $c_d \in (0, 1]$, depending only on d and the fixed bump, with

$$\phi_{\text{lin}} \geq c_d \quad \text{on } [-3/4, 3/4]^d. \quad (107)$$

Put $x_k = x_0 + rk$ and $\psi_k(x) = \phi_{\text{lin}}((x - x_k)/r)$, so $0 \leq \psi_k \leq 1$, $\sum_k \psi_k \equiv 1$, and $\psi_k \geq c_d$ on $x_k + [-3r/4, 3r/4]^d$. Enlarge each support to a cube S_k of side Cr , identified with $\mathbb{T}^d$ by an affine chart θ_k ; $\text{supp } \psi_k$ lies in a fixed compact subcube. Let N_1 be the maximal number of enlarged cubes S_k containing a given point (a fixed number, $\leq C^d$); a fortiori at most N_1 supports $\text{supp } \psi_k$ meet a point. Active blocks are those meeting $\text{supp } G_h$; there are $O(h^d r^{-d})$. Lagrange vectors $c_{ki}(T) = c_i(T)$ for tuples in block k are those of Lemma 4.1, and the taper is $\chi_Q(T) = \prod_{i=1}^Q \vartheta(P_i(T)/t_n)$ with the fixed cutoff ϑ of Section 4 ($\vartheta = 0$ on $[0, 1]$, $= 1$ on $[2, \infty)$).

Rough packets. The rough packets ψ_m^b , the rough field Λ_λ , the packet sets $J(x)$, the constants N_0, R_0 , and the moments $M_j = \mathbb{E} \Lambda_\lambda^j$ (in particular $M_2 = G_h^2$, $M_4 = \mu_4 = G_h^4 \{3 - 2q_4\}$) are those of Section 2, (7)–(8), at the fine scale $\ell = tr$ of (12). Case (a) uses only M_0, M_1, M_2 ; case (b) also uses $M_3 = 0$ and M_4 .

13.3 Assignment of sites to blocks

The block partition $\{\psi_k\}$ is smooth with constant overlap, which is required for the smoothness of the carried field below. The carrier of Section 15, however, needs every site to receive its virtual jitter from a *single* block, up to a set of sites of small relative volume. Let $C_k = x_k + [-r/2, r/2]^d$ be the cubes of the lattice $\{x_k\}$, so that $\{C_k\}$ partitions $\mathbb{R}^d$; recall from (107) that $\psi_k \geq c_d$ on the enlarged cube $x_k + [-3r/4, 3r/4]^d$. Let

$$w = t^d \quad (108)$$

be the relative transition width, and construct the assignment partition as a tensor product of one-dimensional ones: let ϱ_w be a fixed smooth probability density on $\mathbb{R}$ supported in $[-w/2, w/2]$ (a fixed bump rescaled to width w), $\sigma = \varrho_w * \mathbf{1}_{[-1/2, 1/2]}$, and

$$\phi_k(x) = \prod_{j=1}^d \sigma\left(\frac{x^{(j)} - x_k^{(j)}}{r}\right), \quad x \in \mathbb{R}^d. \quad (109)$$

Since $\sum_{l \in \mathbb{Z}} \mathbf{1}_{[-1/2, 1/2]}(y - l) \equiv 1$, also $\sum_l \sigma(y - l) \equiv 1$, hence $\sum_k \phi_k \equiv 1$; moreover $0 \leq \phi_k \leq 1$, $\text{supp } \phi_k \subseteq x_k + [-(1+w)r/2, (1+w)r/2]^d \subseteq x_k + [-3r/4, 3r/4]^d$, and $\phi_k = 1$ on $\{x \in C_k : \text{dist}_\infty(x, \partial C_k) \geq wr/2\}$. Sites with $\phi_k(x) = 1$ for some k are *interior* and are *assigned* to that block, $k(x) = k$; the remaining set $\mathcal{L} = \{x : 0 < \phi_k(x) < 1 \text{ for some } k\}$ is the *transition layer*. It is contained in the union over k of the wr -neighbourhoods of ∂C_k , so its relative volume in every cube C_k is at most $2dw = 2dt^d$. Put

$$m_k(x) = \frac{\phi_k(x)}{\psi_k(x)} \mathbf{1}_{\{\phi_k(x) > 0\}}, \quad 0 \leq m_k \leq c_d^{-1}, \quad (110)$$

which is smooth because $\psi_k \geq c_d$ on $\text{supp } \phi_k$ by (107); c_d^{-1} is a fixed constant and enters the bounds below only through the constant N_m of (124).

Lemma 13.1 (Wiener norm of the multiplier). *In the chart of block k , $\|m_k \circ \theta_k^{-1}\|_{\mathcal{W}_d} \leq C_d \{1 + \log(1/w)\}^d \leq C_d \Lambda^d$.*

Proof. One-dimensional step. On the circle of length one identify $\sigma(\cdot)$ (after the affine chart, the cube S_k of side Cr becomes $\mathbb{T}^d$ and $x \mapsto (x - x_k)/r$ becomes a fixed dilation, so we may work with σ on an interval of fixed relative length and transition width $w' = w/C$). Its Fourier coefficients are $\hat{\sigma}(\nu) = \hat{\mathbf{1}}(\nu) \hat{\varrho}(\nu)$ with $|\hat{\mathbf{1}}(\nu)| \leq C/|\nu|$ ($\nu \neq 0$) and, by smoothness of the bump, $|\hat{\varrho}(\nu)| \leq C_K(1 + w'|\nu|)^{-K}$ for every K . Hence

$$\sum_\nu |\hat{\sigma}(\nu)| \leq C + \sum_{1 \leq |\nu| \leq 1/w'} \frac{C}{|\nu|} + \sum_{|\nu| > 1/w'} \frac{C_K}{|\nu|(w'|\nu|)^K} \leq C \{1 + \log(1/w')\}.$$

Tensor product. In the chart, $\phi_k \circ \theta_k^{-1}$ is the product over the d coordinates of such one-dimensional functions, and the Wiener norm of a tensor product is the product of the Wiener norms; so $\|\phi_k \circ \theta_k^{-1}\|_{\mathcal{W}_d} \leq C_d \{1 + \log(1/w)\}^d$. *The factor $1/\psi_k$.* Let ζ be a fixed smooth function on $\mathbb{T}^d$ equal to $1/(\psi_k \circ \theta_k^{-1})$ on $\{\psi_k \circ \theta_k^{-1} \geq c_d\}$ (which contains the support of $\phi_k \circ \theta_k^{-1}$); ζ does not depend on k or n because $\psi_k \circ \theta_k^{-1}$ is the same fixed bump for every block, so $\|\zeta\|_{\mathcal{W}_d} \leq C$. Then $m_k \circ \theta_k^{-1} = (\phi_k \circ \theta_k^{-1})\zeta$ and the algebra property of $\mathcal{W}_d$ gives the claim. Finally, by (12),

$$\log(1/w) = d \log(1/t) = \frac{d(z - \xi A_0)}{2s_1} \log n = O(\log n) = O(\Lambda).$$

□

13.4 Fields, virtual shift, and the two hypotheses

Only active blocks carry coefficients: $U = (U_k)_{k \text{ active}}$, $U_k \in \mathbb{R}^M$, and the *legal ball* is $\{\|U_k\| \leq R_U \forall k\}$ for a fixed R_U with $2\mathbb{B}_0 \subseteq \{\|U\| \leq R_U\}$, where $\mathbb{B}_0$ is the support ball of Π_0 . The *carried field* (unit amplitude) is

$$\mathfrak{c}(x) = \sum_{k \text{ active}} \psi_k(x) v_Q(\theta_k(x))^\top U_k. \quad (111)$$

Let $T_k = (x_1, \dots, x_Q)$ be a full Q -tuple in block k , with $\chi_Q(\theta_k(T_k)) > 0$; the Lagrange vectors below are evaluated at its chart coordinates. The *virtual shift* of block k is

$$\tilde{U}_k = U_k + t \sum_{i=1}^Q m_k(x_i) \Lambda_\lambda(x_i) c_{ki}(T_k) + (\text{moment corrections, Section 15.5}). \quad (112)$$

For $q < Q$ retained sites, the remaining $Q - q$ sites are auxiliary integration variables, removed after likelihood evaluation as in Section 15.6. In particular, χ_Q is never applied to a tuple of length less than Q . $\tilde{U}_k$ is a bookkeeping expression, not a random variable of the construction: the actual parameters below contain only U_k , and (112) records which moments the carrier reproduces. By $v_Q(\theta_k(x_j))^\top c_{ki}(T_k) = \mathbf{1}\{i = j\}$, the first sum changes $\mathbf{c}(x_j)$ by

$$\psi_k(x_j) t m_k(x_j) \Lambda_\lambda(x_j) = t \phi_k(x_j) \Lambda_\lambda(x_j);$$

summing over the blocks meeting x_j and using $\sum_k \phi_k = 1$, the total change of $\mathbf{c}(x_j)$ at a fully shifted site is $t \Lambda_\lambda(x_j)$, coming from the assigned block alone when the site is interior. The fields and the two hypotheses are, in coded coordinates:

Case (a) ($\alpha \leq 1 < \beta$): $p_\lambda = \mathbf{j}_a \Lambda_\lambda$, $y_0 = b \mathbf{c}$, site jitter $\delta = bt \Lambda_\lambda = \mathbf{j}_b \Lambda_\lambda$, and

	$p = 2\pi - 1$	$y = 2\mu_0 - 1$	τ
P	p_λ	$y_0 - d_*$	d_*
Q	p_λ	y_0	0
Q (virtual)	p_λ	$y_0 + \delta$	0

(113)

Case (b) ($\beta \leq 1 < \alpha$): $p_0 = a \mathbf{c}$, site jitter $\delta = at \Lambda_\lambda = \mathbf{j}_a \Lambda_\lambda$, $y_\lambda(\cdot; p) = \mathbf{j}_b \Lambda_\lambda \{1 + \eta - p^2\}$ with the fourth-moment correction

$$\eta(x) := \mathbf{j}_a^2 G_h(x)^2 \{3 - 2q_4(x)\} \geq 0, \quad \|\eta\|_\infty \leq 3\mathbf{j}_a^2, \quad \|\eta\|_{C^\beta} \leq C \mathbf{j}_a^2 \ell^{-\beta}, \quad d_* \eta = \mathbf{j}_a^3 \mathbf{j}_b \mu_4, \quad (114)$$

defined for every x and equal to $\mathbf{j}_a^2 \mu_4(x) / G_h(x)^2$ wherever $G_h(x) \neq 0$ (by (8)); the last identity holds everywhere, both sides vanishing where $G_h = 0$, and

	$p = 2\pi - 1$	$y = 2\mu_0 - 1$	τ
P	p_0	$y_\lambda(\cdot; p_0) - d_*(1 + 3p_0)$	d_*
Q	p_0	$y_\lambda(\cdot; p_0)$	0
Q (virtual)	$p_0 + \delta$	$y_\lambda(\cdot; p_0 + \delta)$	0

(115)

In both tables the rows P and Q are the *actual* parameters: under both hypotheses the coefficients U_k of the carried field are drawn from the laws of Proposition 15.7, the rough signs from the same Rademacher prior, and no parameter contains δ . The row “ Q (virtual)” is not a parameter of any law in the mixture; it is the *jittered* parameter used in the single-site identities of Section 13.5 and in the target of Section 15.5. Its role is the following. The two carrier laws differ only in the law of U_k given the design, and Proposition 15.7(iv) states that this difference, tested against any likelihood cell, equals the tapered, design-weighted moment increment $\mathcal{J}(\mathbb{D})$ built from the jittered parameter. This is a moment identity with a taper and with the moment corrections of Section 15.5; it does not say that the shifted coefficients (112) are the values of a random vector under Q . In case (b) the Q -side outcome is a deterministic quadratic function of the propensity and of the rough field: this is the cubic bridge of Part B with the rough field in place of one power of the propensity.

13.5 The single-site identities

Fix a site x and write $\Lambda = \Lambda_\lambda(x)$, $G = G_h(x)$, $d = d_*(x)$, $\eta = \eta(x)$, $M_j = \mathbb{E}\Lambda^j$. The *shift variable* ζ is the coded value of the smooth nuisance at the site, $\zeta = y$ in case (a) and $\zeta = p$ in case (b); the unshifted Q -likelihood, as a function of ζ for a cell $(R, T) \in \{\pm 1\}^2$, and the *real field* are

	$L(\zeta)$	$\text{real}(\zeta)$
(a)	$(1 + Rj_a\Lambda)(1 + T\zeta)$	$dj_a\Lambda T + dRT$
(b)	$(1 + R\zeta)(1 + Tj_b\Lambda(1 + \eta - \zeta^2))$	$-2d\zeta T + d(1 - 3\zeta^2)RT$

(116)

L is a polynomial of degree $\mathfrak{d}$ in ζ and affine in Λ , $L = A + B\Lambda$ with B carrying the factor $j_\bullet$. By (5), $L(\zeta_0)$ is the Q -likelihood at the unshifted parameter ($\zeta_0 = y_0$, resp. p_0), $L(\zeta_0 + j_\bullet\Lambda)$ the Q -likelihood at the jittered parameter, and the P -likelihood is

$$L^P = L(\zeta_0) + \text{real}(\zeta_0) : \quad (117)$$

in case (a), $y + \tau(1 + p) = y_0 + dj_a\Lambda$ and $py + \tau(1 + p) = j_a\Lambda y_0 + d$; in case (b), $y + \tau(1 + p) = y_\lambda - d(1 + 3p_0) + d(1 + p_0) = y_\lambda - 2dp_0$ and $py + \tau(1 + p) = p_0y_\lambda - dp_0(1 + 3p_0) + d(1 + p_0) = p_0y_\lambda + d(1 - 3p_0^2)$. Define the *increments*

$$\Delta^{(e)}(\zeta) = \frac{j_\bullet^e}{e!} \partial_\zeta^e L(\zeta), \quad e = 1, \dots, \mathfrak{d}, \quad \text{so that} \quad L(\zeta + j_\bullet\Lambda) - L(\zeta) = \sum_{e=1}^{\mathfrak{d}} \Lambda^e \Delta^{(e)}(\zeta), \quad (118)$$

and the *substitution polynomials* in Λ , for $e \in \{1, \dots, \mathfrak{d}\}$, $f \in \{0, \dots, \mathfrak{d}\}$:

$$\begin{aligned} \text{(a)} \quad & \left| \begin{array}{l} \mathsf{K}^{(1,0)}(\Lambda) = \Lambda, \quad \mathsf{K}^{(1,1)}(\Lambda) = \varkappa := j_a^2 G^4 \quad (\text{so that } \varkappa j_b = dj_a G^2), \\ \mathsf{K}^{(e,f)}(\Lambda) = M_f \Lambda^e : \quad \mathsf{K}^{(e,0)} = \Lambda^e, \quad \mathsf{K}^{(e,f)} = 0 \text{ (} f \text{ odd)}, \quad \mathsf{K}^{(e,2)} = G^2 \Lambda^e. \end{array} \right. \end{aligned} \quad (119)$$

In both cases every nonzero $\mathsf{K}^{(e,f)}$ is a Walsh polynomial all of whose monomials have degree $\equiv e + f \pmod{2}$ (in case (a): Λ for $(1, 0)$, the constant $\varkappa$ for $(1, 1)$), and $\|\mathsf{K}^{(e,f)}\|_W \leq N_0^{\mathfrak{d}}$.

Lemma 13.2 (Bridge with design exponents). *In either case, for every real ζ , every cell (R, T) , and every $f \in \{0, \dots, \mathfrak{d}\}$,*

$$\sum_{e=1}^{\mathfrak{d}} \mathbb{E}_\Lambda [\mathsf{K}^{(e,f)}(\Lambda) \Delta^{(e)}(\zeta)] = \mathbb{E}_\Lambda [\Lambda^f \text{real}(\zeta)], \quad (120)$$

where $\mathbb{E}_\Lambda$ averages the rough field at the site (the other quantities being fixed). In particular ($f = 0$) $\mathbb{E}_\Lambda L(\zeta + j_\bullet\Lambda) = \mathbb{E}_\Lambda L(\zeta) + \mathbb{E}_\Lambda \text{real}(\zeta)$.

Proof. Case (a). $\Delta^{(1)}(y) = j_b \partial_y L = j_b T(1 + Rj_a\Lambda)$. For $f = 0$: $\mathbb{E}[\Lambda j_b T(1 + Rj_a\Lambda)] = j_a j_b G^2 RT = dRT$, and $\mathbb{E}[\text{real}] = \mathbb{E}[dj_a\Lambda T + dRT] = dRT$. For $f = 1$: $\mathbb{E}[\varkappa j_b T(1 + Rj_a\Lambda)] = \varkappa j_b T = dj_a G^2 T$, and $\mathbb{E}[\Lambda \text{real}] = dj_a G^2 T$.

Case (b). Write $c = Tj_b\Lambda$, so that $L(p) = 1 + c(1 + \eta) + Rp + Rpc(1 + \eta) - cp^2 - Rcp^3$ and

$$\partial_p L = R + Rc(1 + \eta) - 2cp - 3Rcp^2, \quad \partial_p^2 L = -2c - 6Rcp, \quad \partial_p^3 L = -6Rc.$$

With $\mathbb{E}[\Lambda^j c] = M_{j+1} j_b T$, for every $j \geq 0$:

$$\mathbb{E}[\Lambda^j \Delta^{(1)}(p)] = j_a [M_j R + M_{j+1} j_b T \{R(1 + \eta) - 2p - 3Rp^2\}], \quad (121)$$

$$\mathbb{E}[\Lambda^j \Delta^{(2)}(p)] = -j_a^2 M_{j+1} j_b T (1 + 3Rp), \quad (122)$$

$$\mathbb{E}[\Lambda^j \Delta^{(3)}(p)] = -j_a^3 M_{j+1} j_b RT. \quad (123)$$

For $f = 0$ the left side of (120) is $\sum_e \mathbb{E}[\Lambda^e \Delta^{(e)}]$. By (121) with $j = 1$ ($M_1 = 0$, $M_2 = G^2$): $j_a j_b G^2 T \{R(1 + \eta) - 2p - 3Rp^2\} = d \{RT(1 + \eta) - 2pT - 3RTp^2\}$. By (122) with $j = 2$: $M_3 = 0$, so the term vanishes. By (123) with $j = 3$: $-j_a^3 \mu_4 j_b RT = -d\eta RT$ by (114). The sum is $d(1 - 3p^2)RT - 2dpT = \text{real}(p) = \mathbb{E}[\Lambda^0 \text{real}(p)]$ (real does not contain Λ). For $f \geq 1$, $K^{(e,f)} = M_f \Lambda^e$, so the left side is M_f times the left side for $f = 0$, i.e. $M_f \text{real}(p) = \mathbb{E}[\Lambda^f \text{real}(p)]$. $\square$

The identity with $f = 0$ is the idea behind the construction: at a single site, averaging the rough field, the jittered Q -likelihood equals the P -likelihood; in case (a) the Q -side RT -cell produces d through $\mathbb{E}[j_a \Lambda \cdot j_b \Lambda] = j_a j_b G^2$, in case (b) the fourth-moment correction η cancels the cubic term of the jitter and the P -side control outcome absorbs the T -cell $-2dp_0$, which is legal because p_0 is smooth. The identities with $f \geq 1$ are what the design weights of the carrier require (Proposition 16.4, Step 3): when the design density carries a factor Λ^f at the site, the jitter Λ^e of the site must be replaced by $K^{(e,f)}(\Lambda)$, and Lemma 13.2 says that this reproduces the design-weighted P -side $\mathbb{E}[\Lambda^f \text{real}]$ exactly. In case (a) the real field contains Λ and the correction replaces the jitter by the constant $\varkappa$; in case (b) it does not, and the correction is a rescaling of the jitter by $\mathbb{E}[\Lambda^f]$.

14 Case C: legality

Lemma 14.1. *For sufficiently small fixed $c_a, c_b > 0$, all sufficiently large n , every λ , every U in the legal ball, and every design density produced by the carrier, the two actual rows P and Q of (113), resp. (115), define laws in $\mathcal{P}$, and $\tau_P(x_0) - \tau_Q(x_0) = c_a c_b \Delta$.*

Proof. The rough field. $\Lambda_\lambda = G_h \cdot \sum_m \lambda_m \phi((\cdot - x_0)/\ell - m)$ is G_h times a bounded-overlap sum of translates of $\phi(\cdot/\ell)$. By Lemma 2.2 with $t = \alpha \wedge \beta \leq 1$, $\|\Lambda_\lambda\|_{C^{\alpha \wedge \beta}} \leq C\ell^{-\alpha \wedge \beta} + Ch^{-\alpha \wedge \beta} \leq C\ell^{-\alpha \wedge \beta}$, and $|\Lambda_\lambda| \leq N_0$.

The carried field. $\mathbf{c} = \sum_k \psi_k v_Q(\theta_k)^\top U_k$ is a bounded-overlap sum of functions smooth at scale r with bounded coefficients, so by Lemma 2.2 (any $t > 0$) $\|\mathbf{c}\|_{C^{\alpha \vee \beta}} \leq C_Q r^{-\alpha \vee \beta}$ and $|\mathbf{c}| \leq C_Q$. The carrier draws every U_k inside the fixed legal ball by Proposition 15.7(iii), and legality of the actual parameters involves only U_k . (The shifted coefficients (112) are never drawn and are not parameters of any law. The jittered parameter is used only in the single-site identities and in Lemma 16.2, where the jitter is the first-order term of (112) alone, of size $\|t \sum_i m_k(x_i) c_{ki} \Lambda_i\| \leq C_Q N_0 t / t_n = C_Q \Lambda^{-B} \rightarrow 0$ on the support of χ_Q . The base point there is $U \in \mathbb{B}_0 = \text{supp } \Pi_0$, and $2\mathbb{B}_0$ lies inside the legal ball, so the fixed margin between $\mathbb{B}_0$ and the boundary of the legal ball absorbs the shift for large n ; the statement is not made for arbitrary U on the boundary of the legal ball. The moment corrections of Section 15.5 are terms of the algebra element $\mathcal{J}(\mathbb{D})$, controlled through $\|\mathbb{D} - 1^{\otimes Q}\|_{\mathcal{A}}$, and are not shifts of any parameter.)

Case (a). Propensity: $\|p_\lambda\|_{C^\alpha} \leq Cc_a$ and $|p_\lambda| \leq N_0 c_a$; choose c_a so that $\pi = (1 + p_\lambda)/2$ lies in $\mathcal{H}^\alpha(L_\pi)$ (the constant $1/2$ is an interior point) and $N_0 c_a \leq 1 - 2\kappa$. Outcome, Q side: $y_0 = b\mathbf{c}$ has $\|y_0\|_{C^\beta} \leq C_Q b r^{-\beta} = C_Q c_b$, $|y_0| \leq C_Q c_b$. Outcome, P side: $y_P = y_0 - d_*$, and $\|d_*\|_{C^\beta} \leq C\Delta h^{-\beta} = Ch^{\gamma-\beta} \rightarrow 0$ since $\gamma > A_0 \geq \beta$.

Case (b). Propensity: $\|p_0\|_{C^\alpha} \leq C_Q a r^{-\alpha} = C_Q c_a$, $|p_0| \leq C_Q c_a$; choose c_a accordingly. Outcome, Q side: $y = j_b \Lambda_\lambda (1 + \eta - p_0^2)$; by (114) and the bound on p_0 , $\|1 + \eta - p_0^2\|_\infty \leq 2$ and $\|1 + \eta - p_0^2\|_{C^\beta} \leq C j_a^2 \ell^{-\beta} + C_Q c_a^2 r^{-\beta} \leq C\ell^{-\beta}$, so $\|y\|_{C^\beta} \leq C_Q j_b \ell^{-\beta} = C_Q c_b$ and $|y| \leq 2N_0 c_b$. Outcome, P side: $y_P = y - d_*(1 + 3p_0)$, and $\|d_*(1 + 3p_0)\|_{C^\beta} \leq C\Delta h^{-\beta} + C_Q \Delta a r^{-\beta} = Ch^{\gamma-\beta} + C_Q c_a \Delta r^{\alpha-\beta} \rightarrow 0$, since $\gamma > A_0 > \beta$ and $\alpha > \beta$.

CATE, ranges, design, separation (both cases). $\tau_P = d_* = c_a c_b \Delta G_h^2$ has $\|\tau_P\|_{C^\gamma} \leq Cc_a c_b \Delta h^{-\gamma} = Cc_a c_b$; $\tau_Q = 0$. $\pi = (1 + p)/2$, $\mu_0 = (1 + y)/2$ and $\mu_1 = \mu_0 + \tau$ lie in $[\kappa, 1 - \kappa]$ because $|p| \leq C_Q c_a$ and $|y| + |\tau| \leq C_Q c_b + o(1)$, both $< 1 - 2\kappa$ after choosing c_a, c_b . The design density is f_H of

Section 15, which lies in $[f, \bar{f}]$ by (144), the only requirement that $\mathcal{P}$ places on the design. Finally $\tau_P(x_0) - \tau_Q(x_0) = c_a c_b \Delta \bar{G}(0)^2 = c_a c_b \Delta$. $\square$

15 Case C: the carrier with exact design-weighted matching

Fix an active block k . The chart θ_k identifies S_k with $\mathbb{T}^d$; functions on S_k become functions on the torus only after a cutoff, since a rough packet or the bump G_h may cross ∂S_k and its restriction to S_k , read periodically, would have a jump there. Let $\chi_0 \in C_c^\infty$ on the open chart cube be a fixed cutoff with $0 \leq \chi_0 \leq 1$, equal to one on the fixed subcube containing $\text{supp}(\phi_k \circ \theta_k^{-1})$ (hence on $\text{supp } m$, with m as below) and vanishing near the boundary of the cube; χ_0 is the same function for every block. In the chart write $m(u) = m_k(\theta_k^{-1}(u))$ and

$$\Lambda(u) = \chi_0(u) \Lambda_\lambda(\theta_k^{-1}(u)), \quad \psi_m(u) = \chi_0(u) \psi_m^b(\theta_k^{-1}(u)), \quad G(u) = G_h(\theta_k^{-1}(u)),$$

so that $\Lambda(u) = \sum_{m \in \mathcal{J}_k} \lambda_m \psi_m(u)$, where $\mathcal{J}_k = \{m : \text{supp } \psi_m^b \cap S_k \neq \emptyset\}$ is the set of packets *read by block k* . Since $\chi_0 = 1$ on $\text{supp } m$, $m\Lambda$ and $m\psi_m$ are the chart expressions of $m_k \Lambda_\lambda$ and $m_k \psi_m^b$, so the cutoff does not change any quantity that carries the multiplier m ; it only makes the packets smooth on the torus. Then $|\mathcal{J}_k| \leq Ct^{-d}$, at most N_0 of the ψ_m are nonzero at any point, and each ψ_m is a smooth function supported in a cube of side $\asymp t$ with

$$\|\psi_m\|_{\mathcal{W}_d} \leq C_\psi, \quad \|\psi_m\|_{L^1(\mathbb{T}^d)} \leq Ct^d :$$

by (7), $\psi_m^b(x) = G_h(x) \phi((x - x_m^b)/\ell)$ with $x_m^b = x_0 + \ell m$, and the chart is affine with $\ell \mapsto t' \asymp t$, so $\psi_m(u) = b_{k,m,n}((u - u_m)/t')$ with the scaled profile

$$b_{k,m,n}(v) = \chi_0(u_m + t'v) G_h(x_m^b + \ell v) \phi(v), \quad u_m = \theta_k(x_m^b).$$

This profile is not a fixed function, but the family is supported in $\text{supp } \phi$ and has C^{d+1} norms bounded uniformly in k, m and n : χ_0 is fixed and $t' \leq 1$, and $\partial_v^\nu \{G_h(x_m^b + \ell v)\} = (\ell/h)^{|\nu|} (\partial^\nu G)((x_m^b - x_0 + \ell v)/h)$ with $\ell/h \leq 1$. The Wiener norm of a dilate $v \mapsto b((u - u_m)/t')$ of a bump b is bounded by a constant times the C^{d+1} norm of b , uniformly in the dilation, which gives the first bound; the second is $|\psi_m| \leq |\phi(\cdot/t')|$ with $0 \leq \chi_0 \leq 1$ and $|G_h| \leq 1$. G is not cut off here: it enters the algebra only through the products $\chi_0 G^2, \chi_0 G^4$ of (132). All constants are uniform in k and in the signs. The likelihood cells are polynomials of degree $\leq \mathfrak{d}$ in each site's evaluation of the carried field, so the carrier must reproduce moments of degree $\leq \mathfrak{d}$ per site; accordingly the algebra, the ideal increment, the polarization and the target below are of degree $\leq \mathfrak{d}$ per slot ($\mathfrak{d} = 1$: multilinear).

15.1 The m -adapted algebra of degree $\mathfrak{d}$

Let $X_1, \dots, X_Q$ be commuting formal symbols and let $\{\lambda_m\}_{m \in \mathcal{J}_k}$ be a second family of commuting symbols with $\lambda_m^2 = 1$; monomials are $\lambda^S = \prod_{m \in S} \lambda_m$, $S \subseteq \mathcal{J}_k$, with $\lambda^S \lambda^{S'} = \lambda^{S \Delta S'}$. Put $\mathcal{E} = \{0, 1, \dots, \mathfrak{d}\}^Q$ and

$$\begin{aligned} N_m &:= c_d^{-1} N_0 \max\{1, \|m\|_{\mathcal{W}_d}\} \leq C_d N_0 \Lambda^d \quad (\text{Lemma 13.1}), \\ |m(u) \Lambda(u)| &\leq c_d^{-1} N_0 \leq N_m. \end{aligned} \tag{124}$$

An element of $\mathcal{A}$ (the *m-adapted representative algebra*) is an array $W = (W_{e,S})$, $e \in \mathcal{E}$, $S \subseteq \mathcal{J}_k$, with $W_{e,S} \in \mathcal{W}_{dQ}$. Its *realization* as a polynomial in the bare symbols X_i , and its norm, are

$$\begin{aligned} \mathfrak{r}(W)(T; X, \lambda) &= \sum_{e \in \mathcal{E}} \sum_S m^e(T) W_{e,S}(T) X^e \lambda^S, \quad m^e(T) := \prod_i m(u_i)^{e_i}, \\ \|W\|_{\mathcal{A}} &= \sum_{e,S} \|W_{e,S}\|_{\mathcal{W}_{dQ}} N_m^{|e|} < \infty, \quad |e| := \sum_i e_i. \end{aligned} \quad (125)$$

Equality and the norm in $\mathcal{A}$ are defined on the arrays, not on their realizations. No quotient by the kernel of $\mathfrak{r}$ is taken. In particular, $\mathfrak{r}$ need not be injective: a nonzero representative supported where $m^e = 0$ can have zero realization. Thus $\mathcal{A}$ is not identified with a subspace of bare-coefficient polynomials. The notation $m^e W_{e,S} X^e \lambda^S$ below can denote a *tagged representative term*; when read as an ordinary polynomial, it denotes its realization only.

Addition is coefficientwise. Multiplication is convolution of the representatives, with $\lambda^S \lambda^{S'} = \lambda^{S \triangle S'}$, followed by the projection $P_{\mathfrak{d}}$ discarding exponents greater than $\mathfrak{d}$ in any slot. Thus the product of tagged terms uses representative $W_{e,S} V_{e',S'}$ at $(e + e', S \triangle S')$, before projection. Vector-valued elements carry the sum of the coordinate norms. All polarization identities and the fixed point are formed in this representative space.

The *evaluation* is $\text{ev}_{\lambda} W = \mathfrak{r}(W)(T; \Lambda_{\lambda}(u.), \lambda)$. The *reflection* is the isometric involution $(RW)_{e,S} = (-1)^{|e|+|S|} W_{e,S}$, corresponding after realization to $X_i \mapsto -X_i$, $\lambda_m \mapsto -\lambda_m$. Hence $\text{ev}_{-\lambda} W = \text{ev}_{\lambda}(RW)$. For a symbol $m \in \mathcal{J}_k$, the *symbol seminorm* is

$$\|W\|_{[m]} = \sum_e \sum_{S \ni m} \|W_{e,S}\|_{\mathcal{W}_{dQ}} N_m^{|e|}, \quad (126)$$

the mass of the representative terms containing the explicit symbol λ_m , as distinct from dependence entering through site values.

Lemma 15.1. *$\mathcal{A}$ is a commutative Banach algebra with unit; R is an isometric involution; moreover*

$$\sup_T \|\lambda \mapsto \text{ev}_{\lambda} W(T)\|_W \leq \|W\|_{\mathcal{A}}, \quad \|\text{ev}_{\lambda} W\|_{\infty} \leq \|W\|_{\mathcal{A}} \quad \text{for every } \lambda.$$

Also $\text{ev}_{\lambda}(WV) = \text{ev}_{\lambda}(W) \text{ev}_{\lambda}(V)$ whenever the projection $P_{\mathfrak{d}}$ discards no term, i.e. whenever every monomial of WV has exponents $\leq \mathfrak{d}$ (this is the case in every product used below: the slot factors of an atom power $F^{\otimes Q} = \prod_s (1 + \theta t_r g(u_s))$ involve only X_s ; the ambient expression in (135) also has degree $\leq \mathfrak{d}$ per slot, and its representative lift is defined explicitly in Section 15.5; and the polarization coefficients $a_m(A)$ are X -free); multiplying by a scalar Wiener element $w(u_i)$ of one slot has norm $\leq \|w\|_{\mathcal{W}_d}$; and for every symbol m ,

$$\|WV\|_{[m]} \leq \|W\|_{[m]} \|V\|_{\mathcal{A}} + \|W\|_{\mathcal{A}} \|V\|_{[m]}.$$

Moreover, in the frozen-field evaluation $W(\lambda; \xi) = \sum_{e,S} m^e(T) W_{e,S}(T) \xi^e \lambda^S$, in which the site symbols X_i are replaced by fixed numbers ξ_i with $|m(u_i) \xi_i| \leq N_m$ (for instance $\xi_i = \Lambda_{\lambda_0}(u_i)$ for a fixed sign vector λ_0), the Walsh monomials containing λ_m have total mass $\leq \|W\|_{[m]}$ uniformly in T , so that $\sup_T |W(\lambda; \xi) - W(\lambda^{(m)}; \xi)| \leq 2\|W\|_{[m]}$, where $\lambda^{(m)}$ is λ with the sign of λ_m reversed. The full evaluation $\text{ev}_{\lambda} W = W(\lambda; \Lambda_{\lambda}(u.))$ depends on λ_m in addition through the site values $\Lambda_{\lambda}(u_i)$ with $m \in J(u_i)$; this dependence is not controlled by the seminorm and is treated separately where it matters (Lemma 15.9).

Proof. Let $\mathcal{A}^{\text{full}}$ be the same construction with $e \in \mathbb{N}^Q$ (and $m^e = \prod m(u_i)^{e_i}$); it is a commutative Banach algebra with unit by submultiplicativity of $\mathcal{W}_{dQ}$, $N_m^{|e'|} N_m^{|e''|} = N_m^{|e'+e''|}$, $\|\lambda^S \lambda^{S'}\| = 1$, and completeness of weighted ℓ^1 -sums. $\|P_{\mathfrak{d}}\| \leq 1$. The product on $\mathcal{A}$ is $W \cdot V = P_{\mathfrak{d}}(WV)$; it is commutative, bilinear, submultiplicative, and associative because exponents add: a monomial of WVU exceeds degree $\mathfrak{d}$ in a slot iff one of the partial products already does or the total does, so $P_{\mathfrak{d}}(P_{\mathfrak{d}}(WV)U) = P_{\mathfrak{d}}(WVU) = P_{\mathfrak{d}}(WP_{\mathfrak{d}}(VU))$. For the Walsh-norm evaluation bound, at each fixed tuple T , $\|\Lambda_{\lambda}(u_i)\|_W \leq N_0$. The Walsh algebra inequality therefore gives

$$\|\text{ev}_{\lambda} W(T)\|_W \leq \sum_{e,S} |W_{e,S}(T)| (c_d^{-1} N_0)^{|e|} \leq \|W\|_{\mathcal{A}}.$$

For the pointwise bound, $\|W_{e,S}\|_{\infty} \leq \|W_{e,S}\|_{\mathcal{W}}$, $|m^e \prod \Lambda(u_i)^{e_i}| \leq (c_d^{-1} N_0)^{|e|} \leq N_m^{|e|}$, $|\lambda^S| = 1$; ev_{λ} is multiplicative on $\mathcal{A}^{\text{full}}$ (substitution of commuting values into absolutely convergent series), hence on $\mathcal{A}$ whenever $P_{\mathfrak{d}}(WV) = WV$. A product monomial contains λ_m only if at least one factor does ($\lambda_m^2 = 1$ can only remove it), which gives the seminorm inequality. In $W(\lambda; \xi)$ the monomials containing λ_m are those with $m \in S$, of total mass $\leq \sum_e \sum_{S \ni m} \|W_{e,S}\|_{\infty} N_m^{|e|} \leq \|W\|_{[m]}$, and reversing λ_m changes exactly their sign. $\square$

An element is *symmetric* if invariant under simultaneous permutation of tensor slots and of the indices of e .

Coefficient extraction on representatives. For $S' \subseteq [Q]$ and $f \in \{0, \dots, \mathfrak{d}\}^{S'}$, let $\hat{f}$ be its extension by zero to $[Q]$. Define the specific representative lift $W^{(f;S')}$ by

$$(W^{(f;S')})_{a,S} = \begin{cases} m^{\hat{f}}(T) W_{a+\hat{f},S}(T), & a|_{S'} = 0, \\ 0, & \text{otherwise.} \end{cases}$$

This is a linear operation on the given array and involves no division by m . Its realization is the bare $X_{S'}^f$ -coefficient of $\mathfrak{r}(W)$:

$$\mathfrak{r}(W) = \sum_{f \in \{0, \dots, \mathfrak{d}\}^{S'}} X_{S'}^f \mathfrak{r}(W^{(f;S')}), \quad X_{S'}^f = \prod_{i \in S'} X_i^{f_i}, \quad (127)$$

$$W^{(f;S')} \text{ has zero site degree on } S', \quad \|W^{(f;S')}\|_{\mathcal{A}} \leq \|W\|_{\mathcal{A}}.$$

Indeed, multiplication of the representative by $m^{\hat{f}}$ costs at most $\|m\|_{\mathcal{W}_d}^{|f|} \leq N_m^{|f|}$, exactly compensated by removing $|f|$ from its weight. The same argument gives $\|W^{(f;S')}\|_{[m]} \leq \|W\|_{[m]}$. If $RW = W$, then $RW^{(f;S')} = (-1)^{|f|} W^{(f;S')}$. The ordinary coefficients of $\mathfrak{r}(W)$ are unique, but their representative lifts here are the ones just specified; no uniqueness of an arbitrary lift, or of an expansion after rough-field evaluation, is asserted.

Ambient module and bare symbols. Let $\mathcal{A}^{\text{amb}}$ be the space of bare-coefficient polynomials $\sum_{e,S} V_{e,S}(T) X^e \lambda^S$, $e \in \mathcal{E}$, with norm $\sum_{e,S} \|V_{e,S}\|_{\mathcal{W}_{dQ}} (c_d^{-1} N_0)^{|e|}$. The realization map is bounded:

$$\|\mathfrak{r}(W)\|_{\mathcal{A}^{\text{amb}}} \leq \|W\|_{\mathcal{A}}, \quad \text{ev}_{\lambda} \mathfrak{r}(W) = \text{ev}_{\lambda} W.$$

Bare powers X_i , the substitution polynomials $\mathsf{K}_i^{(e,f)}$, and coefficient vectors such as $J^{(e)}/X^e := [X^e] \mathfrak{r}(J^{(e)})$ are interpreted in this ambient module. Products used below never exceed the allowed degree. An ambient identity is not used to infer an identity in $\mathcal{A}$. Instead, the target in Section 15.5 is assigned an explicit representative array, with surplus nonnegative powers of m absorbed in its representatives. This array is the object to which polarization, the norm estimates, and the fixed-point theorem apply.

15.2 The ideal increment

Let $\Psi = \Psi_{\mathfrak{d}Q}(U)$ list the nonconstant monomials of degree $\leq \mathfrak{d}Q$ in $U \in \mathbb{R}^M$, $N = N_{\mathfrak{d}Q}$ its length. For a tuple T and $0 \neq e \in \mathcal{E}$ define

$$J^{(e)}(T) = \chi_Q(T) t^{|e|} \prod_i \frac{m(u_i)^{e_i}}{e_i!} \mathbb{E}_{\Pi_0} \left[\prod_i (c_i(T) \cdot \nabla_U)^{e_i} \Psi(U) \right] X^e, \quad (128)$$

extended by zero across the diagonals, and $J = \sum_{e \neq 0} J^{(e)} \in \mathcal{A}^N$. J is symmetric, $RJ^{(e)} = (-1)^{|e|} J^{(e)}$, and contains no Walsh symbols. By Taylor's formula for the polynomial Ψ , $\Psi(U+w) = \sum_{e \in \mathbb{N}^Q} \prod_i (w_i \cdot \nabla)^{e_i} \Psi(U) / e_i!$ for $w = \sum_i w_i$ (the operators commute), so $\sum_{e \in \mathcal{E} \setminus 0} J^{(e)}$ is the part of degree $\leq \mathfrak{d}$ per site of $\chi_Q \{ \mathbb{E}_{\Pi_0} \Psi(U + t \sum_i m(u_i) X_i c_i) - \mathbb{E}_{\Pi_0} \Psi(U) \}$, the moment increment of the virtual shift (112). For $\mathfrak{d} = 1$ it is the multilinear part.

Lemma 15.2 (Cells see only degree $\mathfrak{d}$ per slot). *For $n \in \mathcal{E}$ let $\ell_{n,T}$ be the linear functional on $\mathbb{R}^N$ with $\ell_{n,T}(\Psi(U)) = \prod_i (v_Q(u_i)^\top U)^{n_i}$ ($\ell_{0,T} = 0$). Then*

$$\ell_{n,T}(\text{ev}_\lambda J(T)) = \chi_Q \left\{ \mathbb{E}_{\Pi_0} \prod_i (v_Q(u_i)^\top U + t m(u_i) \Lambda(u_i))^{n_i} - \mathbb{E}_{\Pi_0} \prod_i (v_Q(u_i)^\top U)^{n_i} \right\},$$

and $\ell_{n,T}(\text{ev}_\lambda J^{(e)}) = \chi_Q \prod_i \binom{n_i}{e_i} (t m(u_i) \Lambda(u_i))^{e_i} \mathbb{E}_{\Pi_0} \prod_i (v_Q(u_i)^\top U)^{n_i - e_i}$, zero unless $e \leq n$ coordinatewise.

Proof. $v_Q(u_i)^\top c_j = \mathbf{1}\{i = j\}$ gives $\prod_i (v_Q(u_i)^\top (U + t \sum_j m(u_j) c_j \Lambda(u_j)))^{n_i} = \prod_i (v_Q(u_i)^\top U + t m(u_i) \Lambda(u_i))^{n_i}$, a polynomial of degree $n_i \leq \mathfrak{d}$ in each $\Lambda(u_i)$; so the terms of the Taylor expansion with some exponent $> \mathfrak{d}$ lie in $\ker \ell_{n,T}$, and the e -th term is the binomial expansion. $\square$

Lemma 15.3 (Wiener estimate). $J^{(e)} \in \mathcal{A}^N$ with representatives free of the factors $m(u_i)$, and

$$\|J^{(e)}\|_{\mathcal{A}} \leq C_Q \Lambda^{E_Q} N_m^{|e|} (t/t_n)^{|e|};$$

hence, with

$$B > E_Q + \mathfrak{d}dQ + 1, \quad \|J\|_{\mathcal{A}} \leq C_Q \Lambda^{E_Q + \mathfrak{d}dQ - B} =: \delta_n \leq C_Q \Lambda^{-1} \rightarrow 0. \quad (129)$$

Proof. Each coordinate of (128) is $m^e(T)$ times a constant times $\chi_Q t^{|e|}$ times a product of $|e| \leq \mathfrak{d}Q$ Lagrange coordinates (with repetitions: e_i coordinates of c_i) times X^e , so the representative is the latter. Insert the shell partition $\sum_{\mathbf{m}} \rho_{\mathbf{m}} = 1$; by Corollary 5.2 (repetitions of Lagrange coordinates allowed) and Lemma 4.4, each of the $\leq C_Q \{1 + \log(1/t_n)\}^{E_Q}$ shells meeting the taper contributes $\leq C_Q (t/t_n)^{|e|}$ to the $\mathcal{W}_{dQ}$ norm. Sum over shells, coordinates, and e , multiply by the weight $N_m^{|e|} \leq C_Q \Lambda^{d|e|}$, and use $t/t_n = \Lambda^{-B}$, $\log(1/t_n) = O(\Lambda)$, $|e| \leq \mathfrak{d}Q$. $\square$

15.3 One-slot algebra, centring, polarization

Let $\mathcal{B} = \mathcal{B}_k$ be the *one-slot algebra*: the construction (125) with $Q = 1$, i.e. elements

$$g = \sum_{e \leq \mathfrak{d}} \sum_S m(u)^e g_{e,S}(u) X^e \lambda^S, \quad g_{e,S} \in \mathcal{W}_d, \quad \|g\|_{\mathcal{B}} = \sum_{e,S} \|g_{e,S}\|_{\mathcal{W}_d} N_m^e,$$

with the truncated product, evaluation, and reflection. For $g \in \mathcal{B}$ and a slot $s \in [Q]$, $g(u_s)$ denotes the element of $\mathcal{A}$ obtained by placing g in slot s (with $X \mapsto X_s$) and 1 in the other slots;

$\|g(u_s)\|_{\mathcal{A}} = \|g\|_{\mathcal{B}}$. For $g_1, \dots, g_Q \in \mathcal{B}$, $\bigotimes_s g_s := \prod_s g_s(u_s) \in \mathcal{A}$, formed without truncation since the slots are distinct, and $g^{\otimes Q} = \bigotimes_s g$.

Let $\{w_\nu\}$ be the real trigonometric basis of $L^2(\mathbb{T}^d)$: $w_0 = 1$, $\|w_\nu\|_\infty \leq 1$, $\int w_\nu = 0$ for $\nu \neq 0$; $\|\cdot\|_{\mathcal{W}_{dQ}}$ is (up to a constant) the ℓ^1 norm of the coefficients in $\{w_{\nu_1} \otimes \dots \otimes w_{\nu_Q}\}$. Expanding every representative $W_{e,S}$ in this basis, every $W \in \mathcal{A}$ is an absolutely convergent sum of *basis tensors* $\lambda^S \bigotimes_i g_i$ with the *m-adapted slot functions*

$$g_i = m(u)^{e_i} w_{\nu_i}(u) X^{e_i} \in \mathcal{B}, \quad e_i \in \{0, \dots, \mathfrak{d}\},$$

with weight $N_m^{|e|}$, and $\sum(\text{coefficient} \times \text{weight}) \leq C\|W\|_{\mathcal{A}}$. The multiplier m is kept inside the slot function and is never expanded into Fourier modes; this is what makes the atoms below *m-adapted*. For $e_i \geq 1$ the evaluated slot function $m^{e_i} w_{\nu_i} \Lambda^{e_i}$ is not mean zero; its mean is the Walsh polynomial

$$\bar{g}_i = \int_{\mathbb{T}^d} m(u)^{e_i} w_{\nu_i}(u) \Lambda(u)^{e_i} du = \sum_{S: |S| \leq e_i} \lambda^S \bar{g}_{i,S}, \quad (130)$$

an element of $\mathcal{B}$ (free of X : $e = 0$, constant in u), regarded as an element of $\mathcal{A}$ through the slot embedding. Expanding Λ^{e_i} into products of packets and using $|m| \leq c_d^{-1}$: $\|\bar{g}_i\|_W \leq \int c_d^{-e_i} (\sum_m |\psi_m|)^{e_i} \leq (c_d^{-1} N_0)^{e_i} \leq N_m^{e_i}$, and for every symbol m' , $\|\bar{g}_i\|_{[m']} \leq e_i c_d^{-e_i} N_0^{e_i-1} \int |\psi_{m'}| \leq C t^d$. Put $g_i^\circ = (g_i - \bar{g}_i)/(2N_m^{e_i})$ for $e_i \geq 1$ and $g_i^\circ = g_i/2$ for $e_i = 0$, $\nu_i \neq 0$; then $\|g_i^\circ\|_{\mathcal{B}} \leq 1$, $\|\text{ev}_\lambda g_i^\circ\|_\infty \leq 1$ (by (124)), $\int \text{ev}_\lambda g_i^\circ du = 0$ for every λ , $\|g_i^\circ\|_{[m']} \leq C t^d$, and $Rg_i^\circ = (-1)^{e_i} g_i^\circ$.

Lemma 15.4 (Polarization). *Fix $\eta_f \in (0, 1)$, $\theta = \eta_f/(2 \max_r |t_r|)$, and a finite-dimensional normed space V . There is a countable family of one-slot atoms $F_m = 1 + \theta t_{r(m)} \bar{g}_m \in \mathcal{B}$, closed under R , with $\|\bar{g}_m\|_{\mathcal{B}} \leq 1$, $\|\text{ev}_\lambda \bar{g}_m\|_\infty \leq 1$, $\int \text{ev}_\lambda \bar{g}_m = 0$ for all λ , and linear maps $a_m : \mathcal{A}^{\text{sym}} \otimes V \rightarrow \mathcal{A}^{\text{sc}} \otimes V$ ($\mathcal{A}^{\text{sc}}$: Walsh polynomials with constant coefficients) such that every symmetric $A \in \mathcal{A} \otimes V$ satisfies $A = \sum_m a_m(A) F_m^{\otimes Q}$ absolutely in $\mathcal{A}$, and*

$$\sum_m \|a_m(A)\| \leq C_{\text{pol}} \|A\|, \quad \sum_m \left| \|a_m(A)\| - \|a_m(A')\| \right| \leq C_{\text{pol}} \|A - A'\|. \quad (131)$$

Each $\text{ev}_\lambda F_m$ is a probability density on $\mathbb{T}^d$ with values in $[1 - \eta_f/2, 1 + \eta_f/2]$, and $\|F_m^{\otimes Q} - 1^{\otimes Q}\|_{\mathcal{A}} \leq (1 + \eta_f/2)^Q - 1 =: C_F$.

Proof. Treat one basis tensor $\lambda^S \bigotimes_i g_i$; keep λ^S in the coefficient. Multilinearity of the tensor product gives $\bigotimes_i g_i = \sum_{I \subseteq [Q]} \prod_{i \notin I} \bar{g}_i \prod_{i \in I} 2N_m^{e_i} \bigotimes_{i \in I} g_i^\circ \bigotimes_{i \notin I} 1$ (with $\bar{g}_i = 0$ for $e_i = 0$, $\nu_i \neq 0$, and the slot $w_0 = 1$ treated as the unit), 2^Q terms whose coefficients are Walsh polynomials of norm $\leq 2^Q \prod_i N_m^{e_i}$, i.e. $\leq 2^Q$ times the weight of the basis tensor. Symmetrise, and apply Lemma 4.5 in the commutative algebra $\mathcal{B}$ with the symmetrised tensor product on $\mathcal{A}$: it expresses $g_{i_1}^\circ \odot \dots \odot g_{i_j}^\circ \odot 1^{\odot(Q-j)}$ through the tensor powers $(1 + \theta t_r g_\epsilon)^{\otimes Q} = \prod_s (1 + \theta t_r g_\epsilon(u_s))$, $g_\epsilon = j^{-1} \sum_{i \in I} \epsilon_i g_i^\circ \in \mathcal{B}$; only sums within $\mathcal{B}$ and products across distinct slots occur, so no truncation is active and the identity holds in $\mathcal{A}$. The atoms $1 + \theta t_r g_\epsilon$ are sums of *m-adapted slot functions*, hence in $\mathcal{B}$ and of degree $\leq \mathfrak{d}$ in X , and satisfy the three bounds; the coefficients are $C_j \theta^{-j} \lambda_{j,r} \prod \epsilon_i$ times the Walsh-polynomial coefficient above. Since $Rg_i^\circ = \pm g_i^\circ$, $Rg_\epsilon = g_{\epsilon'}$ for $\epsilon'_i = (-1)^{e_i} \epsilon_i$, so the family is *R-closed*. Summing over the expansion of A gives (131); linearity of a_m holds because the centring elements are attached to the basis element, not to A . The density statement follows from $|\theta t_r \text{ev}_\lambda g_\epsilon| \leq \eta_f/2$ and $\int \text{ev}_\lambda g_i^\circ = 0$. $\square$

15.4 The coefficient function in the chart

With the cutoff χ_0 fixed at the start of this section, put

$$(a) \quad \varkappa_k(u) = \chi_0(u) j_a^2 G(u)^4, \quad (b) \quad G_k^2(u) = \chi_0(u) G(u)^2. \quad (132)$$

Lemma 15.5. $\|\varkappa_k\|_{W_d} \leq C j_a^2 \leq 1$ and $\|G_k^2\|_{W_d} \leq C$, uniformly in k and n .

Proof. $G = G_h \circ \theta_k^{-1}$ has all derivatives bounded (it varies at scale $h/r \geq 1$ in the chart), so $\chi_0 G^2$ and $\chi_0 G^4$ are smooth functions on $\mathbb{T}^d$ with bounded C^{d+1} norm, hence bounded Wiener norm. $\square$

Since $\chi_0 = 1$ on $\text{supp } m$, multiplying a coefficient that carries $m(u_i)$ by $\varkappa_k(u_i)$ or $G_k^2(u_i)$ is the same as multiplying by $\varkappa$ or G^2 at the site.

15.5 Substitution operators and the target

In the chart, the substitution polynomials (119) at slot i read

$$\begin{aligned} \text{(a)} \quad & \left| \begin{array}{l} K_i^{(1,0)} = X_i, \quad K_i^{(1,1)} = \varkappa_k(u_i), \\ K_i^{(e,0)} = X_i^e, \quad K_i^{(e,f)} = 0 \text{ (} f \text{ odd),} \quad K_i^{(e,2)} = G_k^2(u_i) X_i^e, \end{array} \right. \end{aligned} \quad (133)$$

In both cases $K_i^{(e,f)} = \varkappa^{(e,f)}(u_i) X_i^{e'(e,f)}$ with $e'(e, f) \in \{0, e\}$ and $\|\varkappa^{(e,f)}\|_{W_d} \leq C$ (Lemma 15.5). For $0 \neq e \in \mathcal{E}$ with $S' = \text{supp } e$ define the linear map $\mathcal{S}_e : \mathcal{A} \rightarrow \mathcal{A}^{\text{amb}}$ using the prescribed extraction (127):

$$\mathcal{S}_e(W) = \sum_{f \in \{0, \dots, d\}^{S'}} \mathfrak{r}(W^{(f; S')}) \prod_{i \in S'} K_i^{(e_i, f_i)}. \quad (134)$$

In words: in the $X_{S'}$ -expansion of W , the monomial $X_{S'}^f$ is replaced by $\prod_i K_i^{(e_i, f_i)}$; for $f_i = 0$ this inserts the jitter $X_i^{e_i}$, and for $f_i \geq 1$ the corrected jitter of (119). The *target* is the representative-valued linear map described by

$$\mathcal{J}(\mathbb{D}) = \sum_{0 \neq e \in \mathcal{E}} \mathcal{J}^{(e)}(\mathbb{D}), \quad \mathfrak{r}(\mathcal{J}^{(e)}(\mathbb{D})) = \frac{J^{(e)}}{X^e} \mathcal{S}_e(\mathbb{D}), \quad (135)$$

where $J^{(e)}/X^e = [X^e] \mathfrak{r}(J^{(e)})$ is the X -free vector coefficient in (128). The second equality describes only the realization; the following formula *defines* its lift. Let $B_e(T)$ denote the sole vector representative of $J^{(e)}$, so $\mathfrak{r}(J^{(e)}) = m^e B_e X^e$. For an input representative $\mathbb{D}_{a,S}$, put $S' = \text{supp } e$, $f = a|_{S'}$, and

$$g_i = \begin{cases} e'(e_i, f_i), & i \in S', \\ a_i, & i \notin S'. \end{cases}$$

If any substitution factor is zero, its contribution is zero. Otherwise add to the (g, S) -representative of $\mathcal{J}^{(e)}(\mathbb{D})$ the vector

$$\left\{ \prod_{i \in S'} m(u_i)^{e_i + f_i - g_i} \varkappa^{(e_i, f_i)}(u_i) \right\} B_e(T) \mathbb{D}_{a,S}(T).$$

Sum these contributions over (a, S) . Every exponent $e_i + f_i - g_i$ is nonnegative, because $g_i \in \{0, e_i\}$, and $g \in \mathcal{E}$. Thus this defines the representative without division by m or an inverse of $\mathfrak{r}$, even on the zero set of m . Realizing the formula gives precisely (135).

In case (a), $e = \mathbf{1}_{S'}$; in the ambient notation the formula reads

$$\mathfrak{r}(\mathcal{J}^{(S')}(\mathbb{D})) = \frac{J^{(S')}}{X^{S'}} \sum_{A \subseteq S'} \mathfrak{r}(\mathbb{D}^{(A; S')}) \prod_{i \in A} \varkappa_i \prod_{i \in S' \setminus A} X_i.$$

The term $A = \emptyset$ is the jitter weighted by the extracted zero-degree design coefficient. The remaining terms are the *moment corrections* in which the site field is replaced by $\varkappa_i$ together with its prescribed design coefficient.

Lemma 15.6. *For symmetric R -invariant $\mathbb{D} \in \mathcal{A}$: $\mathcal{J}(\mathbb{D}) \in \mathcal{A}^N$ is symmetric, linear in $\mathbb{D}$, satisfies $R\mathcal{J}^{(e)}(\mathbb{D}) = (-1)^{|e|}\mathcal{J}^{(e)}(\mathbb{D})$, and*

$$\|\mathcal{J}(\mathbb{D})\|_{\mathcal{A}} \leq C_Q \delta_n \|\mathbb{D}\|_{\mathcal{A}}, \quad \|\mathcal{J}(\mathbb{D}) - \mathcal{J}(\mathbb{D}')\|_{\mathcal{A}} \leq C_Q \delta_n \|\mathbb{D} - \mathbb{D}'\|_{\mathcal{A}}. \quad (136)$$

Proof. Membership and norm. Use the representative rule preceding the lemma. For one input (a, S) , the output degree is $g \in \mathcal{E}$; on S' its surplus power is $e_i + f_i - g_i \geq 0$, and outside S' the original degree a_i is unchanged. The Wiener algebra inequality and Lemma 15.5 give

$$\begin{aligned} N_m^{|g|} \left\| \left\{ \prod_{i \in S'} m(u_i)^{e_i + f_i - g_i} \chi^{(e_i, f_i)}(u_i) \right\} B_e \mathbb{D}_{a, S} \right\|_{\mathcal{W}_{dQ}} \\ \leq C_Q N_m^{|e| + |a|} \|B_e\|_{\mathcal{W}_{dQ}} \|\mathbb{D}_{a, S}\|_{\mathcal{W}_{dQ}}. \end{aligned}$$

Here $\|m\|_{\mathcal{W}_d} \leq N_m$ and $|g| + \sum_{i \in S'} (e_i + f_i - g_i) = |e| + |a|$. Since $N_m^{|e|} \|B_e\| = \|\mathcal{J}^{(e)}\|_{\mathcal{A}}$, summing first over (a, S) , then over $e \neq 0$ and vector coordinates, gives (136) by (129). The representative formula is linear, so the difference estimate follows identically.

Symmetry and parity. The rule for g and for the surplus powers commutes with simultaneous permutations of slots and the increment pattern; summing over e proves symmetry. If $\mathbb{D}$ is R -invariant, each nonzero representative has $|a| + |S|$ even. For every nonzero substitution, $g_i \equiv e_i + f_i \pmod{2}$ on S' . Thus the output representative satisfies

$$|g| + |S| \equiv |a| + |S| + |e| \equiv |e| \pmod{2}.$$

This proves $R\mathcal{J}^{(e)}(\mathbb{D}) = (-1)^{|e|}\mathcal{J}^{(e)}(\mathbb{D})$ in the representative space, not merely after realization. $\square$

Proposition 15.7 (Carrier). *For every active block k and all large n there is a symmetric, R -invariant $\mathbb{D} = \mathbb{D}_k \in \mathcal{A}$ with $\|\mathbb{D} - 1^{\otimes Q}\|_{\mathcal{A}} \leq C_0 \delta_n$, and for every sign vector λ laws $\nu_{P,k}(\cdot \mid \lambda)$, $\nu_{Q,k}(\cdot \mid \lambda)$ of a pair (H_k, U_k) such that:*

- (i) *the law of H_k is the same under both and does not depend on λ ;*
- (ii) *each label H carries a density $F_H(\cdot; \lambda) = \text{ev}_\lambda F$ on $\mathbb{T}^d$, $F \in \mathcal{B}$, with $\int F_H = 1$, $1 - \eta_f \leq F_H \leq 1 + \eta_f$, and $D(T; \lambda) := \mathbb{E} F_{H_k}^{\otimes Q}(T; \lambda) = \text{ev}_\lambda \mathbb{D}(T)$, which is even in λ ;*
- (iii) *under both laws U_k is supported on the fixed finite support of Π_0 ; under $\nu_{P,k}$, $U_k \sim \Pi_0$ independently of H_k ;*
- (iv) *pointwise on $(\mathbb{T}^d)^Q$,*

$$\mathbb{E}_{\nu_{Q,k}} [F_{H_k}^{\otimes Q}(T) \Psi(U_k)] - \mathbb{E}_{\nu_{P,k}} [F_{H_k}^{\otimes Q}(T) \Psi(U_k)] = \text{ev}_\lambda \mathcal{J}(\mathbb{D})(T). \quad (137)$$

The kernels depend on λ only through $(\lambda_m)_{m \in \mathcal{J}_k}$ and are measurable: the label set is countable, and for each fixed label the kernel depends on finitely many signs and takes finitely many values. Moreover $\mathbb{D} \in \mathcal{A}$, so for $1 \leq e \leq \mathfrak{d}$ the realized coefficient of X_i^e carries $m(u_i)^e$ and vanishes where $m(u_i) = 0$; and

$$\|\mathbb{D}\|_{[m]} \leq C_Q \delta_n t^d \quad \text{for every symbol } m \in \mathcal{J}_k, \quad (138)$$

so that, in the frozen-field evaluation of Lemma 15.1, the Walsh monomials of $\mathbb{D}$ containing a given symbol λ_m have total mass $\leq C_Q \delta_n t^d$ uniformly on $(\mathbb{T}^d)^Q$. (The dependence of $\text{ev}_\lambda \mathbb{D}$ on λ_m through the site values $\Lambda_\lambda(u_i)$ is a separate matter, handled in Lemma 15.9.)

Proof. For symmetric R -invariant $\mathbb{D}$ put $A = \mathcal{J}(\mathbb{D})$, decompose $A = \sum_m a_m(A) F_m^{\otimes Q}$ and $RA = \sum_m a_m(RA) F_m^{\otimes Q}$ (Lemma 15.4); applying R to the second, $A = R(RA) = \sum_m (Ra_m(RA)) (RF_m)^{\otimes Q}$. Set

$$\omega_m = K_0(\|a_m(A)\| + \|a_m(RA)\|), \quad K_0 > 4/\delta_Q,$$

$$\Phi(\mathbb{D}) = \left(1 - \sum_m \omega_m\right) 1^{\otimes Q} + \sum_m \omega_m \frac{1}{2} \{F_m^{\otimes Q} + (RF_m)^{\otimes Q}\}.$$

$\Phi(\mathbb{D})$ is symmetric and R -invariant. By (131) and (136), $\|\Phi(\mathbb{D}) - \Phi(\mathbb{D}')\| \leq C_0 \delta_n \|\mathbb{D} - \mathbb{D}'\|$ with $C_0 = 2K_0 C_F C_{\text{pol}} C_Q$, and $\sum_m \omega_m \leq C_1 \|\mathbb{D}\| \delta_n$, which is ≤ 1 for large n on the ball below ($\delta_n \rightarrow 0$), so that the label 0 introduced below has nonnegative probability. For large n , Φ maps the ball $\|\mathbb{D} - 1\| \leq 2C_0 \delta_n$ into itself with contraction constant $\leq 1/2$; let $\mathbb{D}$ be its fixed point.

Labels: for each m with $\omega_m > 0$ (labels with $\omega_m = 0$ are omitted), m^+ with density F_m and m^- with density RF_m , each of probability $\omega_m/2$, and a label 0 with density 1; the ω_m are λ -free. This gives (i), (ii) (using Lemma 15.4, $\text{ev}_{-\lambda} F_m = \text{ev}_{\lambda} RF_m$, and $\mathbb{E} F_H^{\otimes Q} = \text{ev}_{\lambda} \Phi(\mathbb{D}) = \text{ev}_{\lambda} \mathbb{D}$, even in λ by R -invariance). Tilts: under Q and label $m^{\pm}$, $U_k \sim \Pi_{v_m^{\pm}(\lambda)}$ with $v_m^+(\lambda) = \text{ev}_{\lambda} a_m(A)/\omega_m$, $v_m^-(\lambda) = \text{ev}_{\lambda} Ra_m(RA)/\omega_m$, so $\|v_m^{\pm}(\lambda)\| \leq K_0^{-1} < \delta_Q$ (the evaluation of a Walsh polynomial is bounded by its norm) and Lemma 4.3 applies (with Ψ embedded in Ψ_{4Q}); under P , $U_k \sim \Pi_0$. Then

$$\begin{aligned} \mathbb{E}_{\nu_Q}[F^{\otimes Q} \Psi] - \mathbb{E}_{\nu_P}[F^{\otimes Q} \Psi] &= \sum_m \frac{\omega_m}{2} [\text{ev}_{\lambda} F_m^{\otimes Q} v_m^+ + \text{ev}_{\lambda} (RF_m)^{\otimes Q} v_m^-] \\ &= \frac{1}{2} \text{ev}_{\lambda} A + \frac{1}{2} \text{ev}_{\lambda} R(RA) = \text{ev}_{\lambda} A, \end{aligned}$$

which is (iv). The kernels read only the symbols occurring in $\mathcal{A}$. The support statement is the definition of $\mathcal{A}$: $\mathcal{J}(\mathbb{D}) \in \mathcal{A}$ for $\mathbb{D} \in \mathcal{A}$ (Lemma 15.6), the atoms are products of m -adapted slot functions, and Φ maps $\mathcal{A}$ into $\mathcal{A}$; hence the fixed point lies in $\mathcal{A}$. Finally (138) is Lemma 15.8 below. $\square$

Lemma 15.8 (Symbol seminorm of the fixed point). *For every atom F of Lemma 15.4 and every symbol m , $\|F^{\otimes Q}\|_{[m]} \leq C_Q t^d$. Consequently, for every $\mathbb{D}$ in the ball of Proposition 15.7, $\|\Phi(\mathbb{D})\|_{[m]} \leq C_Q \delta_n t^d$, and the fixed point satisfies (138).*

Proof. The only symbols in an atom $F = 1 + \theta t_r g_{\epsilon}$ come from the centring polynomials $\bar{g}_i$ in g_i° , and $\|g_i^{\circ}\|_{[m]} \leq C t^d$ by (130); hence $\|g_{\epsilon}\|_{[m]} \leq C t^d$ and $\|1 + \theta t_r g_{\epsilon}(u_s)\|_{[m]} \leq C t^d$ for each slot s , while $\|1 + \theta t_r g_{\epsilon}(u_s)\|_{\mathcal{A}} \leq 1 + \eta_f/2$. By the derivation inequality of Lemma 15.1 applied to the product over the Q slots, $\|F^{\otimes Q}\|_{[m]} \leq Q(1 + \eta_f/2)^{Q-1} C t^d$. The same holds for RF , since R preserves the seminorm. Now $\Phi(\mathbb{D}) - (1 - \sum \omega_{m'}) 1 = \sum_{m'} \omega_{m'} \frac{1}{2} \{F_{m'}^{\otimes Q} + (RF_{m'})^{\otimes Q}\}$ with λ -free numbers $\omega_{m'} \geq 0$, $\sum_{m'} \omega_{m'} \leq C_1 \|\mathbb{D}\| \delta_n$, and the atom family does not depend on $\mathbb{D}$; therefore $\|\Phi(\mathbb{D})\|_{[m]} \leq \sum_{m'} \omega_{m'} C_Q t^d \leq C_Q \delta_n t^d$. The fixed point $\mathbb{D} = \Phi(\mathbb{D})$ inherits the bound. (No contraction argument in the seminorm is needed, because Φ depends on its argument only through the numbers $\omega_{m'}$.) $\square$

15.6 Projectivity and the integrated identity

For $q \leq Q$, $T_q = (u_1, \dots, u_q)$ and $Z \in (\mathbb{T}^d)^{Q-q}$, all atoms integrate to one for every λ , so $\mathbb{E}_{\nu_S}[F^{\otimes q}(T_q) \Psi] = \int \mathbb{E}_{\nu_S}[F^{\otimes Q}(T_q, Z) \Psi] dZ$ and $D_q(T_q; \lambda) = \int D(T_q, Z; \lambda) dZ$. Write $\mathbb{D}_q = \int \mathbb{D}(\cdot, Z) dZ$ as shorthand for evaluated slot integration: the integrated site variables are replaced by their rough-field values and then integrated, giving Walsh-polynomial coefficients. The retained coefficient lifts are always those specified in (127) before this integration. This notation does not assert that slot integration is an algebra homomorphism on $\mathcal{A}$, or require a uniformly bounded representative Wiener

norm for $\mathbb{D}_q$; the pointwise Walsh bounds below are sufficient. Integrating (137) and applying ℓ_{n,T_q} , $n \in \{0, \dots, \mathfrak{d}\}^q$, by Lemma 15.2:

$$\begin{aligned} \ell_{n,T_q} \left(\mathbb{E}_{\nu_Q} [F^{\otimes q} \Psi] - \mathbb{E}_{\nu_P} [F^{\otimes q} \Psi] \right) \\ = \sum_{0 \neq e \leq n} \sum_f \mathbb{D}_q^{(f;S'_e)} \prod_{i \in S'_e} \binom{n_i}{e_i} (tm(u_i))^{e_i} \mathbf{K}_i^{(e_i, f_i)}(\Lambda(u_i)) \mathbb{E}_{\Pi_0} \prod_i (v_Q(u_i)^\top U)^{n_i - e_i}, \end{aligned} \quad (139)$$

$S'_e = \text{supp } e$, $f \in \{0, \dots, \mathfrak{d}\}^{S'_e}$, $\mathbf{K}_i^{(e_i, f_i)}(\Lambda(u_i))$ the evaluated substitution polynomial, and the *tapered integrated coefficients*

$$\mathbb{D}_q^{(f;S')}(T_q; \lambda) = \int \text{ev}_\lambda \mathbb{D}^{(f;S')}(T_q, Z) \chi_Q(T_q, Z) dZ. \quad (140)$$

Compare with the untapered decomposition of the design weight, $D_q = \sum_f \prod_{i \in S'} \Lambda(u_i)^{f_i} D_q^{(f;S')}$, $D_q^{(f;S')} = \int \text{ev}_\lambda \mathbb{D}^{(f;S')} dZ$:

$$\begin{aligned} D_q^{(f;S')} - \mathbb{D}_q^{(f;S')} &= \int \text{ev}_\lambda \mathbb{D}^{(f;S')}(T_q, Z) \{1 - \chi_Q(T_q, Z)\} dZ, \\ \|D_q^{(f;S')}(T_q; \cdot) - \mathbb{D}_q^{(f;S')}(T_q; \cdot)\|_W &\leq C_Q g_q(T_q), \end{aligned} \quad (141)$$

where $g_q(T_q) := \int \{1 - \chi_Q(T_q, Z)\} dZ$. The bound is in Walsh norm, not only absolute value. Indeed, Lemma 15.1 and (127) give $\|\text{ev}_\lambda \mathbb{D}^{(f;S')}(T_q, Z)\|_W \leq \|\mathbb{D}^{(f;S')}\|_{\mathcal{A}} \leq C_Q$ uniformly in (T_q, Z) . Since the taper is nonnegative and independent of λ , the triangle inequality under the integral gives

$$\|D_q^{(f;S')} - \mathbb{D}_q^{(f;S')}\|_W \leq \int (1 - \chi_Q) \|\text{ev}_\lambda \mathbb{D}^{(f;S')}\|_W dZ \leq C_Q g_q(T_q).$$

The leakage also satisfies

$$\int_{(\mathbb{T}^d)^q} g_q(T_q) dT_q \leq C_Q t^d \Lambda^{M_Q}, \quad M_Q = Bd + Q - 2, \quad (142)$$

because $\{\chi_Q \neq 1\} \subseteq \{\min_i P_i \leq 2t_n\}$, whose volume is $\leq C_Q t_n^d \{1 + \log(1/t_n)\}^{Q-2}$ by Lemma 4.2.

Lemma 15.9 (Structure of the design weight). *Let T_q be a tuple of distinct points and $S' \subseteq [q]$.*

- (a) D_q is even in λ ; $D_q^{(f;S')}$ has parity $(-1)^{|f|}$ in λ ; $\|D_q^{(f;S')}(T_q; \cdot)\|_W \leq C_Q$ and $\|D_q^{(f;S')}(T_q; \cdot)\|_W \leq C_Q \delta_n$ for $f \neq 0$ (Walsh norm at fixed T_q).
- (b) $D_q^{(f;S')}$ does not depend on the site values $\Lambda(u_i)$, $i \in S'$. Regard $D_q^{(f;S')}(T_q; \cdot)$ as a Walsh polynomial in λ at fixed T_q , and let $J_{S'} = \bigcup_{i \in S'} J(u_i)$. A sign $\lambda_{m'}$, $m' \in J_{S'}$, can enter it only through the explicit symbols of $\mathbb{D}$ (the centring polynomials), through integrated slots, and through retained slots u_j , $j \notin S'$, that share a packet with some u_i , $i \in S'$. Outside this last (collision) case, the Walsh monomials of $D_q^{(f;S')}$ containing $\lambda_{m'}$ have total mass $\leq C_Q \delta_n t^d$, uniformly in T_q . Consequently, writing $\mathbb{E}_{J_{S'}}$ for the average over the signs in $J_{S'}$ (at most N_{0q} of them), outside the collision case

$$\|D_q^{(f;S')} - \mathbb{E}_{J_{S'}} D_q^{(f;S')}\|_W \leq C_Q \delta_n t^d,$$

$\mathbb{E}_{J_{S'}} D_q^{(f;S')}$ does not depend on the signs in $J_{S'}$, and both terms have parity $(-1)^{|f|}$.

(c) If $m(u_i) = 0$ (site i not assigned to this block) then D_q does not contain $\Lambda(u_i)$ at all.

Proof. (a) R -invariance of $\mathbb{D}$ and (127); the norms are $\|\mathbb{D}\|$ and $\|\mathbb{D} - 1\|$ since every X -monomial of $\mathbb{D} - 1$ has norm $\leq C_0\delta_n$. (b) $D_q^{(f;S')}$ is free of $X_{S'}$ by the decomposition. Fix $i \in S'$ and a sign $\lambda_{m'}$, $m' \in J(u_i)$. It enters $D_q^{(f;S')}$ in three ways. Write $\mathbb{D}^{(f;S')} = \sum_{e,S} m^e W_{e,S} X^e \lambda^S$ ($e|_{S'} = 0$), expand every evaluated site value into products of packets,

$$\Lambda(u_i)^{e_i} = \sum_{m_1, \dots, m_{e_i}} \lambda_{m_1} \cdots \lambda_{m_{e_i}} \psi_{m_1}(u_i) \cdots \psi_{m_{e_i}}(u_i),$$

and integrate over Z . Then $D_q^{(f;S')}$ is a sum, over (e, S) and packet choices, of Walsh monomials with coefficients $\int m^e W_{e,S} \prod \psi dZ$, and the mass of the monomials containing $\lambda_{m'}$ is at most the sum of the absolute coefficients of the contributions in which $\lambda_{m'}$ occurs explicitly, i.e. $m' \in S$ or m' is one of the chosen packets (a product may cancel $\lambda_{m'}$ through $\lambda_{m'}^2 = 1$, which only decreases the mass). *Explicit symbol* $m' \in S$: the packet expansion of the site values contributes mass $\leq (c_d^{-1} N_0)^{|e|} \leq N_m^{|e|}$ per term, so these contributions have mass $\leq \sum_e \sum_{S \ni m'} \|W_{e,S}\|_\infty N_m^{|e|} \leq \|\mathbb{D}^{(f;S')}\|_{[m']} \leq \|\mathbb{D}\|_{[m']} \leq C_Q \delta_n t^d$ by (138). *Chosen packet* m' at an integrated slot u_z : such terms have $e_z \geq 1$, hence belong to $\mathbb{D} - 1$, so $\sum_{e_z \geq 1, S} \|W_{e,S}\|_\mathcal{W} N_m^{|e|} \leq \|\mathbb{D} - 1\|_\mathcal{A} \leq C_0 \delta_n$. For one such term the sum of the absolute coefficients over the packet choices in which one of the e_z factors at slot z reads m' is at most

$$\begin{aligned} c_d^{-|e|} e_z \|W_{e,S}\|_\infty \int |\psi_{m'}(u_z)| \left(\sum_a |\psi_a(u_z)| \right)^{e_z-1} \prod_{j \neq z} \left(\sum_a |\psi_a(u_j)| \right)^{e_j} dZ \\ \leq c_d^{-|e|} e_z N_0^{|e|-1} \|\psi_{m'}\|_{L^1(\mathbb{T}^d)} \|W_{e,S}\|_\mathcal{W} \leq C t^d N_m^{|e|} \|W_{e,S}\|_\mathcal{W}, \end{aligned}$$

(the product over $j \neq z$ runs over the integrated and the retained slots, the latter contributing their sup norms), where $\|\psi_{m'}\|_{L^1} \leq C t^d$ because $\psi_{m'}$ is bounded by one and supported in a set of diameter $\asymp t$ in the chart $(\psi_{m'}^b = G_h \phi((\cdot - x_0)/\ell - m'))$ and $\ell/r = t$). Summing over the $\leq Q$ integrated slots and over (e, S) with $e_z \geq 1$, these contributions have mass $\leq C_Q t^d \cdot C_0 \delta_n$. *Chosen packet* m' at a retained slot u_j , $j \notin S'$: $\psi_{m'}(u_j) \neq 0$ requires $J(u_j) \cap J(u_i) \neq \emptyset$, i.e. the collision case, which is excluded wherever (b) is used. Altogether the mass of the monomials containing $\lambda_{m'}$ is $\leq C_Q \delta_n t^d$ outside the collision case. Finally, $D_q^{(f;S')} - \mathbb{E}_{J_{S'}} D_q^{(f;S')}$ consists exactly of the monomials of $D_q^{(f;S')}$ containing at least one sign of $J_{S'}$ (averaging kills these and keeps the others), so its Walsh norm is at most the sum over the $\leq N_0 q$ signs $m' \in J_{S'}$ of the masses just bounded; the parity statement holds because averaging over a set of signs removes monomials without changing the degrees of the remaining ones. (c) The coefficient of X_i^e in $\mathbb{D}$ carries $m(u_i)^e$ (Proposition 15.7), and slot integration does not affect the retained slot u_i . $\square$

Lemma 15.10 (Local-sign removal across all incident blocks). *Let $\mathcal{I}$ be a set of at most Q distinct physical sites, all interior for the assignment partition, with pairwise disjoint packet sets $J(x_i)$, $i \in \mathcal{I}$. Fix $S' \subseteq \mathcal{I}$, and put $J_{S'} = \bigcup_{i \in S'} J(x_i)$. For any incident active block k , let*

$$\mathcal{I}_k = \{i \in \mathcal{I} : x_i \in S_k\}, \quad T_k = (\theta_k(x_i))_{i \in \mathcal{I}_k}, \quad q_k = |\mathcal{I}_k|, \quad S'_k = \{i \in S' : k(i) = k\}.$$

Identify physical site labels with their slots in T_k . For any $f_k \in \{0, \dots, \mathfrak{d}\}^{S'_k}$, set $A_k = D_{k, q_k}^{(f_k; S'_k)}(T_k; \cdot)$, with $A_k = D_{k, q_k}$ when $S'_k = \emptyset$. Then

$$\|A_k - \mathbb{E}_{J_{S'}} A_k\|_W \leq C_Q \delta_n t^d.$$

Both A_k and $\mathbb{E}_{J_{S'}} A_k$ have parity $(-1)^{|f_k|}$, and the latter is independent of all signs in $J_{S'}$. The conclusion includes blocks with $S'_k = \emptyset$ and signs read at sites assigned to other blocks.

Proof. Fix a packet index $a \in J_{S'}$. If $a \notin \mathcal{J}_k$, the block kernel and all its coefficients are independent of λ_a . Otherwise the collision-free assumption gives a unique $i \in S'$ with $a \in J(x_i)$. We first check that no retained site-value factor in A_k can read this sign.

If $i \in \mathcal{I}_k$ and $k(i) = k$, the slot belongs to S'_k , and coefficient extraction removes its site variable. If $i \in \mathcal{I}_k$ and $k(i) \neq k$, interior assignment gives $\phi_k(x_i) = 0$, hence $m_k(x_i) = 0$; every positive site-degree term vanishes by the representative formula. If $i \notin \mathcal{I}_k$, there is no such retained slot in this block. Every other retained point x_j has $a \notin J(x_j)$ by collision-freeness. The chart cutoff can only remove packet support, not introduce a new packet at a retained point.

Consequently λ_a can enter only as an explicit symbol or through a ghost slot. For the explicit-symbol contributions, the coefficient-extraction bound and Lemma 15.8 give mass at most

$$\|\mathbb{D}_k^{(f_k; S'_k)}\|_{[a]} \leq \|\mathbb{D}_k\|_{[a]} \leq C_Q \delta_n t^d.$$

For a ghost-slot contribution, expand its rough-field power into packets as in Lemma 15.9(b). A term using a at a ghost slot has positive degree there and hence comes from $\mathbb{D}_k - 1$. For each such representative, integrate one factor $|\psi_a|$, using $\int |\psi_a| \leq C t^d$, and bound the remaining packet sums by N_0 , the multipliers by c_d^{-1} , and the representative by its Wiener norm. Summing over the at most Q ghost slots costs

$$C_Q t^d \|\mathbb{D}_k - 1\|_A \leq C_Q \delta_n t^d.$$

Coefficient extraction is contractive and introduces only deterministic powers of m , so the same estimate applies for every f_k , including the empty extraction. This proves that the Walsh monomials of A_k containing λ_a have total mass at most $C_Q \delta_n t^d$.

Averaging over $J_{S'}$ deletes exactly the monomials containing at least one of those signs. There are at most $N_0 Q$ such signs, so summing the last bound proves the assertion. Averaging removes monomials without changing the parity of any monomial that remains; Lemma 15.9(a) therefore gives the parity statement. $\square$

15.7 Design density and tilt

Given λ and labels $H = (H_k)$, put

$$\tilde{f}_H(x) = \prod_{k: x \in S_k} F_{H_k}(\theta_k(x); \lambda), \quad Z(H, \lambda) = \int \tilde{f}_H, \quad f_H = \tilde{f}_H / Z. \quad (143)$$

At most N_1 blocks meet a point, so

$$\left(\frac{1 - \eta_f}{1 + \eta_f} \right)^{N_1} \leq f_H \leq \left(\frac{1 + \eta_f}{1 - \eta_f} \right)^{N_1}, \quad (144)$$

and η_f is fixed so that these lie in $[f, \bar{f}]$. Let $\nu_S = (\bigotimes_k \nu_{S,k}(\cdot \mid \lambda)) \otimes \text{Rad}(\lambda)$ be the law of (H, λ, U) on side S ; the (H, λ) -marginals coincide (Proposition 15.7(i)). Apply Lemma 2.6: here the density depends on λ as well as on H , the tilted priors (9) induce the same law of f_H and the same X^n -marginal, and the mixed joint density is (10), in which the untilted law has independent Rademacher packet signs and conditionally independent block draws given those signs. Unconditional independence is asserted only for components with disjoint enlarged dependence sets, as verified in Lemma 16.1.

16 Case C: proof

16.1 Mixtures and the two-point reduction

Let $\mathbb{P}_n$ and $\mathbb{Q}_n$ be the laws of $(X_i, A_i, Y_i)_{i \leq n}$ obtained by drawing $(H, U, \lambda) \sim \Pi_{S,n}$ ($S = P, Q$) and then n iid triples from the P -row, respectively the Q -row, of (113) (case (a)) or (115) (case (b)). By Lemma 14.1 every law in either mixture belongs to $\mathcal{P}$, and $\tau(x_0)$ equals $c_a c_b \Delta$ on the P -side and 0 on the Q -side. By Lemma 2.3, Theorem 1.1 in case (C) follows from $\text{TV}(\mathbb{P}_n, \mathbb{Q}_n) \leq 1/2$ for $n \geq n_\varepsilon$, since $\Delta = n^{-z} = n^{-(\rho+\varepsilon)}$.

16.2 Components and factorisation

By (10) the X^n -marginals coincide, so $\text{TV}(\mathbb{P}_n, \mathbb{Q}_n) \leq \mathbb{E} \text{TV}(\mathbb{P}_n(\cdot | X^n), \mathbb{Q}_n(\cdot | X^n))$. Fix x^n . For a site i let $\mathcal{K}_i = \{k \text{ active} : x_i \in S_k\}$ and let the *dependence set* be

$$\tilde{\mathcal{J}}_i = J(x_i) \cup \bigcup_{k \in \mathcal{K}_i} \mathcal{J}_k,$$

the packets read at x_i or by any block meeting x_i . Declare $i \sim j$ if $\mathcal{K}_i \cap \mathcal{K}_j \neq \emptyset$ or $\tilde{\mathcal{J}}_i \cap \tilde{\mathcal{J}}_j \neq \emptyset$, and let $\{C\}$ be the components of the transitive closure; $q_C = |C|$, $\mathcal{K}(C) = \bigcup_{i \in C} \mathcal{K}_i$, $\tilde{\mathcal{J}}(C) = \bigcup_{i \in C} \tilde{\mathcal{J}}_i$.

Lemma 16.1 (Geometry of links). *If $i \sim j$ then $\|x_i - x_j\| \leq C_Q r$. A shared original packet ($J(x_i) \cap J(x_j) \neq \emptyset$) forces $\|x_i - x_j\| \leq R_0 \ell$. Distinct components have disjoint block sets and disjoint dependence sets, and the primitive variables $(H_k, U_k)_{k \in \mathcal{K}(C)}$, $(\lambda_m)_{m \in \tilde{\mathcal{J}}(C)}$ of distinct components are independent under $\nu_S(\cdot | \lambda) \otimes \text{Rad}$.*

Proof. Blocks have diameter $O(r)$ and a packet has diameter $O(\ell) \leq O(r)$; a packet in $\mathcal{J}_k$ lies within $R_0 \ell$ of S_k . So two sites whose blocks share a packet, or whose packets or blocks intersect, are within $C_Q r$. Disjointness follows from the definition of $\sim$, and independence from the product structure (10). $\square$

The site likelihood L_i is a function of $(U_k)_{k \in \mathcal{K}_i}$, $(\lambda_m)_{m \in J(x_i)}$ and $w_i = (R_i, T_i)$ (the functions G_h, η, d_* are deterministic); the kernel of block k reads $(\lambda_m)_{m \in \mathcal{J}_k}$ (Proposition 15.7). With (10) and (143), the mixed joint density of (x^n, w^n) under side S equals

$$\mathcal{Z}_n^{-1} 4^{-n} \int \prod_k \prod_{i: k \in \mathcal{K}_i} F_{H_k}(\theta_k(x_i); \lambda) \prod_i L_i^S(w_i) d\nu_S(H, U | \lambda) d\text{Rad}(\lambda),$$

and by Lemma 16.1 and Fubini the integral factorises over components:

$$\mathbb{P}_n(w^n | x^n) = \prod_C \frac{N_C^P(w_C)}{N_C}, \quad \mathbb{Q}_n(w^n | x^n) = \prod_C \frac{N_C^Q(w_C)}{N_C}, \quad (145)$$

$$N_C^S(w_C) = 4^{-q_C} \int \prod_{k \in \mathcal{K}(C)} \prod_{i \in C: k \in \mathcal{K}_i} F_{H_k}(\theta_k(x_i); \lambda) \prod_{i \in C} L_i^S(w_i) d\nu_S d\text{Rad}, \quad N_C = \sum_{w_C} N_C^S(w_C),$$

where N_C does not depend on S : summing a cell over w_i gives $\sum_{w_i} L_i(w_i) = 4$ for any parameter, so $N_C = \mathbb{E}_\lambda \mathbb{E}_\nu \prod_k \prod_i F_{H_k} = \mathbb{E}_\lambda \prod_{k \in \mathcal{K}(C)} D_{q_k}(T_k; \lambda)$, with T_k the tuple of the q_k sites of C in S_k (in the chart of k). Sites in no active block are singleton components with $L_i^P = L_i^Q = 1$.

Hence, by Lemma 2.4(a),(b), on the event that all components have $q_C \leq Q$,

$$\text{TV}(\mathbb{P}_n(\cdot | x^n), \mathbb{Q}_n(\cdot | x^n)) \leq \left\{ \sum_C H_C^2 \right\}^{1/2}, \quad H_C^2 = H^2 \left(\frac{N_C^P}{N_C}, \frac{N_C^Q}{N_C} \right). \quad (146)$$

16.3 The component identity

Fix a component C with $q_C \leq Q$. For $i \in C$ write $\Lambda_i = \Lambda_\lambda(x_i)$, $G_i = G_h(x_i)$, $d_i = \mathbf{j}_a \mathbf{j}_b G_i^2$, $e_{k,i} = v_Q(\theta_k(x_i))^\top U_k$ for $k \in \mathcal{K}_i$, and $\mathbf{c}_i = \mathbf{c}(x_i) = \sum_{k \in \mathcal{K}_i} \psi_k(x_i) e_{k,i}$; the shift variable at the site is $\zeta_i = \mathbf{a} \mathbf{c}_i$ with the smooth amplitude $\mathbf{a} = b$ (case (a)) or $\mathbf{a} = a$ (case (b)), so that $\mathbf{a} \psi_k(x_i) t m_k(x_i) = \mathbf{j}_o \phi_k(x_i)$. Let $L_i(\zeta)$, $\text{real}_i(\zeta)$, $\Delta_i^{(e)}(\zeta)$, $\mathbf{K}_i^{(e,f)}$ be the objects of Section 13.5 at the site x_i with the cell $w_i = (R_i, T_i)$. Then

$$L_i^{Q,0} = L_i(\zeta_i), \quad L_i^P = L_i(\zeta_i) + \text{real}_i(\zeta_i), \quad (147)$$

where $L_i^{Q,0}$ is the Q -likelihood at the actual (unshifted) parameter. $L_i(\zeta)$ is a polynomial of degree $\mathfrak{d}$ in ζ , hence of degree $\leq \mathfrak{d}$ in each $e_{k,i}$, and affine in Λ_i , $L_i = A_i + B_i \Lambda_i$ with B_i carrying the factor $\mathbf{j}_o$; the same holds for every $\partial_\zeta^e L_i$.

P-side. Under ν_P , $U \sim \Pi_0^\otimes$ independently of H , so

$$N_C^P = 4^{-q_C} \mathbb{E}_\lambda \left[D_C \mathbb{E}_{\Pi_0^\otimes} \prod_{i \in C} (L_i(\zeta_i) + \text{real}_i(\zeta_i)) \right], \quad D_C := \prod_{k \in \mathcal{K}(C)} D_{q_k}(T_k; \lambda). \quad (148)$$

Q-side. $\prod_{i \in C} L_i(\zeta_i)$ is a polynomial in the variables $\{e_{k,i}\}$ of degree $\leq \mathfrak{d}$ in each, and for each block k of degree $\leq \mathfrak{d} q_k \leq \mathfrak{d} Q$ in U_k . Expand it into monomials $\prod_k \prod_i e_{k,i}^{n_{k,i}}$; the blocks are independent given λ , so the ν_Q -expectation of a monomial times the design factors is the product over k of $\mathbb{E}_{\nu_{Q,k}}[F^{\otimes q_k} \prod_i e_{k,i}^{n_{k,i}}]$, each given by (139). Substituting and re-summing over the monomials, the binomial coefficients reassemble into Taylor coefficients ($\sum_n (\text{coefficient}) \binom{n}{e} x^{n-e} = \partial_x^e g(x)/e!$ for a polynomial g), and $\partial_{e_{k,i}} = \mathbf{a} \psi_k(x_i) \partial_{\zeta_i}$. This gives the exact expansion

$$N_C^Q = 4^{-q_C} \mathbb{E}_\lambda \sum_{(e_k, f_k)_k} \prod_{k \in \mathcal{K}(C)} \Omega_k(f_k, e_k) \times \mathbb{E}_{\Pi_0^\otimes} \left[\prod_{i \in C \setminus S'} L_i(\zeta_i) \prod_{i \in S'} \left\{ \prod_{k \in \mathcal{K}_i: e_{k,i} \geq 1} \frac{\phi_k(x_i)^{e_{k,i}} \mathbf{K}_i^{(e_{k,i}, f_{k,i})}(\Lambda_i)}{e_{k,i}!} \right\} \mathbf{j}_o^{e(i)} \partial_\zeta^{e(i)} L_i(\zeta_i) \right], \quad (149)$$

where $e_k \in \{0, \dots, \mathfrak{d}\}^{T_k}$ is the increment pattern of block k on its sites, $f_k \in \{0, \dots, \mathfrak{d}\}^{\text{supp } e_k}$ the design exponents, $e(i) = \sum_k e_{k,i}$, $S' = \{i : e(i) \geq 1\}$, and the weights are the tapered integrated coefficients (140) of block k ,

$$\Omega_k(f_k, e_k) = \overline{\mathbb{D}}_{k, q_k}^{(f_k; \text{supp } e_k)}(T_k; \lambda), \quad \Omega_k(\emptyset, 0) = D_{q_k}(T_k; \lambda).$$

(If $e(i) > \mathfrak{d}$ the factor $\partial_\zeta^{e(i)} L_i$ vanishes.) Which blocks can act at a site is read off from the explicit factor $\prod_{i: e_{k,i} \geq 1} \phi_k(x_i)^{e_{k,i}}$ in (149): a summand with $e_{k,i} \geq 1$ at a site with $\phi_k(x_i) = 0$ vanishes. (The weight $\Omega_k(f_k, e_k)$ itself need not vanish: for $f_k = 0$ it is the integrated X -free coefficient, which is bounded but in general not small, Lemma 16.2. When $f_{k,i} \geq 1$ the weight does vanish, because by Proposition 15.7 the coefficient of X_i^f in $\mathbb{D}_k$ carries $m_k(u_i)^f = 0$; this is consistent, since $f_{k,i} \geq 1$ requires $e_{k,i} \geq 1$.) Hence every nonzero summand has, at each site with $e_{k,i} \geq 1$, $\phi_k(x_i) > 0$, i.e. the site is assigned to k or lies in the transition layer; for an interior site assigned to k , $\phi_k(x_i) = 1$ and k is the only block that can carry a nonzero increment at i . In case (a) each $e_{k,i} \in \{0, 1\}$, a site has only one factor ζ_i in the multilinear cell, and (149) is the sum over pairwise disjoint $S'_k \subseteq C \cap S_k$ and $A_k \subseteq S'_k$ (the sites with $f = 1$) of

$$\prod_k \Omega_k(A_k, S'_k) \mathbb{E}_{\Pi_0^\otimes} \left[\prod_{i \notin S'} L_i \prod_k \left\{ \prod_{i \in S'_k \setminus A_k} \mathbf{j}_b \phi_k(x_i) \Lambda_i \partial_y L_i \prod_{i \in A_k} \mathbf{j}_b \phi_k(x_i) \varkappa_i \partial_y L_i \right\} \right].$$

16.4 Matching

Let $\text{cl}(C) = \mathbf{1}\{\exists i \neq j \in C : J(x_i) \cap J(x_j) \neq \emptyset\}$, $\text{tr}(C) = \mathbf{1}\{C \cap \mathcal{L} \neq \emptyset\}$, and $g_C = \sum_{k \in \mathcal{K}(C)} g_{q_k}(T_k)$ with g_q from (141).

Lemma 16.2 (Walsh bounds). *For U in the legal ball and n large: $\|L_i(\zeta_i)\|_W, \|\partial_\zeta^e L_i(\zeta_i)\|_W \leq C$ ($e \leq \mathfrak{d}$); $\|\text{real}_i(\zeta_i)\|_W \leq C\Delta$; $j_\circ \leq j_\bullet$, $j_\circ^2 \leq c_a c_b \Delta$; $\|\mathbf{K}_i^{(e,f)}(\Lambda_i)\|_W \leq C$; $\|\Omega_k(f, e)\|_W \leq C_Q$ for all (f, e) , and $\leq C_Q \delta_n$ when $f \neq 0$ (for $f = 0 \neq e$ the weight $\Omega_k(0, e) = \int \text{ev}_\lambda \mathbb{D}^{(0;S')} \chi_Q dZ = \int \chi_Q dZ + O(\delta_n)$ is uniformly $O(1)$ and need not be $O(\delta_n)$; it can vanish, e.g. when the tuple lies where the taper is zero, and only the upper bound is used); the same for the untapered $D_{q_k}^{(f;S')}$; and $\Omega_k(f, e)$, $D_{q_k}^{(f;S')}$ have parity $(-1)^{|f|}$ in λ (the taper does not affect parity or norms).*

Proof. (116), $|\zeta_i| \leq C_Q c$, $\|\Lambda_i\|_W \leq N_0$, $j_a, j_b \leq 1$, $|\eta_i| \leq 3j_a^2$, (106), $\varkappa_i \leq j_a^2 \leq 1$, $G_i^2 \leq 1$, and Lemma 15.9(a). $\square$

Lemma 16.3 (Parity count). *Let Θ be a Walsh polynomial in λ of parity $(-1)^{|f|}$, $f = (f_k)_k$, and consider a term of the form appearing in (149),*

$$\Xi = \Theta \cdot \prod_{i \in C \setminus S'} L_i(\zeta_i) \prod_{i \in S'} \left\{ \prod_{k: e_{k,i} \geq 1} \phi_k(x_i)^{e_{k,i}} \mathbf{K}_i^{(e_{k,i}, f_{k,i})}(\Lambda_i) \right\} j_\circ^{e(i)} \partial_\zeta^{e(i)} L_i(\zeta_i), \quad S' \neq \emptyset.$$

Then $|\mathbb{E}_\lambda \Xi| \leq C_Q \|\Theta\|_W \Delta$, uniformly in U in the legal ball and in the cells.

Proof. Expand each cell factor L_i or $\partial_\zeta^e L_i$ as $A + B\Lambda_i$ and choose, for each, either the λ -free part or the part $B\Lambda_i$ (which carries the factor $j_\bullet$); let $\#y$ be the number of factors $B\Lambda_i$ chosen. Every monomial of Θ has degree $\equiv |f|$, every monomial of a nonzero $\mathbf{K}_i^{(e,f)}(\Lambda_i)$ has degree $\equiv e + f \pmod{2}$ (Section 13.5), and Λ_i has degree one. Since the parity of the degree of a product of Walsh monomials is the sum of the parities, every monomial of the chosen term has degree $\equiv |f| + \sum_{k,i} (e_{k,i} + f_{k,i}) + \#y \equiv |e| + \#y \pmod{2}$, where $|e| = \sum_i e(i) \geq 1$. A Walsh polynomial all of whose monomials have odd degree has zero λ -mean. Hence only the choices with $|e| + \#y$ even contribute, and for these the Walsh norm of the term is $\leq C_Q \|\Theta\|_W j_\circ^{|e|} j_\bullet^{\#y}$ by Lemma 16.2. If $|e| = 1$ then $\#y \geq 1$ and $j_\circ j_\bullet = c_a c_b \Delta$; if $|e| \geq 2$ then $j_\circ^2 \leq c_a c_b \Delta$. Summing over the $\leq 2^Q$ choices proves the claim. $\square$

Proposition 16.4 (Component matching). *For every component C with $q_C \leq Q$ and every cell w_C ,*

$$|N_C^Q(w_C) - N_C^P(w_C)| \leq 4^{-q_C} C_Q \Delta [g_C + \text{cl}(C) + \text{tr}(C) + \delta_n t^d]. \quad (150)$$

Proof. Write $\widehat{N}_C^Q$ for (149) with every tapered weight $\Omega_k(f_k, e_k)$, $(f_k, e_k) \neq (\emptyset, 0)$, replaced by the untapered $D_{q_k}^{(f_k; \text{supp } e_k)}$ (and $\Omega_k(\emptyset, 0) = D_{q_k}$ unchanged).

Step 1: taper. $N_C^Q - \widehat{N}_C^Q$ is a sum of C_Q terms, each with $S' \neq \emptyset$ and containing at least one factor $\Omega_k - D_{q_k}^{(f_k; \cdot)}$, of Walsh norm $\leq C_Q g_{q_k}(T_k)$ and parity $(-1)^{|f_k|}$ by (141) and Lemma 15.9(a). The product Θ of the weights and differences has parity $(-1)^{|f|}$ and Walsh norm $\leq C_Q g_{q_k}(T_k)$, so Lemma 16.3 bounds each term by $C_Q \Delta g_{q_k}(T_k)$: $|N_C^Q - \widehat{N}_C^Q| \leq 4^{-q_C} C_Q \Delta g_C$.

Step 2: crude bound when $\text{cl}(C) = 1$ or $\text{tr}(C) = 1$. In (148), every term with at least one real_i has Walsh norm $\leq C_Q \Delta$, and the term with none equals the $(f_k, e_k) = (\emptyset, 0)$ term of $\widehat{N}_C^Q$. Every other term of $\widehat{N}_C^Q$ has $S' \neq \emptyset$ and is bounded by Lemma 16.3 with $\Theta = \prod_k D_{q_k}^{(f_k; \cdot)}$ by $C_Q \Delta$. Hence $|\widehat{N}_C^Q - N_C^P| \leq 4^{-q_C} C_Q \Delta$.

Step 3: exact matching when $\text{cl}(C) = \text{tr}(C) = 0$. All sites are interior; site i is assigned to $k(i)$. By Lemma 15.9(c) and the remark after (149), only $k(i)$ can have a nonzero increment or positive retained site degree at i . Other blocks can still read its signs through symbols or ghost slots; this is controlled below. If $k(i)$ is inactive, then $G_i = \Lambda_i = d_i = 0$, so every term choosing an increment or real_i at that site is zero. Such zero terms may be omitted. Hence in (149) the increment patterns are $e = (e_i)_{i \in S'}$ with $e_i \in \{1, \dots, \mathfrak{d}\}$, $\phi_{k(i)}(x_i) = 1$, the exponents are $f = (f_i)_{i \in S'}$, and the untapered weights multiply to

$$\prod_k D_{q_k}^{(f_k; S'_k)} = D_C^{(f; S')}, \quad D_C = \sum_{f \in \{0, \dots, \mathfrak{d}\}^{S'}} \Lambda_{S'}^f D_C^{(f; S')},$$

the coefficients prescribed by the representative extraction (127), multiplied across blocks and then evaluated. Only $k(i)$ contributes a positive retained site degree at i . This is an identity for the specified expansion, not a claim of uniqueness after rough-field evaluation. Thus

$$\hat{N}_C^Q = 4^{-q_C} \mathbb{E}_\lambda \sum_{S' \subseteq C} \sum_f D_C^{(f; S')} \mathbb{E}_{\Pi_0^\otimes} \left[\prod_{i \notin S'} L_i(\zeta_i) \prod_{i \in S'} \sum_{e_i=1}^{\mathfrak{d}} \mathsf{K}_i^{(e_i, f_i)}(\Lambda_i) \Delta_i^{(e_i)}(\zeta_i) \right], \quad (151)$$

with $\Delta_i^{(e)}(\zeta) = \mathsf{j}_0^e \partial_\zeta^e L_i(\zeta) / e!$ as in (118), and by (148)

$$N_C^P = 4^{-q_C} \mathbb{E}_\lambda \sum_{S' \subseteq C} \sum_f \Lambda_{S'}^f D_C^{(f; S')} \mathbb{E}_{\Pi_0^\otimes} \left[\prod_{i \notin S'} L_i(\zeta_i) \prod_{i \in S'} \text{real}_i(\zeta_i) \right]. \quad (152)$$

The terms with $S' = \emptyset$ agree exactly and need no replacement. Fix $S' \neq \emptyset$ and f for a remaining pair of terms, and let $J_{S'} = \bigcup_{i \in S'} J(x_i)$. For every incident block, including one with $S'_k = \emptyset$, define

$$A_k = D_{k, q_k}^{(f_k; S'_k)}(T_k; \cdot), \quad \tilde{D}_k = \mathbb{E}_{J_{S'}} A_k, \quad \tilde{D}_C = \prod_{k \in \mathcal{K}(C)} \tilde{D}_k.$$

Lemma 15.10, applied with $\mathcal{I} = C$, gives

$$\|A_k - \tilde{D}_k\|_W \leq C_Q \delta_n t^d, \quad \|A_k\|_W + \|\tilde{D}_k\|_W \leq C_Q,$$

and both coefficients have parity $(-1)^{|f_k|}$. In particular, $\tilde{D}_C$ is independent of every sign in $J_{S'}$. Averaging is performed factor by factor; no factorization of $\mathbb{E}_{J_{S'}}$ over the dependent block coefficients is assumed.

Exact matching of the replaced terms. Replace $D_C^{(f; S')}$ by the same $\tilde{D}_C$ in (151) and (152). Since $\text{cl}(C) = 0$, the fields Λ_i , $i \in S'$, are independent of each other, of all remaining retained fields, of $\tilde{D}_C$, and of the baseline coefficients $U \sim \Pi_0^\otimes$. At fixed U and fixed remaining signs, average these fields one site at a time. The Q -side site factor is

$$\sum_{e=1}^{\mathfrak{d}} \mathbb{E}_{\Lambda_i} [\mathsf{K}_i^{(e, f_i)}(\Lambda_i) \Delta_i^{(e)}(\zeta_i)],$$

whereas the P -side factor is $\mathbb{E}_{\Lambda_i} [\Lambda_i^{f_i} \text{real}_i(\zeta_i)]$. They agree by Lemma 13.2 for every baseline value ζ_i and every binary cell. All other factors are the same, so the replaced (S', f) -terms agree exactly.

Residual dependence. The difference $D_C^{(f; S')} - \tilde{D}_C = \prod_k A_k - \prod_k \tilde{D}_k$ is a telescoping sum of at most $|\mathcal{K}(C)| \leq N_1 Q$ products. Each product contains one factor $A_k - \tilde{D}_k$ of Walsh norm at

most $C_Q \delta_n t^d$, all other factors have Walsh norm at most C_Q , and the product has parity $(-1)^{|f|}$. Lemma 16.3 therefore bounds the replacement error in the Q -side term by $C_Q \Delta \delta_n t^d$. On the P -side, $S' \neq \emptyset$ guarantees at least one real_i factor of Walsh norm at most $C\Delta$; $\|\Lambda_{S'}^f\|_W \leq C_Q$, so its replacement error has the same bound directly. There are only C_Q choices of (S', f) , whence

$$|\widehat{N}_C^Q - N_C^P| \leq 4^{-q_C} C_Q \Delta \delta_n t^d \quad \text{when } \text{cl}(C) = \text{tr}(C) = 0.$$

Combining the three steps gives (150). $\square$

Lemma 16.5. *For every component with $q_C \leq Q$,*

$$H_C^2 \leq C_Q \Delta^2 [g_C + \text{cl}(C) + \text{tr}(C) + \delta_n t^d]. \quad (153)$$

Proof. Normalisation. $N_C = \mathbb{E}_\lambda D_C \in [(1 - \eta_f)^{QK}, (1 + \eta_f)^{QK}]$ with $K = |\mathcal{K}(C)| \leq N_1 Q$. *Cell lower bound.* For every λ, U in the legal ball and w_i , (5) gives $L_i^P \geq 1 - |p| - |y + \tau(1 + p)| - |py + \tau(1 + p)| \geq 1 - C_Q c_a - C_Q c_b - o(1) \geq \frac{1}{2}$ for small c_a, c_b and large n (Lemma 14.1); hence $\mathbb{E}_{\Pi_0^\otimes} \prod_i L_i^P(w_i) \geq 2^{-q_C}$ and $N_C^P(w_C)/N_C \geq 8^{-Q} =: c_Q$ by (148). *Conclusion.* By Lemma 2.4(c) and (150), with $\varrho_C = g_C + \text{cl} + \text{tr} + \delta_n t^d \leq C_Q$, $H_C^2 \leq c_Q^{-1} 4^Q N_C^{-2} (4^{-q_C} C_Q \Delta \varrho_C)^2 \leq C_Q \Delta^2 \varrho_C^2 \leq C_Q \Delta^2 \varrho_C$. $\square$

16.5 Hellinger sums

Let m_I be the marginal density of $(X_i)_{i \in I}$ under the common design law; $m_I \leq \bar{f}^{|I|}$ since $f_H \leq \bar{f}$. Let $\mathcal{A}_n = \bigcup_{k \text{ active}} S_k$ be the active region; $|\mathcal{A}_n| \leq Ch^d$ (there are $O(h^d r^{-d})$ active blocks of volume Cr^d), and $\text{supp } G_h \subseteq \mathcal{A}_n$. A site outside $\mathcal{A}_n$ has $\mathcal{K}_i = \emptyset$ and $J(x_i) = \emptyset$, hence is a singleton component with $L_i^P = L_i^Q = 1$ and $H_C^2 = 0$; only components meeting $\mathcal{A}_n$ contribute. Call I *active* if it is connected (for $\sim$) and $x_I \cap \mathcal{A}_n \neq \emptyset$; this is a property of x_I , and the right side of (153) for $C = I$ depends only on x_I . Since components are connected,

$$\mathbb{E} \sum_{C: q_C \leq Q} H_C^2 \leq C_Q \Delta^2 \sum_{q=1}^Q \binom{n}{q} \int \mathbf{1}\{I \text{ active}\} [g_I + \text{cl}(I) + \text{tr}(I) + \delta_n t^d] m_I dx_I. \quad (154)$$

Lemma 16.6. *For $|I| = q \leq Q$, with $V_q := \int \mathbf{1}\{I \text{ active}\} m_I dx_I \leq C_Q h^d (C_Q r)^{d(q-1)}$,*

$$\begin{aligned} n^q \int \mathbf{1}\{I \text{ active}\} g_I m_I dx_I &\leq C_Q n h^d t^d \Lambda^{M_Q} (nr^d)^{q-1}, \\ n^q \int \mathbf{1}\{I \text{ active}\} \text{cl}(I) m_I dx_I &\leq C_Q n^2 h^d \ell^d (nr^d)^{q-2} \quad (q \geq 2), \\ n^q \int \mathbf{1}\{I \text{ active}\} \text{tr}(I) m_I dx_I &\leq C_Q n h^d t^d (nr^d)^{q-1}, \\ n^q \delta_n t^d V_q &\leq C_Q n h^d t^d \delta_n (nr^d)^{q-1}. \end{aligned}$$

Consequently $\mathbb{E} \sum_{C: q_C \leq Q} H_C^2 \leq C_Q \Delta^2 \{n h^d t^d \Lambda^{M_Q} + n^2 h^d \ell^d\} \rightarrow 0$.

Proof. By Lemma 16.1 every link has length $\leq C_Q r$. Order the sites of an active I so that the first site lies in $\mathcal{A}_n$ and each later site is linked to an earlier one (C_Q orderings; a connected set meeting $\mathcal{A}_n$ admits such an ordering); the first site has volume $|\mathcal{A}_n| \leq Ch^d$, each later one $(C_Q r)^d$; this gives V_q (for $q = 1$ it is $|\mathcal{A}_n|$).

Ghost. Fix the block k_0 in g_I and its sites I_0 , $q_0 = |I_0| \geq 1$; order I with I_0 first. There are $O(h^d r^{-d})$ active blocks; given k_0 , the sites of I_0 lie in S_{k_0} and contribute $\int_{(\mathbb{T}^d)^{q_0}} g_{q_0} dT \leq C_Q t^d \Lambda^{M_Q}$ by (142), i.e. $C_Q r^{dq_0} t^d \Lambda^{M_Q}$ in physical units; the other $q - q_0$ sites contribute $(C_Q r)^d$ each.

Collision. Choose the pair with a shared packet: the first site in $\text{supp } G_h$ (Ch^d), the second within $R_0\ell$ ($C\ell^d$), the rest $(C_Qr)^d$ each.

Transition. Choose the site in $\mathcal{L}$; since I is active, connected, and all links have length $\leq C_Qr$, this site lies within C_QQr of $\mathcal{A}_n$, i.e. in the transition layer of one of the $O(h^dr^{-d})$ blocks within that distance of the active region, a set of volume $\leq C_d t^d \cdot O(h^d)$ in total; the rest $(C_Qr)^d$ each.

Residual. V_q times $\delta_n t^d$.

Conclusion. $nr^d = n^{-cd\varepsilon} \leq 1$ and $\delta_n \leq 1$, so all four bounds are $\leq C_Q\{nh^dt^d\Lambda^{M_Q} + n^2h^d\ell^d\}$; apply (14)–(15). $\square$

16.6 High components

Let $\mathcal{E}_n$ be the event that some component has more than Q sites. Such a component meets $\mathcal{A}_n$ (components outside $\mathcal{A}_n$ are singletons), so some active set I of exactly $Q+1$ sites exists, and by the ordering argument

$$\mathbb{P}(\mathcal{E}_n) \leq \binom{n}{Q+1} \int \mathbf{1}\{I \text{ active}\} m_I dx_I \leq C_Q \bar{f}^{Q+1} n^{Q+1} h^d (C_Q r)^{dQ} \rightarrow 0$$

by (16).

16.7 Conclusion

By Lemma 2.5 (the conditional laws factor over the components on $\mathcal{E}_n^c$), Lemma 16.6 and Section 16.6,

$$\text{TV}(\mathbb{P}_n, \mathbb{Q}_n) \leq \mathbb{P}(\mathcal{E}_n) + \left\{ \mathbb{E} \sum_{C: q_C \leq Q} H_C^2 \right\}^{1/2} \rightarrow 0,$$

so $\text{TV}(\mathbb{P}_n, \mathbb{Q}_n) \leq 1/2$ for $n \geq n_\varepsilon$, and the two-point reduction (Lemma 2.3) proves Theorem 1.1 in case (C) for all sufficiently small fixed $\varepsilon > 0$. For an arbitrary prescribed $\varepsilon > 0$, apply the construction with a fixed $0 < \varepsilon_0 \leq \varepsilon$ in that range; since $n^{-(\rho+\varepsilon_0)} \geq n^{-(\rho+\varepsilon)}$, the stated bound follows after changing the constant. $\square$

17 Case C: remark

Remark 17.1 (What the carrier does). *The common design density may depend on the rough field: a label law independent of λ does not make its evaluated density atoms independent of λ . The construction chooses the Q -side conditional law of the smooth coefficients to reproduce the design-weighted moment increments. The substitution polynomials (119) supply the design-weighted part of the P -side real field. For an interior, collision-free component, the untapered cells match exactly after the block coefficients have been averaged over the selected local signs. Lemma 15.10 controls the error of this replacement, including in blocks with no increment.*

Only upper bounds on design coefficients and their residual local-sign dependence are used: Lemma 15.9 and Lemma 15.10 give the required bounds. No sharp asymptotic order for a particular pair-design coefficient is claimed. The remaining discrepancy is charged to the taper, collision, transition, and residual terms in Proposition 16.4. The decisive properties are bounded site degree, assignment of each interior site's jitter to one block, and reflection symmetry for the parity count. The representative-space formulation keeps these properties valid without identifying representative arrays that have the same realization. None of these corrections changes the scales or proves the endpoint $\varepsilon = 0$ in case (C).

References

- [1] E. H. Kennedy, S. Balakrishnan, J. M. Robins, L. Wasserman. Minimax rates for heterogeneous causal effect estimation. *Annals of Statistics* 52(2), 793–816, 2024.
- [2] A. B. Tsybakov. *Introduction to Nonparametric Estimation*. Springer, 2009.
- [3] R. O'Donnell. *Analysis of Boolean Functions*. Cambridge University Press, 2014.